

QCM NSE4

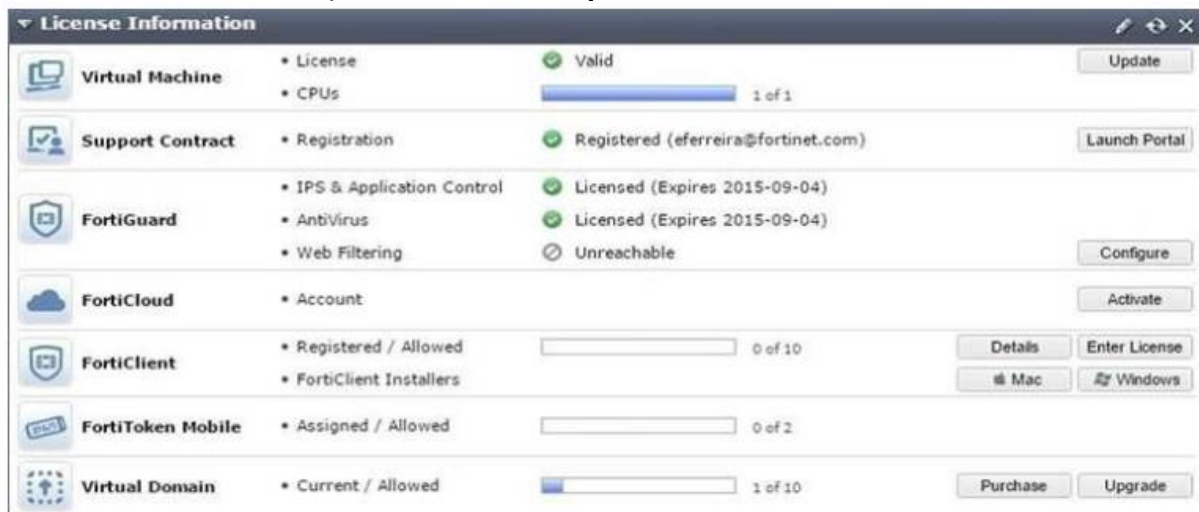
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QCM_NSE4_Base

1. Examine the exhibit; then answer the question below.



Which statement describes the green status indicators that appear next to the different FortiGuard Distribution Network services as illustrated in the exhibit?

- A. They indicate that the FortiGate has the latest updates available from the FortiGuard Distribution Network.
- B. They indicate that updates are available and should be downloaded from the FortiGuard Distribution Network to the FortiGate unit.
- C. They indicate that the FortiGate is in the process of downloading updates from the FortiGuard Distribution Network.
- D. They indicate that the FortiGate is able to connect to the FortiGuard Distribution Network.

2. Which network protocols are supported for administrative access to a FortiGate unit? (Choose three.)

- A. SMTP
- B. WINS
- C. HTTP
- D. HTTPS
- E. SSH

3. What is the FortiGate password recovery process?

- A. Interrupt boot sequence, modify the boot registry and reboot. After changing the password, reset the boot registry.
- B. Log in through the console port using the "maintainer" account within several seconds of physically power cycling the FortiGate.
- C. Hold down the CTRL + Esc (Escape) keys during reboot, then reset the admin password.
- D. Interrupt the boot sequence and restore a configuration file for which the password has been modified.

4. What methods can be used to deliver the token code to a user who is configured to use two-factor authentication? (Choose three.)

- A. Code blocks
- B. SMS phone message
- C. FortiToken
- D. Browser pop-up window
- E. Email

5. Which statements correctly describe transparent mode operation? (Choose three.)

- A. All interfaces of the transparent mode FortiGate device must be on different IP subnets.
- B. The transparent FortiGate is visible to network hosts in an IP traceroute.
- C. It permits inline traffic inspection and firewalling without changing the IP scheme of the network.
- D. Ethernet packets are forwarded based on destination MAC addresses, not IP addresses.
- E. The FortiGate acts as transparent bridge and forwards traffic at Layer-2.

6. View the exhibit.

The screenshot shows the 'Edit Interface' configuration window for 'port1'. The 'Interface Name' is 'port1 (00:0C:29:29:38:DA)'. The 'Alias' field is empty. The 'Link Status' is 'Up' with a green up arrow icon. The 'Type' is 'Physical Interface'. The 'Virtual Domain' is 'root'. The 'Role' is set to 'Undefined' in a dropdown menu.

When **Role** is set to **Undefined**, which statement is true?

- A. The GUI provides all the configuration options available for the **port1** interface.
- B. You cannot configure a static IP address for the **port1** interface because it allows only DHCP addressing mode.
- C. Firewall policies can be created from only the **port1** interface to **any** interface.
- D. The **port1** interface is reserved for management only.

7. Which of the following settings and protocols can be used to provide secure and restrictive administrative access to FortiGate? (Choose three.)

- A. Trusted host
- B. HTTPS
- C. Trusted authentication
- D. SSH
- E. FortiTelemetry

8. Which antivirus and attack definition update options are supported by FortiGate units? (Choose two.)

- A. Manual update by downloading the signatures from the support site.
- B. FortiGuard pull updates.
- C. Push updates from a FortiAnalyzer.
- D. execute fortiguard-AV-AS command from the CLI.

9. In which process states is it impossible to interrupt/kill a process? (Choose two.)

- A. S – Sleep
- B. R – Running
- C. D – Uninterruptable Sleep
- D. Z – Zombie

10. Which is not a FortiGate feature?

- A. Database auditing
- B. Intrusion prevention
- C. Web filtering
- D. Application control

11. When an administrator attempts to manage FortiGate from an IP address that is not a trusted host, what happens?

- A. FortiGate will still subject that person's traffic to firewall policies; it will not bypass them.
- B. FortiGate will drop the packets and not respond.
- C. FortiGate responds with a block message, indicating that it will not allow that person to log in.
- D. FortiGate responds only if the administrator uses a secure protocol. Otherwise, it does not respond

12. A backup file begins with this line: Can you restore it to a FortiWiFi 60D?

```
#config-version=FGVM64-5.02-FW-build589-140613:opmode=0:vdom=0:user=admin  
#conf_file_ver=3881503152630288414 #buildno=0589 #global_vdom=1
```

- A. Yes
- B. Yes, but only if you replace the "#conf_file_ver" line so that it contains the serial number of that specific FortiWiFi 60D.
- C. Yes, but only if it is running the same version of FortiOS, or a newer compatible version.
- D. No

13. You have configured the DHCP server on a FortiGate's port1 interface (or internal, depending on the model) to offer IPs in a range of 192.168.1.65-192.168.1.253. When the first host sends a DHCP request, what IP will the DHCP offer?

- A. 192.168.1.99
- B. 192.168.1.253
- C. 192.168.1.65
- D. 192.168.1.66

14. You have created a new administrator account, and assign it the prof_admin profile. Which is false about that account's permissions?

- A. It cannot upgrade or downgrade firmware.
- B. It can create and assign administrator accounts to parts of its own VDOM.
- C. It can reset forgotten passwords for other administrator accounts such as "admin".
- D. It has a smaller permissions scope than accounts with the "super_admin" profile.

15. A new version of FortiOS firmware has just been released. When you upload new firmware, which is true?

- A. If you upload the firmware image via the boot loader's menu from a TFTP server, it will not preserve the configuration. But if you upload new firmware via the GUI or CLI, as long as you are following a supported upgrade path, FortiOS will attempt to convert the existing configuration to be valid with any new or changed syntax.
- B. No settings are preserved. You must completely reconfigure.
- C. No settings are preserved. After the upgrade, you must upload a configuration backup file. FortiOS will ignore any commands that are not valid in the new OS. In those cases, you must reconfigure settings that are not compatible with the new firmware.
- D. You must use FortiConverter to convert a backup configuration file into the syntax required by the new FortiOS, then upload it to FortiGate.

16. If you have lost your password for the "admin" account on your FortiGate, how should you reset it?

- A. Log in with another administrator account that has "super_admin" profile permissions, then reset the password for the "admin" account.
- B. Reboot the FortiGate. Via the local console, during the boot loader, use the menu to format the flash disk and reinstall the firmware. Then you can log in with the default password.
- C. Power off the FortiGate. After several seconds, restart it. Via the local console, within 30 seconds after booting has completed, log in as "maintainer" and enter the CLI commands to set the password for the "admin" account.
- D. Reboot the FortiGate. Via the local console, during the boot loader, use the menu to log in as "maintainer" and enter the CLI commands to set the password for the "admin" account.

17. Which methods can FortiGate use to send a One Time Password (OTP) to Two-Factor Authentication users? (Choose three.)

- A. Hardware FortiToken
- B. Web Portal
- C. Email
- D. USB Token
- E. Software FortiToken (FortiToken mobile)

18. Which does FortiToken use as input when generating a token code? (Choose two.)

- A. User password
- B. Time
- C. User name
- D. Seed

19. Which of the following are operating mode supported in FortiGate devices? (Choose two)

- A. Proxy
- B. Transparent
- C. NAT/route
- D. Offline inspection

20. What types of troubleshooting can you do when uploading firmware? (Choose two.)

- A. Investigate corrupted firmware
- B. Investigate current runtime state
- C. Investigate damaged hardware
- D. Investigate configuration history

21. Which are outputs for the command 'diagnose hardware deviceinfo nic'? (Choose two.)

- A. ARP cache
- B. Physical MAC address
- C. Errors and collisions
- D. Listening TCP ports

22. Which commands are appropriate for investigating high CPU? (Choose two.)

- A. diag sys top
- B. diag hardware sysinfo mem
- C. diag debug flow
- D. get system performance status

23. An administrator wants to configure a FortiGate as a DNS server. The FortiGate must use its DNS database first, and then relay all irresolvable queries to an external DNS server. Which of the following DNS method must you use?

- A. Non-recursive
- B. Recursive
- C. Forward to primary and secondary DNS
- D. Forward to system DNS

24. Which of the following Fortinet products can receive updates from the FortiGuard Distribution Network?

- A. FortiGate
- B. FortiClient
- C. FortiMail
- D. FortiAnalyzer

25. A FortiGate is configured to receive push updates from the FortiGuard Distribution Network, however, updates are not being received. Which are two reasons for this problem? (Choose two.)

- A. The FortiGate is connected to multiple ISPs.
- B. There is a NAT device between the FortiGate and the FortiGuard Distribution Network and no override push IP is configured.
- C. The FortiGate is in Transparent mode which does not support push updates.
- D. The external facing interface of the FortiGate is configured to get the IP address from a DHCP server.

26. Which of the following network protocols are supported for administrative access to a FortiGate unit?

- A. HTTPS, HTTP, SSH, TELNET, PING, SNMP
- B. FTP, HTTPS, NNTP, TCP, WINS
- C. HTTP, NNTP, SMTP, DHCP
- D. Telnet, FTP, RLOGIN, HTTP, HTTPS, DDNS
- E. Telnet, UDP, NNTP, SMTP

27. Which of the following statements is correct regarding a FortiGate unit operating in NAT/Route mode?

- A. The FortiGate unit applies NAT to all traffic.
- B. The FortiGate unit functions as a Layer 3 device.
- C. The FortiGate unit functions as a Layer 2 device.
- D. The FortiGate unit functions as a router and the firewall function is disabled.

28. Which of the following statements is correct regarding a FortiGate unit operating in NAT/Route mode?

- A. The FortiGate unit requires only a single IP address for receiving updates and configuring from a management computer.
- B. The FortiGate unit must use public IP addresses on both the internal and external networks.
- C. The FortiGate unit commonly uses private IP addresses on the internal network but hides them using network address translation.
- D. The FortiGate unit uses only DHCP-assigned IP addresses on the internal network.

29. A FortiGate unit can provide which of the following capabilities? (Select all that apply.)

- A. Email filtering
- B. Firewall
- C. VPN gateway
- D. Mail relay
- E. Mail server

30. Which of the following methods can be used to access the CLI? (Select all that apply.)

- A. By using a direct connection to a serial console.
- B. By using the CLI console window in the GUI.
- C. By using an SSH connection.
- D. By using a Telnet connection.

31. FILL BLANK

The _____ CLI command is used on the FortiGate unit to run static commands such as ping or to reset the FortiGate unit to factory defaults.

32. When backing up the configuration file on a FortiGate unit, the contents can be encrypted by enabling the encrypt option and supplying a password.

If the password is forgotten, the configuration file can still be restored using which of the following methods?

- A. Selecting the recover password option during the restore process.
- B. Having the password emailed to the administrative user by selecting the Forgot Password option.
- C. Sending the configuration file to Fortinet Support for decryption.
- D. If the password is forgotten, there is no way to use the file.

33. When creating administrative users which of the following configuration objects determines access rights on the FortiGate unit.

- A. profile
- B. allowaccess interface settings
- C. operation mode
- D. local-in policy

34. Which of the following statements are true of the FortiGate unit's factory default configuration?

- A. 'Port1' or 'Internal' interface will have an IP of 192.168.1.99.
- B. 'Port1' or 'Internal' interface will have a DHCP server set up and enabled (on devices that support DHCP Servers).
- C. Default login will always be the username: admin (all lowercase) and no password.
- D. The implicit firewall action is ACCEPT.

35. Under the System Information widget on the dashboard, which of the following actions are available for the system configuration? (Select all that apply.)

- A. Backup
- B. Restore
- C. Revisions
- D. Export

36. Which of the following products can be installed on a computer running Windows to provide personal firewall protection, antivirus protection, web and mail filtering, spam filtering, and VPN functionality?

- A. FortiGate
- B. FortiAnalyzer
- C. FortiClient
- D. FortiManager
- E. FortiReporter

37. Which statement about FortiGuard services for FortiGate is true?

- A. The web filtering database is downloaded locally on FortiGate.
- B. Antivirus signatures are downloaded locally on FortiGate.
- C. FortiGate downloads IPS updates using UDP port 53 or 8888.
- D. FortiAnalyzer can be configured as a local FDN to provide antivirus and IPS updates.

38. An administrator configures a FortiGate unit in Transparent mode on the 192.168.11.0 subnet. Automatic Discovery is enabled to detect any available FortiAnalyzers on the network.

Which of the following FortiAnalyzers will be detected?

- A. 192.168.11.100
- B. 192.168.11.251
- C. 192.168.10.100
- D. 192.168.10.251

39. What are valid options for handling DNS requests sent directly to a FortiGate's interface IP? (Choose three.)

- A. Conditional-forward.
- B. Forward-only.
- C. Non-recursive.
- D. Iterative.
- E. Recursive.

QCM NSE4 FW rules

**40. Which correctly define "Section View" and "Global View" for firewall policies?
(Choose two.)**

- A. Section View lists firewall policies primarily by their interface pairs
- B. Section View lists firewall policies primarily by their sequence number.
- C. Global View lists firewall policies primarily by their interface pairs.
- D. Global View lists firewall policies primarily by their policy sequence number.
- E. The 'any' interface may be used with Section View.

41. Which is true about incoming and outgoing interfaces in firewall policies?

- A. A physical interface may not be used.
- B. A zone may not be used.
- C. Multiple interfaces may not be used for both incoming and outgoing.
- D. Source and destination interfaces are mandatory.

42. Which traffic can match a firewall policy's "Services" setting? (Choose three.)

- A. HTTP
- B. SSL
- C. DNS
- D. RSS
- E. HTTPS

43. Which is NOT true about source matching with firewall policies?

- A. A source address object must be selected in the firewall policy.
- B. A source user/group may be selected in the firewall policy.
- C. A source device may be defined in the firewall policy.
- D. A source interface must be selected in the firewall policy.
- E. A source user/group and device must be specified in the firewall policy.

44. For traffic that does match any configured firewall policy, what is the default action taken by the FortiGate?

- A. The traffic is allowed and no log is generated.
- B. The traffic is allowed and logged.
- C. The traffic is blocked and no log is generated.
- D. The traffic is blocked and logged.

45. In which order are firewall policies processed on a FortiGate unit?

- A. From top to bottom, according with their sequence number.
- B. From top to bottom, according with their policy ID number.
- C. Based on best match.
- D. Based on the priority value.

46. Which statement is true regarding the policy ID numbers of firewall policies?

- A. Change when firewall policies are re-ordered.
- B. Defined the order in which rules are processed.
- C. Are required to modify a firewall policy from the CLI.
- D. Represent the number of objects used in the firewall policy.

47. Which configuration objects can be selected for the Source field of a firewall policy? (Choose two.)

- A. FQDN address
- B. IP pool
- C. User or user group
- D. Firewall service

48. Which firewall objects can be included in the Destination Address field of a firewall policy? (Choose three.)

- A. IP address pool.
- B. Virtual IP address.
- C. IP address.
- D. IP address group.
- E. MAC address

49. The order of the firewall policies is important. Policies can be re-ordered from either the GUI or the CLI. Which CLI command is used to perform this function?

- A. set order
- B. edit policy
- C. reorder
- D. move

50. Which of the following items is NOT a packet characteristic matched by a firewall service object?

- A. ICMP type and code
- B. TCP/UDP source and destination ports
- C. IP protocol number
- D. TCP sequence number

51. What are the valid sub-types for a Firewall type policy? (Select all that apply)

- A. Device Identity
- B. Address
- C. User Identity
- D. Schedule
- E. SSL VPN

52. Which of the following pieces of information can be included in the Destination Address field of a firewall policy? (Select all that apply.)

- A. An IP address pool.
- B. A virtual IP address.
- C. An actual IP address or an IP address group.
- D. An FQDN or Geographic value(s).

53. You wish to create a firewall policy that applies only to traffic intended for your web server. The web server has an IP address of 192.168.2.2 and a /24 subnet mask. When defining the firewall address for use in this policy, which one of the following addresses is correct?

- A. 192.168.2.0 / 255.255.255.0
- B. 192.168.2.2 / 255.255.255.0
- C. 192.168.2.0 / 255.255.255.255
- D. 192.168.2.2 / 255.255.255.255

54. In NAT/Route mode when there is no matching firewall policy for traffic to be forwarded by the Firewall, which of the following statements describes the action taken on traffic?

- A. The traffic is blocked.
- B. The traffic is passed.
- C. The traffic is passed and logged.
- D. The traffic is blocked and logged.

55. FortiGate units are preconfigured with four default protection profiles. These protection profiles are used to control the type of content inspection to be performed. What action must be taken for one of these profiles to become active?

- A. The protection profile must be assigned to a firewall policy.
- B. The "Use Protection Profile" option must be selected in the Web Config tool under the sections for AntiVirus, IPS, WebFilter, and AntiSpam.
- C. The protection profile must be set as the Active Protection Profile.
- D. All of the above.

56. A FortiGate 60 unit is configured for your small office. The DMZ interface is connected to a network containing a web server and email server. The Internal interface is connected to a network containing 10 user workstations and the WAN1 interface is connected to your ISP.

You want to configure firewall policies so that your users can send and receive email messages to the email server on the DMZ network.

You also want the email server to be able to retrieve email messages from an email server hosted by your ISP using the POP3 protocol.

Which policies must be created for this communication? (Select all that apply.)

- A. Internal > DMZ
- B. DMZ > Internal
- C. Internal > WAN1
- D. WAN1 > Internal
- E. DMZ > WAN1
- F. WAN1 > DMZ

57. In which order are firewall policies processed on a FortiGate unit?

- A. From top to bottom, according with their sequence number.
- B. From top to bottom, according with their policy ID number.
- C. Based on best match.
- D. Based on the priority value.

58. Which header field can be used in a firewall policy for traffic matching?

- A. ICMP type and code.
- B. DSCP.
- C. TCP window size.
- D. TCP sequence number.

59. Which two statements are true regarding firewall policy disclaimers? (Choose two.)

- A. They cannot be used in combination with user authentication.
- B. They can only be applied to wireless interfaces.
- C. Users must accept the disclaimer to continue.
- D. The disclaimer page is customizable.

60. UTM features can be applied to which of the following items?

- A. Firewall policies
- B. User groups
- C. Policy routes
- D. Address groups

61. Each UTM feature has configurable UTM objects such as sensors, profiles or lists that define how the feature will function. How are UTM features applied to traffic?

- A. One or more UTM features are enabled in a firewall policy.
- B. In the system configuration for that UTM feature, you can identify the policies to which the feature is to be applied.
- C. Enable the appropriate UTM objects and identify one of them as the default.
- D. For each UTM object, identify which policy will use it.

62. If no firewall policy is specified between two FortiGate interfaces and zones are not used, which of the following statements describes the action taken on traffic flowing between these interfaces?

- A. The traffic is blocked.
- B. The traffic is passed.
- C. The traffic is passed and logged.
- D. The traffic is blocked and logged.

QCM NSE4 : NAT

63. Which is NOT true about the settings for an IP pool type port block allocation?

- A. A Block Size defines the number of connections.
- B. Blocks Per User defines the number of connection blocks for each user.
- C. An Internal IP Range defines the IP addresses permitted to use the pool.
- D. An External IP Range defines the IP addresses in the pool.

64. Which statements about One-to-One IP pool are true? (Choose two.)

- A. It allows configuration of ARP replies.
- B. It allows fixed mapping of an internal address range to an external address range.
- C. It is used for destination NAT.
- D. It does not use port address translation.

65. How does FortiGate select the central SNAT policy that is applied to a TCP session?

- A. It selects the SNAT policy specified in the configuration of the outgoing interface.
- B. It selects the first matching central-SNAT policy from top to bottom.
- C. It selects the central-SNAT policy with the lowest priority.
- D. It selects the SNAT policy specified in the configuration of the firewall policy that matches the traffic.

66. Parmi les choix suivants, lequel permet à un hôte externe de joindre un hôte interne?

- A Port forwarding
- B NAT overload
- C PPPoE
- D NAT64
- E ATM

67. Quelles implémentation NAT permet de limiter le nombre de connexions par adresse IP ?

- A. IP pool type : Overload
- B. IP pool type : One-to-one
- C. IP pool type : Fixed port range
- D. IP pool type : Port block allocation

68. Which statements are true regarding firewall policy NAT using the outgoing interface IP address with fixed port disabled? (Choose two.)

- A. This is known as many-to-one NAT.
- B. Source IP is translated to the outgoing interface IP.
- C. Connections are tracked using source port and source MAC address.
- D. Port address translation is not used.

69. Examine the exhibit, which contains a virtual IP and a firewall policy configuration.

Virtual IP

Name: VIP

Comments: 0/255

Network

Interface: WAN (port1)

Type: Static NAT

External IP Address/Range: 10.200.1.10 - 10.200.1.10

Mapped IP Address/Range: 10.0.1.10 - 10.0.1.10

Source Address Filter: ☐

Port Forwarding: ☐

OK Cancel

Firewall Policies

Seq.#	Name	Source	Destination	Schedule	Service	Action	NAT
LAN (port2) - WAN (port1) (1 - 1)							
1	Full Access	all	all	always	ALL	ACCEPT	Enabled
WAN (port1) - LAN (port2) (2 - 2)							
2	WebServer	all	VIP	always	ALL	ACCEPT	Disabled

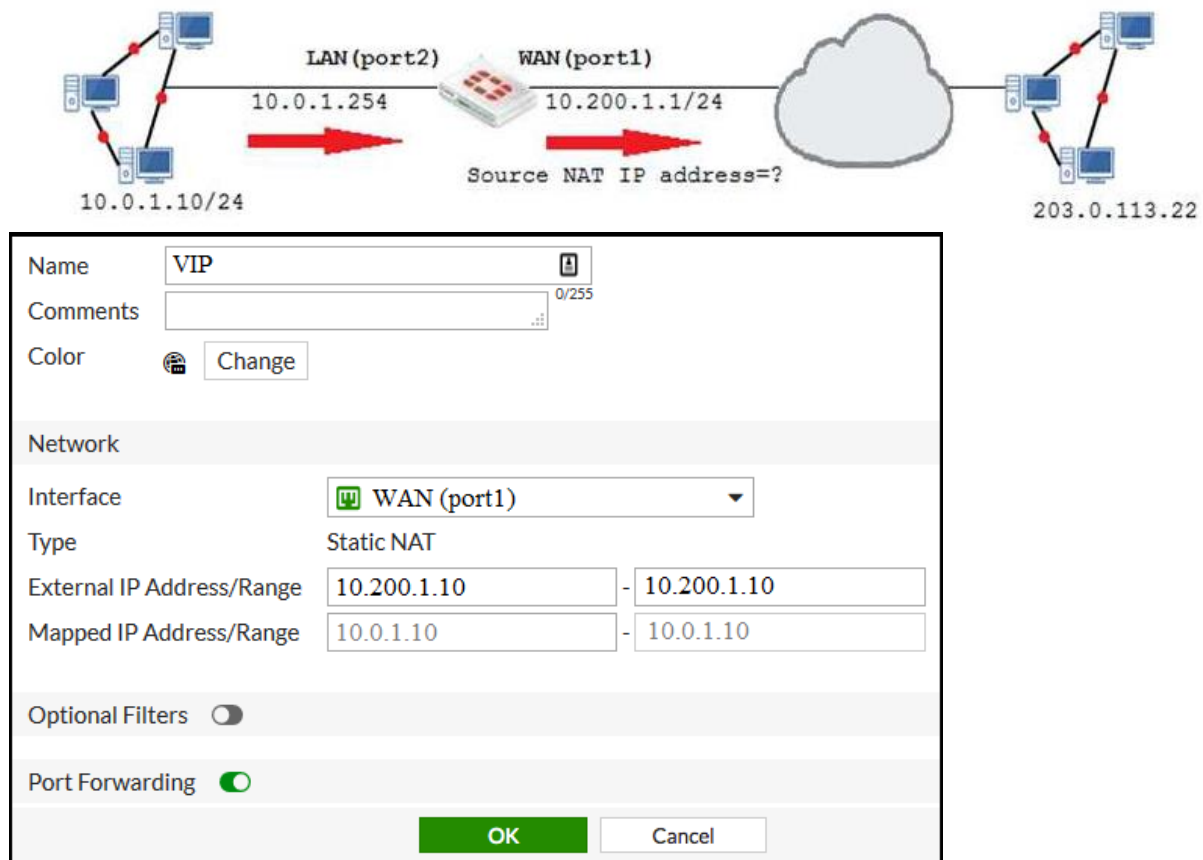
The **WAN(port1)** interface has the IP address 10.200.1.1/24. The **LAN(port2)** interface has the IP address 10.0.1.254/24.

The top firewall policy has NAT enabled using outgoing interface address. The second firewall policy configured with a virtual IP (**VIP**) as the destination address.

Which IP address will be used to source NAT the Internet traffic coming from a workstation with the IP address 10.0.1.10/24?

- A. 10.200.1.1
- B. 10.0.1.254
- C. Any available IP address in the **WAN(port1)** subnet 10.200.1.0/24
- D. 10.200.1.10

70. Examine the exhibit, which contains a virtual IP and firewall policy configuration.
Network Diagram



Firewall Policies

ID	Name	Source	Destination	Schedule	Service	Action	NAT
LAN(port2) → WAN(port1) 1							
1	Full_Access	all	all	always	ALL	✓ ACCEPT	✓ Enabled
LAN(port1) → WAN(port2) 1							
2	WebServer	all	VIP	always	ALL	✓ ACCEPT	✗ Disabled

The WAN (port1) interface has the IP address 10.200.1.1/24. The LAN (port2) interface has the IP address 10.0.1.254/24.

The first firewall policy has NAT enabled on the outgoing interface address. The second firewall policy is configured with a VIP as the destination address.

Which IP address will be used to source NAT the Internet traffic coming from a workstation with the IP address 10.0.1.10/24?

- A. 10.200.1.10
- B. Any available IP address in the WAN (port1) subnet 10.200.1.0/24
- C. 10.200.1.1
- D. 10.0.1.254

QCM NSE4 Routing

71. Examine the two static routes to the same destination subnet 172.20.168.0/24 as shown below; then answer the question following it.

```
config router static
edit 1
set dst 172.20.168.0 255.255.255.0
set distance 20
set priority 10
set device port1
next
edit 2
set dst 172.20.168.0 255.255.255.0
set distance 20
set priority 20
set device port2
next
end
```

Which of the following statements correctly describes the static routing configuration provided above?

- A. The FortiGate evenly shares the traffic to 172.20.168.0/24 through both routes.
- B. The FortiGate shares the traffic to 172.20.168.0/24 through both routes, but the port2 route will carry approximately twice as much of the traffic.
- C. The FortiGate sends all the traffic to 172.20.168.0/24 through port1.
- D. Only the route that is using port1 will show up in the routing table.

72. If there are no changes in the routing table and in the case of TCP traffic, which of the following correctly describes the routing table lookups performed by a FortiGate in NAT /Route mode, when searching for a suitable gateway?

- A. A lookup is done only when the first packet coming from the client (SYN) arrives.
- B. A lookup is done when the first packet coming from the client (SYN) arrives, and a second one is performed when the first packet coming from the server (SYN/ACK) arrives.
- C. Three lookups are done during the TCP 3-way handshake (SYN, SYN/ACK, ACK).
- D. A lookup is always done each time a packet arrives, from either the server or the client side.

73. A static route is configured for a FortiGate unit from the CLI using the following commands. Which of the following conditions are required for this static default route to be displayed in the FortiGate unit's routing table? (Choose two.)

```
config router static
edit 1
set device "wan1"
set distance 20
set gateway 192.168.100.1
next
end
```

- A. The administrative status of the wan1 interface is displayed as down.
- B. The link status of the wan1 interface is displayed as up.
- C. All other default routers should have a lower distance.
- D. The wan1 interface address and gateway address are on the same subnet. **Answer:**

74. Examine the exhibit; then answer the question below.



The Vancouver FortiGate initially had the following information in its routing table:

S 172.20.0.0/16 [10/0] via 172.21.1.2, port2
C 172.21.0.0/16 is directly connected, port2
C 172.11.11.0/24 is directly connected, port1

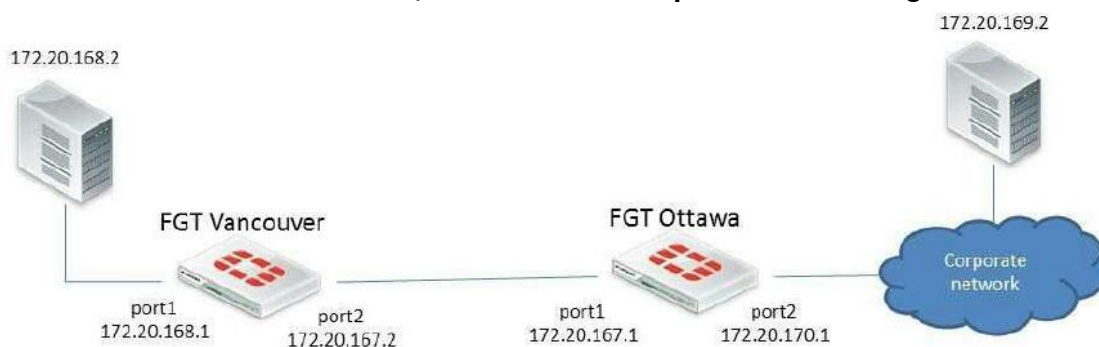
Afterwards, the following static route was added:

```
config router static edit 6
set dst 172.20.1.0 255.255.255.0
set priority 0
set device port1
set gateway 172.11.12.1 next
```

Since this change, the new static route is NOT showing up in the routing table. Given the information provided, which of the following describes the cause of this problem?

- A. The subnet 172.20.1.0/24 is overlapped with the subnet of one static route that is already in the routing table (172.20.0.0/16), so, we need to enable allow-subnet-overlap first.
- B. The 'gateway' IP address is NOT in the same subnet as the IP address of port1.
- C. The priority is 0, which means that the route will remain inactive.
- D. The static route configuration is missing the distance setting.

75. Examine the exhibit below; then answer the question following it.



In this scenario. The FortiGate unit in Ottawa has the following routing table:

s* 0.0.0.0/0 [10/0] via 172.20.170.254, port2
c 172.20.167.0/24 is directly connected, port1
c 172.20.170.0/24 is directly connected, port2

Sniffer tests show that packets sent from the source IP address 170.20.168.2 to the destination IP address 172.20.169.2 are being dropped by the FortiGate located in Ottawa. Which of the following correctly describes the cause for the dropped packets?

- A. The forward policy check.
- B. The reserve path forwarding check.
- C. The subnet 172.20.169.0/24 is NOT in the Ottawa FortiGate's routing table.
- D. The destination workstation 172.20.169.2 does NOT have the subnet 172.20.168.0/24 in its routing table.

76. Examine the static route configuration shown below; then answer the question following it.

```
config router static
edit 1
set dst 172.20.1.0 255.255.255.0
set device port1
set gateway 172.11.12.1
set distance 10
set weight 5
next
edit 2
set dst 172.20.1.0 255.255.255.0
set blackhole enable
set distance 5
set weight 10
next
end
```

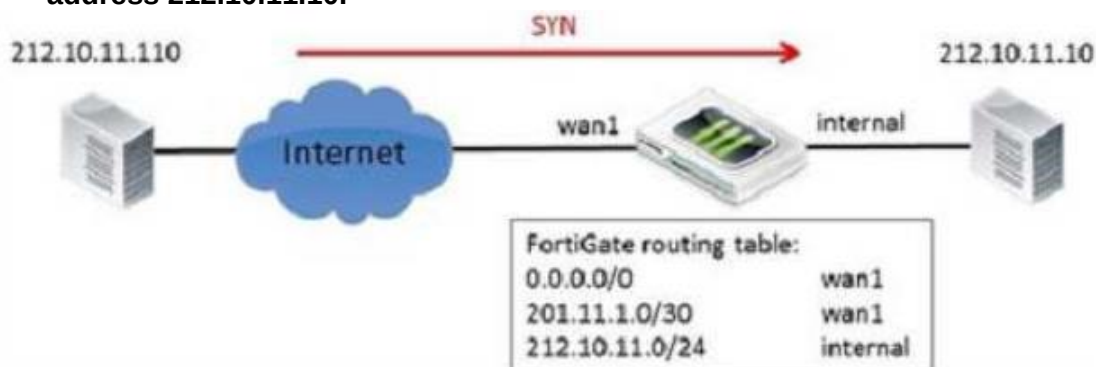
Which of the following statements correctly describes the static routing configuration provided? (Choose two.)

- A. All traffic to 172.20.1.0/24 is dropped by the FortiGate.
- B. As long as port1 is up, all traffic to 172.20.1.0/24 is routed by the static route number 1. if the interface port1 is down, the traffic is routed using the blackhole route.
- C. The FortiGate unit does NOT create a session entry in the session table when the traffic is being routed by the blackhole route.
- D. The FortiGate unit creates a session entry in the session table when the traffic is being routed by the blackhole route.

77. Which of the following fields contained in the IP/TCP/UDP headers can be used to make a routing decision when using policy-based routing? (Choose three)

- A. Source IP address.
- B. TCP flags
- C. Source TCP/UDP ports
- D. Type of service.
- E. Checksum

78. Examine the network topology diagram in the exhibit; the workstation with the IP address 212.10.11.110 sends a TCP SYN packet to the workstation with the IP address 212.10.11.10.



Which of the following sentences best describes the result of the reverse path forwarding (RPF) check executed by the FortiGate on the SYN packets? (Choose two).

- A. Packets is allowed if RPF is configured as loose.
- B. Packets is allowed if RPF is configured as strict.
- C. Packets is blocked if RPF is configured as loose.
- D. Packets is blocked if RPF is configured as strict.

79. Which of the following statements best describe what a FortiGate does when packets match a black hole route?

- A. Packets are dropped.
- B. Packets are routed based on the information in the policy-based routing table.
- C. An ICMP error message is sent back to the originator.
- D. Packet are routed back to the originator.

80. The exhibit shows two static routes to the same destinations subnet 172.20.168.0/24.

```
#config router static
edit 1
    set dst 172.20.168.0 255.255.255.0
    set distance 10
    set priority 20
    set device port1
next

edit 2
    set dst 172.20.168.0 255.255.255.0
    set distance 20
    set priority 20
    set device port2
next
end
```

Which of the following statements correctly describes this static routing configuration? (choose two)

- A. Both routes will show up in the routing table.
- B. The FortiGate unit will evenly share the traffic to 172.20.168.0/24 between routes.
- C. Only one route will show up in the routing table.
- D. The FortiGate will route the traffic to 172.20.168.0/24 only through one route.

81. Which action does the FortiGate take when link health monitor times out?

- A. All routes to the destination subnet configured in the link health monitor are removed from the routing table.
- B. The distance values of all routes using interface configured in the link health monitor are increased.
- C. The priority values of all routes using configured in the link health monitor are increased.
- D. All routes using the next-hop gateway configured in the link health monitor are removed from the routing table.

82. What must be configured in order to keep two static routes to the same destination in the routing table?

- A. The same priority.
- B. The same distance and same priority.
- C. The same distance.
- D. The same metric.

83. The exhibit shows a FortiGate routing table.

```
# get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       E1 - OSPF external type 1, E2 - OSPF external type 2
       * - candidate default
O*E2  0.0.0.0/0 [110/10] via 192.168.11.254, wan1, 01:29:24
C      172.16.78.0/24 is directly connected, wan2
O      192.168.1.0/24 [110/200] via 192.168.11.59, internal, 01:30:28
C      192.168.3.0/24 is directly connected, dmz
C      192.168.11.0/24 is directly connected, internal
```

Which of the following statements are correct?(Choose two)

- A. There is only one active default route.
- B. The distance values for the route to 192.168.1.0/24 is 200
- C. An IP address in the subnet 172.16.78.0/24 has been assigned to the dmz interface.
- D. The FortiGate will route the traffic to 172.17.1.2 to next hop with the IP address 192.168.11.254

84. The exhibit shoes three static routes. Which routes will be used to route the packets to the destination IP address 172.20.168.1?

```

config router static
  edit 1
    set dst 172.20.168.0 255.255.255.0
    set distance 10
    set priority 10
    set device port1
  next
  edit 2
    set dst 172.20.0.0 255.255.0.0
    set distance 5
    set priority 20
    set device port2
  next
  edit 3
    set dst 172.20.0.0 255.255.0.0
    set distance 5
    set priority 20
    set device port3
  next
end

```

- A. The route with the ID number 2 and 3.
- B. Only the route with the ID number 3.
- C. Only the route with the ID number 2.
- D. Only the route with the ID number 1.

85. Examine the routing database. Which of the following statements are correct?
(Choose two.)

```

S    *> 0.0.0.0/0 [10/0] via 172.20.121.2, port1, [20/0]
      *>                [10/0] via 10.0.0.2, port2, [30/0]
S    0.0.0.0/0 [20/0] via 192.168.15.2, port3, [10/0]
C    *> 10.0.0.0/24 is directly connected, port2
S    172.13.24.0/24 [10/0] is directly connected, port4
C    *> 172.20.121.0/24 is directly connected, port1
S    *> 192.167.1.0/24 [10/0] via 10.0.0.2, port2
C    *> 192.168.15.0/24 is directly connected, port3

```

- A. The port3 default route has the lowest metric, making it the best route.
- B. There will be eight routes active in the routing table.
- C. The port3 default has a higher distance than the port1 and port2 default routes.
- D. Both port1 and port2 default routers are active in the routing table.

86. View the exhibit.



This is a sniffer output of a telnet connection request from 172.20.120.186 to the port1 interface of FGT1.

```
FGT1 # di sniff pack any "host 172.20.120.186 and port 23" 4

4.571724 port1 in 172.20.120.186.60584 -> 172.20.121.40.23: syn 273086
7.575327 port1 in 172.20.120.186.60584 -> 172.20.121.40.23: syn 273086
9.571446 port1 in 172.20.120.186.60584 -> 172.20.121.40.23: syn 273086
```

In this scenario, FGT1 has the following routing table:

```
s* 0.0.0.0/0 [10/0] via 172.20.54.254, port2
C 172.20.54.0/24 is directly connected, port2
C 172.20.121.0/24 is directly connected, port1
```

Assuming telnet service is enabled for port1, which of the following statements correctly describes why FGT1 is not responding?

- A. The port1 cable is disconnected.
- B. The connection is dropped due to reverse path forwarding check.
- C. The connection is denied due to forward policy check.
- D. FGT1's port1 interface is administratively down.

87. View the exhibit. Which statements about the exhibit are true? (Choose two.)

Status	Name	VLANID	Type	IP/Netmask
Physical (12)				
+	port1		Physical Interface	10.200.1.1 255.255.255.0
	port1-VLAN1	1	VLAN	10.200.5.1 255.255.255.0
	port1-VLAN10	10	VLAN	10.0.10.1 255.255.255.0
+	port2		Physical Interface	10.200.2.1 255.255.255.0
	port2-VLAN1	1	VLAN	10.0.5.1 255.255.255.0
	port2-VLAN10	10	VLAN	10.0.20.254 255.255.255.0
+	port3		Physical Interface	10.0.1.254 255.255.255.0

- A. port1-VLAN10 and port2-VLAN10 can be assigned to different VDOMs.
- B. port1-VLAN1 is the native VLAN for the port1 physical interface.
- C. Traffic between port1-VLAN1 and port2-VLAN1 is allowed by default.
- D. Broadcast traffic received in port1-VLAN10 will not be forwarded to port2-VLAN10.

88. An administrator has configured two VLAN interfaces: A DHCP server is connected to the VLAN10 interface. A DHCP client is connected to the VLAN5 interface. However, the DHCP client cannot get a dynamic IP address from the DHCP server. What is the cause of the problem?

```
config system interface
  edit "VLAN10"
    set vdom "VDM1"
    set forward-domain 100
    set role lan
    set interface "port9"
    set vlanid 10
  next
  edit "VLAN5"
    set vdom "VDM1"
    set forward-domain 50
    set role lan
    set interface "port10"
    set vlanid 5
  next
end
```

- A. Both interfaces must be in different VDOMs
- B. Both interfaces must have the same VLAN ID.
- C. The `role` of the VLAN10 interface must be set to `server`.
- D. Both interfaces must belong to the same forward domain.

89. View the example routing table.

```
S*  0.0.0.0/0 [10/0] via 172.20.121.2, port1
C   172.20.121.0/24 is directly connected, port1
C   172.20.168.0/24 is directly connected, port2
C   172.20.167.0/24 is directly connected, port3
S   10.20.30.0/26 [10/0] via 172.20.168.254, port2
S   10.20.30.0/24 [10/0] via 172.20.167.254, port3
```

Which route will be selected when trying to reach 10.20.30.254?

- A. 10.20.30.0/26 [10/0] via 172.20.168.254, port2
- B. The traffic will be dropped because it cannot be routed.
- C. 10.20.30.0/24 [10/0] via 172.20.167.254, port3
- D. 0.0.0.0/0 [10/0] via 172.20.121.2, port1

90. When does a FortiGate load-share traffic between two static routes to the same destination subnet?

- A. When they have the same cost and distance.
- B. When they have the same distance and the same weight.
- C. When they have the same distance and different priority.
- D. When they have the same distance and same priority

91. Review the output of the command `get router info routing-table database` shown in the exhibit below; then answer the question following it.

```
STUDENT # get router info routing-table database
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       > - selected route, * - FIB route, p - stale info

S    *> 0.0.0.0/0 [10/0] via 10.200.1.254, port1
      *>      [10/0] via 10.200.2.254, port2, [5/0]
C    *> 10.0.1.0/24 is directly connected, port3
S    10.0.2.0/24 [20/0] is directly connected, Remote_2
S    *> 10.0.2.0/24 [10/0] is directly connected, Remote_1
C    *> 10.200.1.0/24 is directly connected, port1
C    *> 10.200.2.0/24 is directly connected, port2
```

Which two statements are correct regarding this output? (Choose two.)

- A. There will be six routes in the routing table.
- B. There will be seven routes in the routing table.
- C. There will be two default routes in the routing table.
- D. There will be two routes for the 10.0.2.0/24 subnet in the routing table

92. A FortiGate is operating in NAT/Route mode and configured with two virtual LAN (VLAN) sub-interfaces added to the same physical interface. Which one of the following statements is correct regarding the VLAN IDs in this scenario?

- A. The two VLAN sub-interfaces can have the same VLAN ID only if they have IP addresses in different subnets.
- B. The two VLAN sub-interfaces must have different VLAN IDs.
- C. The two VLAN sub-interfaces can have the same VLAN ID only if they belong to different VDOMs.
- D. The two VLAN sub-interfaces can have the same VLAN ID if they are connected to different L2 IEEE 802.1Q compliant switches.

93. Review the output of the command `get router info routing-table all` shown in the Exhibit below; then answer the question following it. Which one of the following statements correctly describes this output?

```
STUDENT # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

S*   0.0.0.0/0 [10/0] via 10.200.1.254, port1
      [10/0] via 10.200.2.254, port2, [5/0]
C    10.0.1.0/24 is directly connected, port3
O    10.0.2.0/24 [110/101] via 172.16.2.1, Remote_1, 00:00:21
      [110/101] via 172.16.2.2, Remote_2, 00:00:21
C    10.200.1.0/24 is directly connected, port1
C    10.200.2.0/24 is directly connected, port2
C    172.16.1.1/32 is directly connected, Remote_1
C    172.16.1.2/32 is directly connected, Remote_2
C    172.16.2.1/32 is directly connected, Remote_1
C    172.16.2.2/32 is directly connected, Remote_2
```

- A. The two routes to the 10.0.2.0/24 subnet are ECMP routes and traffic will be load balanced based on the configured ECMP settings.
- B. The route to the 10.0.2.0/24 subnet via interface Remote_1 is the active and the route via Remote_2 is the backup.
- C. OSPF does not support ECMP therefore only the first route to subnet 10.0.1.0/24 is used.
- D. 172.16.2.1 is the preferred gateway for subnet 10.0.2.0/24.

94. Review the output of the command `config router ospf` shown in the Exhibit below; then answer the question following it. Which one of the following statements is correct regarding this output?

```
STUDENT (ospf) # show
config router ospf
  config area
    edit 0.0.0.0
    next
  end
  config network
    edit 1
      set prefix 10.0.1.0 255.255.255.0
    next
    edit 2
      set prefix 172.16.0.0 255.240.0.0
    next
  end
  config ospf-interface
    edit "R1_OSPF"
      set interface "Remote_1"
      set ip 172.16.1.1
      set mtu 1436
      set network-type point-to-point
    next
    edit "R2_OSPF"
      set cost 20
      set interface "Remote_2"
      set ip 172.16.1.2
      set mtu 1436
      set network-type point-to-point
    next
  end
  config redistribute "connected"
  end
  config redistribute "static"
  end
  config redistribute "rip"
  end
  config redistribute "bgp"
  end
  config redistribute "isis"
  end
```

- A. OSPF Hello packets will only be sent on interfaces configured with the IP addresses 172.16.1.1 and 172.16.1.2.
- B. OSPF Hello packets will be sent on all interfaces of the FortiGate device.
- C. OSPF Hello packets will be sent on all interfaces configured with an address matching the 10.0.1.0/24 and 172.16.0.0/12 networks.
- D. OSPF Hello packets are not sent on point-to-point networks.

95. In order to load-share traffic using multiple static routes, the routes must be configured with ...

- A. the same distance and same priority.
- B. the same distance and the same weight.
- C. the same distance but each of them must be assigned a unique priority.
- D. a distance equal to its desired weight for ECMP but all must have the same priority.

96. If Open Shortest Path First (OSPF) has already been configured on a FortiGate unit, which of the following statements is correct if the routes learned through OSPF need to be announced by Border Gateway Protocol (BGP)?

- A. The FortiGate unit will automatically announce all routes learned through OSPF to its BGP peers if the FortiGate unit is configured as an OSPF Autonomous System Boundary Router (ASBR).
- B. The FortiGate unit will automatically announce all routes learned through OSPF to its BGP peers if the FortiGate unit is configured as an OSPF Area Border Router (ABR).
- C. At a minimum, the network administrator needs to enable Redistribute OSPF in the BGP settings.
- D. The BGP local AS number must be the same as the OSPF area number of the routes learned that need to be redistributed into BGP.
- E. By design, BGP cannot redistribute routes learned through OSPF.

97. When configuring a server load balanced virtual IP, which of the following is the best distribution algorithm to be used in applications where the same physical destination server must be maintained between sessions?

- A. Static
- B. Round robin
- C. Weighted round robin
- D. Least connected

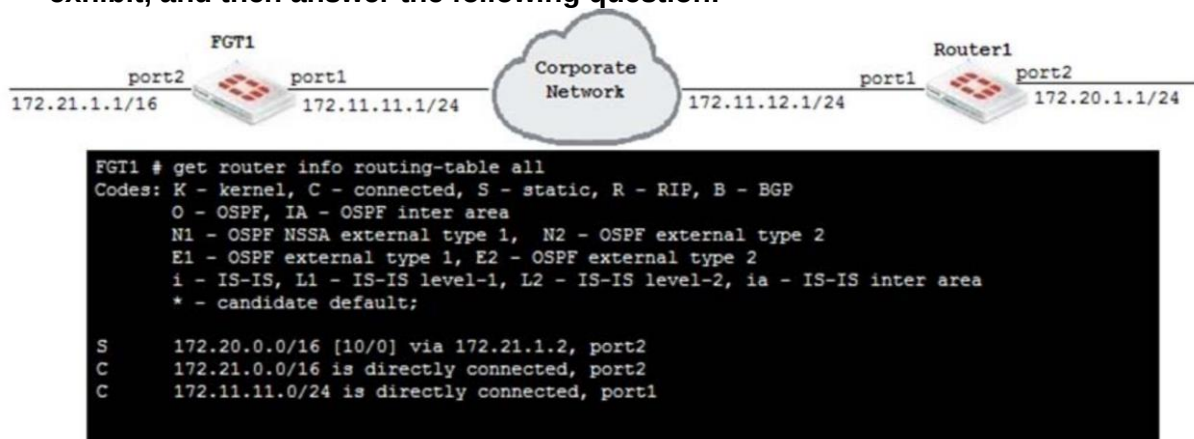
98. If Routing Information Protocol (RIP) version 1 or version 2 has already been configured on a FortiGate unit, which of the following statements is correct if the routes learned through RIP need to be advertised into Open Shortest Path First (OSPF)?

- A. The FortiGate unit will automatically announce all routes learned through RIP v1 or v2 to its OSPF neighbors.
- B. The FortiGate unit will automatically announce all routes learned only through RIP v2 to its OSPF neighbors.
- C. At a minimum, the network administrator needs to enable Redistribute RIP in the OSPF Advanced Options.
- D. The network administrator needs to configure a RIP to OSPF announce policy as part of the RIP settings.
- E. At a minimum, the network administrator needs to enable Redistribute Default in the OSPF Advanced Options.

99. Which of the following route attributes must be equal for static routes to be eligible for equal cost multipath (ECMP) routing? (Choose two.)

- A. Priority
- B. Metric
- C. Distance
- D. Cost

100. Examine the network diagram and the existing FGT1 routing table shown in the exhibit, and then answer the following question:



An administrator has added the following static route on FGT1.

New Static Route

Destination	Subnet	Named Address	Internet Service
	172.20.1.0/24		
Gateway	172.11.12.1		
Interface	port1		
Administrative Distance	10		
Comments			
Status	☒ Enabled ☐ Disabled		

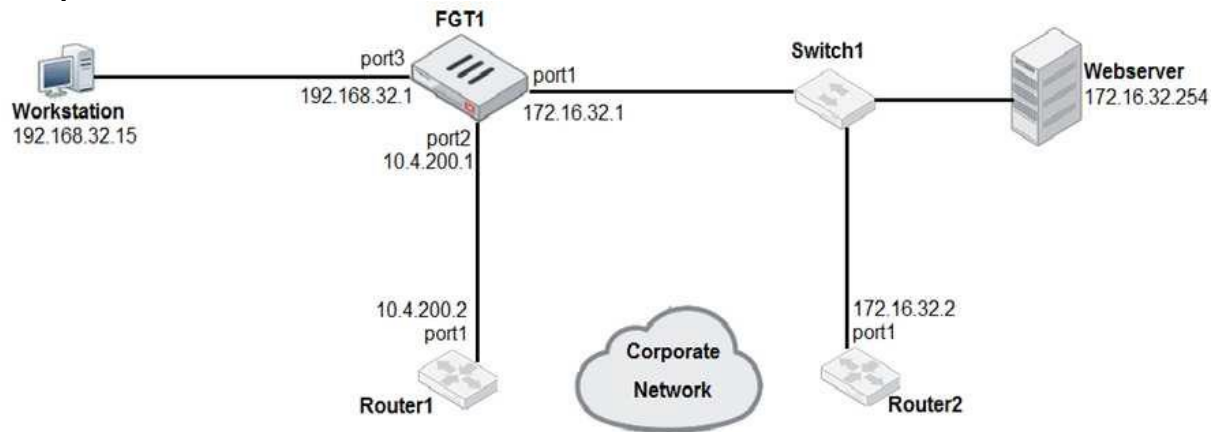
Advanced Options

Priority: 0

Since the change, the new static route is not showing up in the routing table. Given the information provided, which of the following describes the cause of this problem?

- A. The new route's destination subnet overlaps an existing route.
- B. The new route's **Distance** value should be higher than 10.
- C. The **Gateway** IP address is not in the same subnet as **port1**.
- D. The **Priority** is 0, which means that this route will remain inactive.

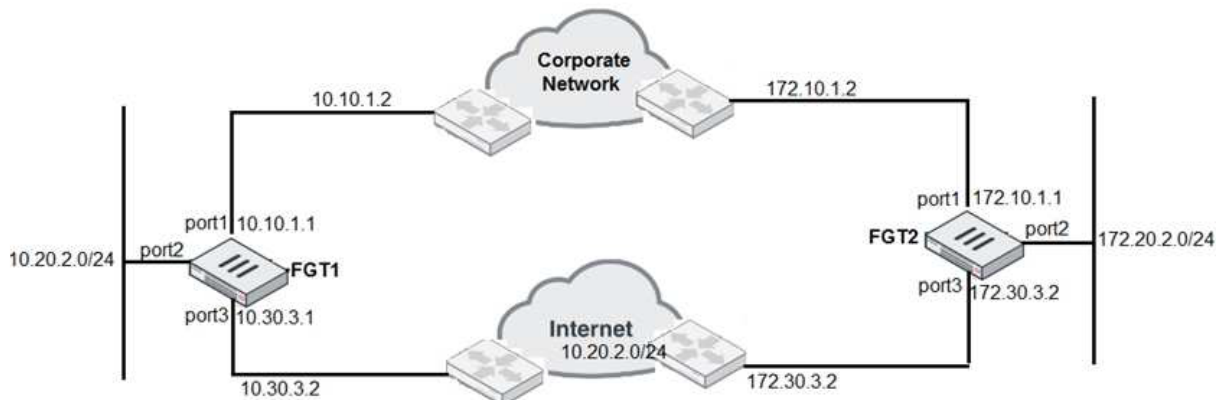
101. Examine the network diagram shown in the exhibit, then answer the following question:



Which one of the following routes is the best candidate route for FGT1 to route traffic from the Workstation to the Web server?

- A. 172.16.0.0/16 [50/0] via 10.4.200.2, port2 [5/0]
- B. 0.0.0.0/0 [20/0] via 10.4.200.2, port2
- C. 10.4.200.0/30 is directly connected, port2
- D. 172.16.32.0/24 is directly connected, port1

102. Examine the network diagram shown in the exhibit, and then answer the following question:



A firewall administrator must configure equal cost multipath (ECMP) routing on FGT1 to ensure both port1 and port3 links are used at the same time for all traffic destined for 172.20.2.0/24. Which of the following static routes will satisfy this requirement on FGT1? (Choose two.)

- A. 172.20.2.0/24 (1/0) via 10.10.1.2, port1 [0/0]
- B. 172.20.2.0/24 (25/0) via 10.10.3.2, port3 [5/0]
- C. 172.20.2.0/24 (1/150) via 10.10.1.2, port1 [10/0]
- D. 172.20.2.0/24 (1/150) via 10.30.3.2, port3 [10/0]

QCM NSE4 : VDOM

103. Which of the following settings can be configured per VDOM? (Choose three)

- A. Operating mode (NAT/route or transparent)
- B. Static routes
- C. Hostname
- D. System time
- E. Firewall Policies

104. A FortiGate administrator with the super_admin profile configures a virtual domain (VDOM) for a new customer. After creating the VDOM, the administrator is unable to reassign the dmz interface to the new VDOM as the option is greyed out in the GUI in the management VDOM. What would be a possible cause for this problem?

- A. The administrator does not have the proper permissions the dmz interface.
- B. The dmz interface is referenced in the configuration of another VDOM.
- C. Non-management VDOMs cannot reference physical interfaces
- D. The dmz interface is in PPPoE or DHCP mode.

105. A FortiGate is configured with three virtual domains (VDOMs). Which of the following statements is correct regarding multiple VDOMs?

- A. The FortiGate must be a model 1000 or above to support multiple VDOMs.
- B. A license has to be purchased and applied to the FortiGate before VDOM mode could be enabled.
- C. Changing the operational mode of a VDOM requires a reboot of the FortiGate.
- D. The FortiGate supports any combination of VDOMs in NAT/Route and transparent modes.

106. A FortiGate devices has two VDOMs in NAT/route mode. Which of the following solutions can be implemented by a network administrator to route traffic between the two VDOMs.(Choose two)

- A. Use the inter-VDOMs links automatically created between all VDOMS.
- B. Manually create and configured an inter-VDOM link between yours.
- C. Interconnect and configure an external physical interface in one VDOM to another physical interface in the second VDOM.
- D. Configure both VDOMs to share the same table.

107. A FortiGate device is configured with two VDOMs. The management VDOM is 'root' , and is configured in transparent mode,'vdom1' is configured as NAT/route mode. Which traffic is generated only by 'root' and not 'vdom1'? (Choose three.)

- A. SNMP traps
- B. FortiGuard
- C. ARP
- D. NTP
- E. ICMP redirect

108. Which of the following statements are correct regarding FortiGate virtual domains (VDOMs)? (Choose two)

- A. VDOMs divide a single FortiGate unit into two or more independent firewall.
- B. A management VDOM handles SNMP. logging, alert email and FortiGuard updates.
- C. Each VDOM can run different firmware versions.
- D. Administrative users with a 'super_admin' profile can administrate only one VDOM.

109. Which of the following statements is correct concerning multiple vdoms configured in a FortiGate device?

- A. FortiGate devices, from the FGT/FWF 60D and above, all support VDOMS.
- B. All FortiGate devices scale to 250 VDOMS.
- C. Each VDOM requires its own FortiGuard license.
- D. FortiGate devices support more NAT/route VDOMs than Transparent Mode VDOMs.

110. A FortiGate unit has multiple VDOMs in NAT/route mode with multiple VLAN interfaces in each VDOM. Which of the following statements is correct regarding the IP addresses assigned to each VLAN interface?

- A. Different VLANs can share the same IP address as long as they have different VLAN IDs.
- B. Different VLANs can share the same IP address as long as they are in different physical interface.
- C. Different VLANs can share the same IP address as long as they are in different VDOMs.
- D. Different VLANs can never share the same IP addresses.

111. A FortiGate device is configured with four VDOMs: 'root' and 'vdom1' are in NAT/route mode; 'vdom2' and 'vdom3' are in transparent mode. The management VDOM is 'root'. Which of the following statements are true? (Choose two.)

- A. An inter-VDOM link between 'root' and 'vdom1' can be created.
- B. An inter-VDOM link between 'vdom1' and 'vdom2' can be created.
- C. An inter-VDOM link between 'vdom2' and 'vdom3' can be created.
- D. Inter-VDOM link links must be manually configured for FortiGuard traffic.

112. Which of the following statements are correct concerning layer 2 broadcast domains in transparent mode VDOMs? (Choose two)

- A. The whole VDOM is a single broadcast domain even when multiple VLAN are used.
- B. Each VLAN is a separate broadcast domain.
- C. Interfaces configured with the same VLAN ID can belong to different broadcast domains.
- D. All the interfaces in the same broadcast domain must use the same VLAN ID.

113. Which of the following statements is correct regarding FortiGate interfaces and spanning tree protocol? (Choose Two)

- A. Only FortiGate switch interfaces participate in spanning tree.
- B. All FortiGate interfaces in transparent mode VDOMs participate in spanning tree.
- C. All FortiGate interfaces in NAT/route mode VDOMs participate in spanning tree.
- D. All FortiGate interfaces in transparent mode VDOMs may block or forward BPDUs.

114. A FortiGate is configured with multiple VDOMs. An administrative account on the device has been assigned a Scope value of VDOM:root.

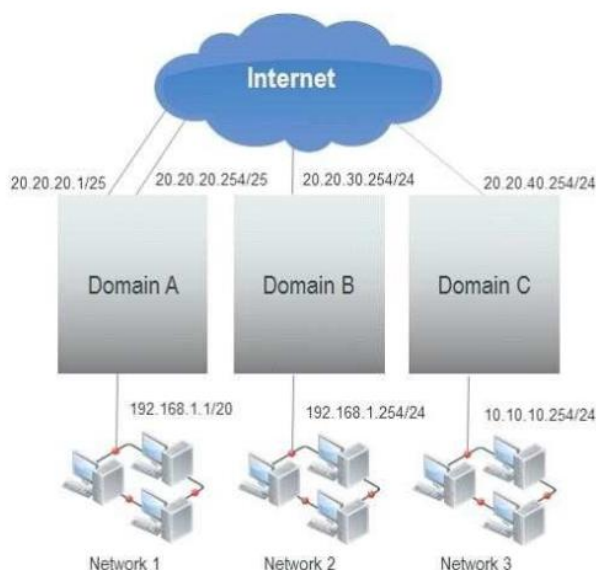
Which of the following settings will this administrator be able to configure? (Choose two.)

- A. Firewall addresses.
- B. DHCP servers.
- C. FortiGuard Distribution Network configuration.
- D. System hostname.

115. Which statements about virtual domains (VDOMs) are true? (Choose two.)

- A. Transparent mode and NAT/Route mode VDOMs cannot be combined on the same FortiGate.
- B. Each VDOM can be configured with different system hostnames.
- C. Different VLAN sub-interfaces of the same physical interface can be assigned to different VDOMs.
- D. Each VDOM has its own routing table.

- 116. A FortiGate unit is configured with three Virtual Domains (VDOMs) as illustrated in the exhibit.**



Which of the following statements are true if the network administrator wants to route traffic between all the VDOMs? (Choose three.)

- A. The administrator can configure inter-VDOM links to avoid using external interfaces and routers.
- B. As with all FortiGate unit interfaces, firewall policies must be in place for traffic to be allowed to pass through any interface, including inter-VDOM links.
- C. This configuration requires a router to be positioned between the FortiGate unit and the Internet for proper routing.
- D. Inter-VDOM routing is automatically provided if all the subnets that need to be routed are locally attached.
- E. As each VDOM has an independent routing table, routing rules need to be set (for example, static routing, OSPF) in each VDOM to route traffic between VDOMs.

- 117. Which of the following statements are correct regarding virtual domains (VDOMs)? (Select all that apply.)**

- A. VDOMs divide a single FortiGate unit into two or more virtual units that function as multiple, independent units.
- B. A management VDOM handles SNMP, logging, alert email, and FDN-based updates.
- C. VDOMs share firmware versions, as well as antivirus and IPS databases.
- D. Only administrative users with a 'super_admin' profile will be able to enter multiple VDOMs to make configuration changes.

- 118. A FortiGate administrator configures a Virtual Domain (VDOM) for a new customer. After creating the VDOM, the administrator is unable to reassign the dmz interface to the new VDOM as the option is greyed out in Web Config in the management VDOM.**

What would be a possible cause for this problem?

- A. The dmz interface is referenced in the configuration of another VDOM.
- B. The administrator does not have the proper permissions to reassign the dmz interface.
- C. Non-management VDOMs can not reference physical interfaces.
- D. The dmz interface is in PPPoE or DHCP mode.
- E. Reassigning an interface to a different VDOM can only be done through the CLI.

- 119. View the exhibit. Why is the administrator getting the error shown in the exhibit ?**

```
192.168.2.1 - PuTTY
login as: admin
Local-FortiGate #
Local-FortiGate # config vdom

Local-FortiGate (vdom) # edit root
current vf=root:0

Local-FortiGate (root) # config system global

command parse error before 'global'
Command fail. Return code 1

Local-FortiGate (root) #
```

- A. The administrator must first enter the command `edit global`.
- B. The administrator `admin` does not have the privileges required to configure global settings.
- C. The global settings cannot be configured from the `root` VDOM context.
- D. The command `config system global` does not exist in FortiGate.

120. In "diag debug flow" output, you see the message "Allowed by Policy-1: SNAT". Which is true?

- A. The packet matched the topmost policy in the list of firewall policies.
- B. The packet matched the firewall policy whose policy ID is 1.
- C. The packet matched a firewall policy, which allows the packet and skips UTM checks
- D. The policy allowed the packet and applied session NAT.

QCM NSE4 : VPN IPsec

121. Review the IPsec diagnostics output of the command `diagnose vpn tunnel list` shown in the exhibit below.

```
STUDENT # diagnose vpn tunnel list
list all ipsec tunnel in vd 0
-----
name=FCClient_0 ver=1 serial=3 10.200.1.1:4500->10.200.3.1:64916 lgwy=static tun=intf mode=dial_inst bound_if=2
parent=FCClient index=0
proxyid_num=1 child_num=0 refcnt=8 ilast=2 olast=2
stat: rxp=59 txp=0 rxb=15192 txb=0
dpd: mode=active on=1 idle=5000ms retry=3 count=0 seqno=10
natt: mode=keepalive draft=32 interval=10 remote_port=64916
proxyid=FCClient proto=0 sa=1 ref=2 auto_negotiate=0 serial=1
  src: 0:0.0.0.0-255.255.255.255:0
  dst: 0:172.20.1.1-172.20.1.1:0
  SA: ref=3 options=00000006 type=00 soft=0 mtu=1280 expire=1717 replaywin=1024 seqno=1
  life: type=01 bytes=0/0 timeout=1791/1800
  dec: spi=a29046e9 esp=3des key=24 0525830c6fd67ca37e9d6dad174d175e24f97c3b87f428fa
  ah=sha1 key=20 982f8ba194f3f797773efc605c0321b728dabf1d
  enc: spi=19be4052 esp=3des key=24 da597cb7fec913528f8598d1aa7ecd17156a2a7a4afeeb4c
  ah=sha1 key=20 9e2c5d0fc055fa0149bc66024732e9a85bbe8016
-----
```

Which statements are correct regarding this output (Choose two.)

- A. The connecting client has been allocated address 172.20.1.1.
- B. In the Phase 1 settings, dead peer detection is enabled.
- C. The tunnel is idle.
- D. The connecting client has been allocated address 10.200.3.1.

122. Review the configuration for FortiClient IPsec shown in the exhibit.

Network

IP Version: IPv4

Incoming Interface: port1

Client Address Range: 172.20.1.1-172.20.1.5

Subnet Mask: 255.255.255.255

Use System DNS: ☒

Enable IPv4 Split Tunnel: ☒

Accessible Networks: student.internal

Which statement is correct regarding this configuration?

- A. The connecting VPN client will install a route to a destination corresponding to the student internal address object.
- B. The connecting VPN client will install a default route.
- C. The connecting VPN client will install a route to the 172.20.1.[1-5] address range.
- D. The connecting VPN client will connect in web portal mode and no route will be installed.

123. Which IPsec mode includes the *peer id* information in the first packet?

- A. Main mode.
- B. Quick mode.
- C. Aggressive mode.
- D. IKEv2 mode.

124. You are the administrator in charge of a point-to-point IPsec VPN between two FortiGate units using route based mode. Users from either side must be able to initiate new sessions with no restrictions. There is only 1 subnet at either end and the FortiGate already has a default route. Which two configuration steps are required in each FortiGate to achieve these objectives? (Choose two.)

- A. Create one firewall policy.
- B. Create two firewall policies.
- C. Add a route to the remote subnet.
- D. Add two IPsec phases 2.

125. An administrator wants to create an IPsec VPN tunnel between two FortiGate devices. Which three configuration steps must be performed on both units to support this scenario? (Choose three.)

- A. Create firewall policies to allow and control traffic between the source and destination IP addresses.
- B. Configure the appropriate user groups to allow users access to the tunnel.
- C. Set the operating mode to IPsec VPN mode.
- D. Define the phase 2 parameters.
- E. Define the Phase 1 parameters.

126. What action does an IPsec Gateway take with the user traffic routed to an IPsec VPN when it does not match any phase 2 quick mode selector?

- A. Traffic is dropped
- B. Traffic is routed across the default phase 2.
- C. Traffic is routed to the next available route in the routing table.
- D. Traffic is routed unencrypted to the interface where the IPsec VPN is terminating.

127. Which of the following authentication methods are supported in an IPsec phase 1? (Choose two.)

- A. Asymmetric Keys
- B. CA root digital certificates
- C. RSA signature
- D. Pre-shared keys

128. Which of the following IPsec configuration modes can be used for implementing L2TP-over-IPSec VPNs?

- A. Policy-based IPsec only.
- B. Route-based IPsec only.
- C. Both policy-based and route-based VPN.
- D. L2TP-over-IPSec is not supported by FortiGate devices.

129. Which of the following IPsec configuration modes can be used when the FortiGate is running in NAT mode?

- A. Policy-based VPN only
- B. Both policy-based and route-based VPN.
- C. Route-based VPN only.
- D. IPSec VPNs are not supported when the FortiGate is running in NAT mode.

130. Which of the following statements is true regarding the differences between route-based and policy-based IPsec VPNs? (Choose two.)

- A. The firewall policies for policy-based are bidirectional. The firewall policies for route-based are unidirectional.
- B. In policy-based VPNs the traffic crossing the tunnel must be routed to the virtual IPsec interface. In route-based, it does not.
- C. The action for firewall policies for route-based VPNs may be Accept or Deny, for policy-based VPNs it is Encrypt.
- D. Policy-based VPN uses an IPsec interface, route-based does not.

131. Which portion of the configuration does an administrator specify the type of IPsec configuration (either policy-based or route-based)?

- A. Under the IPsec VPN global settings.
- B. Under the phase 2 settings.
- C. Under the phase 1 settings.
- D. Under the firewall policy settings.

132. Which of the following options best defines what Diffie-Hellman is?

- A. A symmetric encryption algorithm.
- B. A "key-agreement" protocol.
- C. A "Security-association-agreement" protocol.
- D. An authentication algorithm.

133. How many packets are interchanged between both IPsec ends during the negotiation of a main-mode phase 1?

- A. 5
- B. 3
- C. 2
- D. 6

134. Which of the following IKE modes is the one used during the IPsec phase 2 negotiation?

- A. Aggressive mode
- B. Quick mode
- C. Main mode
- D. Fast mode

135. Which of the following statements are true about IPsec VPNs? (Choose three.)

- A. IPsec increases overhead and bandwidth.
- B. IPsec operates at the layer 2 of the OSI model.
- C. End-user's network applications must be properly pre-configured to send traffic across the IPsec VPN.
- D. IPsec protects upper layer protocols.
- E. IPsec operates at the layer 3 of the OSI model.

136. Which of the following combinations of two FortiGate device configurations (side A and side B), can be used to successfully establish an IPsec VPN between them? (choose two)

- A. Side A: main mode, remote gateway as static IP address, policy based VPN. Side B: aggressive mode, remote gateway as static IP address policy-based VPN.
- B. Side A: main mode, remote gateway as static IP address, policy based VPN. Side B: main mode, remote gateway as static IP address, route-based VPN
- C. Side A: main mode, remote gateway as static IP address, policy based VPN. Side B: main mode, remote gateway as dialup, route-based VPN.
- D. Side A: main mode, remote gateway as dialup policy based VPN, Side B: main mode, remote gateway as dialup, policy based VPN.

137. Which statement is correct concerning an IPsec VPN with the remote gateway setting configured as 'Dynamic DNS'?

- A. The FortiGate will accept IPsec VPN connection from any IP address.
- B. The FQDN resolution of the local FortiGate IP address where the VPN is terminated must be provided by a dynamic DNS provider.
- C. The FortiGate will Accept IPsec VPN connections only from IP addresses included on a dynamic DNS access list.
- D. The remote gateway IP address can change dynamically.

138. Which of the following statements are correct concerning IKE mode config? (Choose two)

- A. It can dynamically assign IP addresses to IPsec VPN clients.
- B. It can dynamically assign DNS settings to IPsec VPN clients.
- C. It uses the ESP protocol.
- D. It can be enabled in the phase 2 configuration.

139. Which of the following statements are correct concerning the IPsec phase 1 and phase 2, shown in the exhibit? (choose two)

The screenshot displays the FortiGate configuration interface for an IPsec VPN. It is divided into four main sections:

- Peer Options:** Contains 'Accept Types' set to 'This peer ID' and 'Peer ID' set to 'fortinet'.
- Phase 1 Proposal:** Includes 'Encryption' set to '3DES' and 'Authentication' set to 'SHA1'. Under 'Diffie-Hellman Groups', groups 14 and 5 are selected. 'Key Lifetime (seconds)' is set to 86400. There is an 'Add' button.
- XAUTH:** The 'Type' is set to 'Disabled'.
- Phase 2 Selectors:** A table with columns 'Name', 'Local Address', and 'Remote Address'. Both addresses are currently '0.0.0.0/0.0.0.0'. There is an 'Add' button and an edit icon.

- A. The quick mode selector in the remote site must also be 0.0.0.0/0 for the source and destination addresses.
- B. Only remote peers with the peer ID 'fortinet' will be able to establish a VPN.
- C. The FortiGate device will automatically add a static route to the source quick mode selector address received from each remote VPN peer.
- D. The configuration will work only to establish FortiClient-to-FortiGate tunnels. A FortiGate tunnel requires a different configuration.

140. The exhibit shows a part output of the diagnostic command 'diagnose debug application ike 255', taken during establishment of a VPN. Which of the following statement are correct concerning this output? (choose two)

```
ike 0:Remote:7:22: responder received first quick-mode message
ike 0:Remote:7:22: peer proposal is: peer:0:0.0.0.0-255.255.255.255:0, me:0:0.0.0.0-255.255.255.255:0
ike 0:Remote:7: sent IKE msg (quick_risend): 172.20.186.222:500->172.20.187.114:500, len=356
ike 0: comes 172.20.187.114:500->172.20.186.222:500, ifindex=2....
ike 0:Remote:7:P2:22: replay protection enabled
ike 0:Remote:7:P2:22: SA life soft seconds=1750.
ike 0:Remote:7:P2:22: SA life hard seconds=1800.
ike 0:Remote:7:P2:22: IPsec SA selectors #src=1 #dst=1
ike 0:Remote:7:P2:22: src 0 7 0:0.0.0.0-255.255.255.255:0
ike 0:Remote:7:P2:22: dst 0 7 0:0.0.0.0-255.255.255.255:0
ike 0:Remote:7:P2:22: add IPsec SA: SPIs=6e13ca19/8f1ce9ae
ike 0:Remote:7:P2:22: added IPsec SA: SPIs=6e13ca19/8f1ce9ae
ike 0:Remote:7:P2:22: sending SNMP tunnel UP trap
```

- A. The quick mode selectors negotiated between both IPsec VPN peers is 0.0.0.0/32 for both source and destination addresses.
- B. The output corresponds to a phase 2 negotiation
- C. NAT-T enabled and there is third device in the path performing NAT of the traffic between both IPsec VPN peers.
- D. The IP address of the remote IPsec VPN peer is 172.20.187.114

141. Which of the following protocols are defined in the IPsec Standard? (Choose two)

- A. AH
- B. GRE
- C. SSL/TLS
- D. ESP

142. What configuration objects are automatically added when using the FortiGate's FortiClient VPN Configurations Wizard?(Choose two)

- A. Static route
- B. Phase 1
- C. Users group
- D. Phase 2

143. What are the purposes of NAT traversal in IPsec? (Choose two.)

- A. To detect intermediary NAT devices in the tunnel path.
- B. To encapsulate ESP packets in UDP packets using port 4500.
- C. To force a new DH exchange with each phase 2 re-key
- D. To dynamically change phase 1 negotiation mode to Aggressive

144. Which of the following statements describe some of the differences between symmetric and asymmetric cryptography? (Choose two.)

- A. In symmetric cryptography, the keys are publicly available. In asymmetric cryptography, the keys must be kept secret.
- B. Asymmetric cryptography can encrypt data faster than symmetric cryptography
- C. Symmetric cryptography uses one pre-shared key. Asymmetric cryptography uses a pair of keys
- D. Asymmetric keys can be sent to the remote peer via digital certificates. Symmetric keys cannot

145. Which of the following statements best describes what a Public Certificate Authority (CA) is?

- A. A service that provides a digital certificate each time a user is authenticating
- B. An entity that certifies that the information contained in a digital certificate is valid and true.
- C. The FortiGate process in charge of generating digital certificates on the fly for SSL inspection purposes
- D. A service that validates digital certificates for certificate-based authentication purposes

146. Bob wants to send Alice a file that is encrypted using public key cryptography. Which of the following statements is correct regarding the use of public key cryptography in this scenario?

- A. Bob will use his private key to encrypt the file and Alice will use her private key to decrypt the file.
- B. Bob will use his public key to encrypt the file and Alice will use Bob's private key to decrypt the file.
- C. Bob will use Alice's public key to encrypt the file and Alice will use her private key to decrypt the file.
- D. Bob will use his public key to encrypt the file and Alice will use her private key to decrypt the file.

147. Which IPsec configuration mode can be used for implementing GRE-over-IPsec VPNs?.

- A. Policy-based only.
- B. Route-based only.
- C. Either policy-based or route-based VPN.
- D. GRE-based only.

148. What is IPsec Perfect Forward Secrecy (PFS)?

- A. A phase-1 setting that allows the use of symmetric encryption.
- B. A phase-2 setting that allows the recalculation of a new common secret key each time the session key expires.
- C. A 'key-agreement' protocol.
- D. A 'security-association-agreement' protocol.

149. An administrator has configured a route-based site-to-site IPsec VPN. Which statement is correct regarding this IPsec VPN configuration?

- A. The IPsec firewall policies must be placed at the top of the list.
- B. This VPN cannot be used as part of a hub and spoke topology.
- C. Routes are automatically created based on the quick mode selectors.
- D. A virtual IPsec interface is automatically created after the Phase 1 configuration is completed.

150. In an IPSec gateway-to-gateway configuration, two FortiGate units create a VPN tunnel between two separate private networks. Which of the following configuration steps must be performed on both FortiGate units to support this configuration? (Select all that apply.)

- A. Create firewall policies to control traffic between the IP source and destination address.
- B. Configure the appropriate user groups on the FortiGate units to allow users access to the IPSec VPN connection.
- C. Set the operating mode of the FortiGate unit to IPSec VPN mode.
- D. Define the Phase 2 parameters that the FortiGate unit needs to create a VPN tunnel with the remote peer.
- E. Define the Phase 1 parameters that the FortiGate unit needs to authenticate the remote peers.

151. You are the administrator in charge of a FortiGate unit which acts as a VPN gateway.

You have chosen to use Interface Mode when configuring the VPN tunnel and you want users from either side to be able to initiate new sessions.

There is only 1 subnet at either end and the FortiGate unit already has a default route.

Which of the following configuration steps are required to achieve these objectives? (Select all that apply.)

- A. Create one firewall policy.
- B. Create two firewall policies.
- C. Add a route for the remote subnet.
- D. Add a route for incoming traffic.
- E. Create a phase 1 definition.
- F. Create a phase 2 definition.

152. Which of the following statements must be true for a digital certificate to be valid? (Choose two.)

- A. It must be signed by a "trusted" CA
- B. It must be listed as valid in a Certificate Revocation List (CRL)
- C. The CA field must be "TRUE"
- D. It must be still within its validity period

153. Why must you use aggressive mode when a local FortiGate IPSec gateway hosts multiple dialup tunnels?

- A. In aggressive mode, the remote peers are able to provide their peer IDs in the first message.
- B. FortiGate is able to handle NATed connections only in aggressive mode.
- C. FortiClient only supports aggressive mode.
- D. Main mode does not support XAuth for user authentication.

154. Which of the following conditions are required for establishing an IPSec VPN between two FortiGate devices? (Choose two.)

- A. If XAuth is enabled as a server in one peer, it must be enabled as a client in the other peer.
- B. If the VPN is configured as route-based, there must be at least one firewall policy with the action set to **IPSec**.
- C. If the VPN is configured as **DialUp User** in one peer, it must be configured as either **Static IP Address** or **Dynamic DNS** in the other peer.
- D. If the VPN is configured as a policy-based in one peer, it must also be configured as policy-based in the other peer.

155. During the digital verification process, comparing the original and fresh hash results satisfies which security requirement?

- A. Authentication.
- B. Data integrity.
- C. Non-repudiation.
- D. Signature verification.

156. Which of the following statements about policy-based IPsec tunnels are true? (Choose two.)

- A. They can be configured in both NAT/Route and transparent operation modes.
- B. They support L2TP-over-IPsec.
- C. They require two firewall policies: one for each directions of traffic flow.
- D. They support GRE-over-IPsec

157. Review the IPsec phase1 configuration in the Exhibit shown below; then answer the question following it. Which of the following statements are correct regarding this configuration? (Select all that apply).

New Phase 1

Name	Remote_1		
Comments	Write a comment...		0/255
Remote Gateway	Static IP Address		
IP Address	10.200.3.1		
Local Interface	port1		
Mode	☐ Aggressive ☒ Main (ID protection)		
Authentication Method	Preshared Key		
Pre-shared Key			
Peer Options			
	☒ Accept any peer ID		
Advanced...		(XAUTH, NAT Traversal, DPD)	
☒ Enable IPsec Interface Mode			
IKE Version	☒ 1 ☐ 2		
Local Gateway IP	☒ Main Interface IP ☐ Specify		
P1 Proposal			
	1 - Encryption	AES192	Authentication SHA1
DH Group	☐ 1 ☐ 2 ☒ 5 ☐ 14		
Keylife	28800 (120-172800 seconds)		
Local ID	(optional)		
XAUTH	☒ Disable ☐ Enable as Client ☐ Enable as Server		
NAT Traversal	☒ Enable		
Keepalive Frequency	10 (10-900 seconds)		
Dead Peer Detection	☒ Enable		

- A. The phase1 is for a route-based VPN configuration.
- B. The phase1 is for a policy-based VPN configuration.
- C. The local gateway IP is the address assigned to port1.
- D. The local gateway IP address is 10.200.3.1.

158. Review the static route configuration for IPsec shown in the exhibit. Which statements are correct regarding this configuration? (Choose two.)

Destination IP/Mask	10.0.2.0/255.255.255	
Device	remote	
Distance	10	(1-255, Default=10)
Priority	0	(0-4294967295)
Comments	VPN: remote (Created by VPN wizard) 35/255	

- A. Interface remote is an IPsec interface.
- B. A gateway address is not required because the interface is a point-to-point connection.
- C. A gateway address is not required because the default route is used.
- D. Interface remote is a zone.

159. Review the IPsec phase 1 configuration in the exhibit; then answer the question below. Which statements are correct regarding this configuration? (Choose two.)

Network

IP Version: IPv4

Remote Gateway: Static IP Address

IP Address: 10.200.3.1

Interface: port1

Mode Config: ☐

NAT Traversal: ☒

Keepalive Frequency: 10

Dead Peer Detection: ☒

- A. The remote gateway address is 10.200.3.1
- B. The local IPsec interface address is 10.200.3.1
- C. The local gateway IP is the address assigned to port1
- D. The local gateway IP is 10.200.3.1

160. Review the IPsec phase 2 configuration shown in the exhibit; then answer the question below. Which statements are correct regarding this configuration? (Choose two.)

Phase 2 Selectors

Name	Local Address	Remote Address
	0.0.0.0/0.0.0.0	0.0.0.0/0.0.0.0

Edit Phase 2

Name: remote

Comments: VPN: remote (Created by VPN wizard)

Local Address: Subnet 0.0.0.0/0.0.0.0

Remote Address: Subnet 0.0.0.0/0.0.0.0

Advanced...

Phase 2 Proposal

Encryption: AES256 Authentication: SHA512 Add

Enable Replay Detection: ☒

Enable Perfect Forward Secrecy (PFS): ☒

Diffie-Hellman Group: ☐ 21 ☐ 20 ☐ 19 ☐ 18 ☐ 17 ☐ 16 ☐ 15 ☒ 14 ☒ 5 ☐ 2 ☐ 1

Local Port: All ☒

Remote Port: All ☒

Protocol: All ☒

Autokey Keep Alive: ☒

Auto-negotiate: ☒

Key Lifetime: Seconds

Seconds: 43200

- A. The Phase 2 will re-key even if there is no traffic.
- B. There will be a DH exchange for each re-key.
- C. The sequence number of ESP packets received from the peer will not be checked.
- D. Quick mode selectors will default to those used in the firewall policy.

QCM_NSE4_LOG

161. What logging options are supported on a FortiGate unit? (choose 2)

- LDAP
- Syslog
- FortiAnalyzer
- SNMP

162. To which remote device can the FortiGate send logs? (Choose three.)

- A. Syslog
- B. FortiAnalyzer
- C. Hard drive
- D. Memory
- E. FortiCloud

163. In a Crash log, what does a status of 0 indicate?

- A. Abnormal termination of a process
- B. A process closed for any reason
- C. Scanunitd process crashed
- D. Normal shutdown with no abnormalities
- E. DHCP process crashed

164. There are eight (8) log severity levels that indicate the importance of an event. Not including Debug, which is only needed to log diagnostic data, what are both the lowest AND highest severity levels?

- A. Notification, Emergency
- B. Information, Critical
- C. Error, Critical
- D. Information, Emergency
- E. Information, Alert

165. Examine this log entry. What does the log indicate? (Choose three.)

```
date=2013-12-04 time=09:30:18 logid=0100032001 type=event subtype=system
level=information vd="root" user="admin" ui=http(192.168.1.112) action=login
status=success reason=none profile="super_admin" msg="Administrator admin
logged in successfully from http(192.168.1.112) "
```

- A. In the GUI, the log entry was located under "Log & Report > Event Log > User".
- B. In the GUI, the log entry was located under "Log & Report > Event Log > System".
- C. In the GUI, the log entry was located under "Log & Report > Traffic Log > Local Traffic".
- D. The connection was encrypted.
- E. The connection was unencrypted.
- F. The IP of the FortiGate interface that "admin" connected to was 192.168.1.112.
- G. The IP of the computer that "admin" connected from was 192.168.1.112.

166. Where are most of the security events logged?

- A. Security log
- B. Forward Traffic log
- C. Event log
- D. Alert log
- E. Alert Monitoring Console

167. What determines whether a log message is generated or not?

- A. Firewall policy setting
- B. Log Settings in the GUI
- C. 'config log' command in the CLI
- D. Syslog
- E. Webtrends

168. Which of the following are considered log types? (Choose three.)

- A. Forward log
- B. Traffic log
- C. Syslog
- D. Event log
- E. Security log

169. What attributes are always included in a log header? (Choose three.)

- A. policyid
- B. level
- C. user
- D. L'horodatage
- E. subtype
- F. duration

170. What log type would indicate whether a VPN is going up or down?

- A. Event log
- B. Security log
- C. Forward log
- D. Syslog

171. In "diag debug flow" output, you see the message "Allowed by Policy-1: SNAT". Which is true?

- A. The packet matched the topmost policy in the list of firewall policies.
- B. The packet matched the firewall policy whose policy ID is 1.
- C. The packet matched a firewall policy, which allows the packet and skips UTM checks
- D. The policy allowed the packet and applied session NAT.

172. If you enable the option "Generate Logs when Session Starts", what effect does this have on the number of traffic log messages generated for each session?

- A. No traffic log message is generated.
- B. One traffic log message is generated.
- C. Two traffic log messages are generated.
- D. A log message is only generated if there is a security event.

173. What is FortiGate's behavior when local disk logging is disabled?

- A. Only real-time logs appear on the FortiGate dashboard.
- B. No logs are generated.
- C. Alert emails are disabled.
- D. Remote logging is automatically enabled.

174. An administrator needs to be able to view logs for application usage on your network. What configurations are required to ensure that FortiGate generates logs for application usage activity? (Choose two.)

- A. Enable a web filtering profile on the firewall policy.
- B. Create an application control policy.
- C. Enable logging on the firewall policy.
- D. Enable an application control security profile on the firewall policy.

175. What is the maximum number of FortiAnalyzer/FortiManager devices a FortiGate unit can be configured to send logs to?

- A. 1
- B. 2
- C. 3
- D. 4

176. Regarding the header and body sections in raw log messages, which statement is correct?

- A. The header and body section layouts change depending on the log type.
- B. The header section layout is always the same regardless of the log type. The body section layout changes depending on the log type.
- C. Some log types include multiple body sections.
- D. Some log types do not include a body section.

177. A FortiAnalyzer device could use which security method to secure the transfer of log data from FortiGate devices?

- A SSL
- B IPSec
- C direct serial connection
- D S/MIME

178. A firewall policy has been configured such that traffic logging is disabled and a UTM function is enabled. In addition, the system setting `utm-incident-traffic-log' has been enabled. In which log will a UTM event message be stored?

- A. Traffic
- B. UTM
- C. System
- D. None

179. Which one of the following statements is correct about raw log messages?

- A. Logs have a header and a body section.
The header will have the same layout for every log message.
The body section will change layout from one type of log message to another.
- B. Logs have a header and a body section.
The header and body will change layout from one type of log message to another.
- C. Logs have a header and a body section.
The header and body will have the same layout for every log message.

180. When performing a log search on a FortiAnalyzer, it is generally recommended to use the Quick Search option.

What is a valid reason for using the Full Search option, instead?

- A. The search items you are looking for are not contained in indexed log fields.
- B. A quick search only searches data received within the last 24 hours.
- C. You want the search to include the FortiAnalyzer's local logs.
- D. You want the search to include content archive data as well.

181. Alert emails enable the FortiGate unit to send email notifications to an email address upon detection of a pre-defined event type. Which of the following are some of the available event types in Web Config? (Select all that apply.)

- A. Intrusion detected.
- B. Successful firewall authentication.
- C. Oversized file detected.
- D. DHCP address assigned.
- E. FortiGuard Web Filtering rating error detected.

182. View the exhibit. What does this raw log indicate? (Choose two.)

```
date=2018-01-30 time=07:21:49 logid="0316013057" type="utm" subtype="webfilter"
eventtype="ftgd_blk" level="warning" vd="root" logtime=1517325709 policyid=1
sessionid=15332 srcip=10.0.1.20 scrport=59538 srcintf="port3" srcintfrole="undefined"
dstip=208.91.112.55 dstport=80 dstintf="port1" dstintfrole="undefined" proto=6
service="HTTP" hostname="lavito.tk" profile="Category-block-and-warning" action="blocked"
reqtype="direct" url="/" sentbyte=140 rcvbyte=0 direction="outgoing" msg="URL belongs
a category with warnings enabled" method="domain" cat=0 catdesc="Unrated" crscore=30
crlevel="high"
```

ID	Name	From	To
2	IPS	port1	port3
1	Full_Access	port3	port1
0	Implicit Deny	☐ any	☐ any

- A. FortiGate blocked the traffic.
- B. type indicates that a security event was recorded.
- C. 10.0.1.20 is the IP address for lavito.tk.
- D. policyid indicates that traffic went through the IPS firewall policy.

183. By default, when logging to disk, when does FortiGate delete logs?

- A. 30 days
- B. 1 year
- C. Never
- D. 7 days

184. Which of the following items does NOT support the Logging feature?

- A. File Filter
- B. Application control
- C. Session timeouts
- D. Administrator activities
- E. Web URL filtering

185. Which of the following statements are correct regarding logging to memory on a FortiGate unit?

- A. When the system has reached its capacity for log messages, the FortiGate unit will stop logging to memory.
- B. When the system has reached its capacity for log messages, the FortiGate unit overwrites the oldest messages.
- C. If the FortiGate unit is reset or loses power, log entries captured to memory will be lost.
- D. None of the above.

186. What are the ways FortiGate can monitor logs? (Choose three.)

- A. MIB
- B. SMS
- C. Alert Emails
- D. SNMP
- E. FortiAnalyzer
- F. Alert Message Console

187. An administrator needs to offload logging to FortiAnalyzer from a FortiGate with an internal hard drive. Which statements are true? (Choose two.)

- A. Logs must be stored on FortiGate first, before transmitting to FortiAnalyzer
- B. FortiGate uses port 8080 for log transmission
- C. Log messages are transmitted as plain text in LZ4 compressed format (store-and-upload method).
- D. FortiGate can encrypt communications using SSL encrypted OFTP traffic.

QCM NSE4 : 7 Web Filter

188. Which UTM feature sends a UDP query to FortiGuard servers each time FortiGate scans a packet (unless the response is locally cached)?

- A. Antivirus
- B. VPN
- C. IPS
- D. Web Filtering

189. Which of the following web filtering modes can inspect the full URL? (Choose two.)

- A. Proxy based
- B. DNS based
- C. Policy based
- D. Flow based

190. Which two web filtering inspection modes inspect the full URL? (Choose two.)

- A. DNS-based
- B. Proxy-based
- C. Flow-based
- D. URL-based

191. Which web filtering inspection mode inspects DNS traffic?

- A. DNS-based
- B. FQDN-based
- C. Flow-based
- D. URL-based

192. What traffic and attacks can be blocked by a web application firewall (WAF) profile? (Choose three.)

- A. Traffic to inappropriate web sites
- B. SQL injection attacks
- C. Server information disclosure attacks
- D. Credit card data leaks
- E. Traffic to botnet command and control (C&C) servers

193. Which statements about DNS filter profiles are true? (Choose two.)

- A. They can inspect HTTP traffic.
- B. They must be applied in firewall policies with SSL inspection enabled.
- C. They can block DNS request to known botnet command and control servers.
- D. They can redirect blocked requests to a specific portal.

194. Which of the following are possible actions for static URL filtering? (Choose three.)

- A. Allow
- B. Block
- C. Exempt
- D. Warning
- E. Shape

195. Which of the following statements are true regarding the web filtering modes? (Choose two.)

- A. Proxy based mode allows for customizable block pages to display when sites are prevented.
- B. Proxy based mode requires more resources than flow-based.
- C. Flow based mode offers more settings under the advanced configuration section of the GUI.
- D. Proxy based mode offers higher throughput than flow-based mode.

196. Examine the following log message attributes and select two correct statements from the list below. (Choose two.)

hostname=www.youtube.com profiletype="Webfilter_Profile" profile="default" status="passthrough"
msg="URL belongs to a category with warnings enabled"

- A. The traffic was blocked.
- B. The user failed authentication.
- C. The category action was set to warning.
- D. The website was allowed

197. Which of the following are possible actions for FortiGuard web category filtering? (Choose three.)

- A. Allow
- B. Block
- C. Exempt
- D. Warning
- E. Shape

198. Which of the following actions can be used with the FortiGuard quota feature? (Choose three.)

- A. Allow
- B. Block
- C. Monitor
- D. Warning
- E. Authenticate

199. Which UTM feature sends a UDP query to FortiGuard servers each time FortiGate scans a packet (unless the response is locally cached)?

- A. Antivirus
- B. VPN
- C. IPS
- D. Web Filtering

200. Which email filter is NOT available on a FortiGate device?

- A. Sender IP reputation database.
- B. URLs included in the body of known SPAM messages.
- C. Email addresses included in the body of known SPAM messages.
- D. Spam object checksums.
- E. Spam grey listing.

201. Users may require access to a web site that is blocked by a policy. Administrators can give users the ability to override the block. Which of the following statements regarding overrides are correct? (Select all that apply.)

- A. A protection profile may have only one user group defined as an override group.
- B. A firewall user group can be used to provide override privileges for FortiGuard Web Filtering.
- C. Authentication to allow the override is based on a user's membership in a user group.
- D. Overrides can be allowed by the administrator for a specific period of time.

202. Which of the following statements is correct regarding URL Filtering on the FortiGate unit?

- A. The available actions for URL Filtering are Allow and Block.
- B. Multiple URL Filter lists can be added to a single Web filter profile.
- C. A FortiGuard Web Filtering Override match will override a block action in the URL filter list.
- D. The available actions for URL Filtering are Allow, Block and Exempt.

203. Which of the following statements is correct regarding URL Filtering on the FortiGate unit?

- A. The FortiGate unit can filter URLs based on patterns using text and regular expressions.
- B. The available actions for URL Filtering are Allow and Block.
- C. Multiple URL Filter lists can be added to a single Web filter profile.
- D. A FortiGuard Web Filtering Override match will override a block action in the URL filter list.

204. Which of the following statements describes the method of creating a policy to block access to an FTP site?

- A. Enable Web Filter URL blocking and add the URL of the FTP site to the URL Block list.
- B. Create a firewall policy with destination address set to the IP address of the FTP site, the Service set to FTP, and the Action set to Deny.
- C. Create a firewall policy with a protection profile containing the Block FTP option enabled.
- D. None of the above.

205. When the SSL proxy is NOT doing man-in-the-middle interception of SSL traffic, which certificate field can be used to determine the rating of a website?

- A. Organizational Unit.
- B. Common name.
- C. Serial Number.
- D. Validity.

206. NGFW mode allows policy-based configuration for most inspection rules. Which security profile's configuration does not change when you enable policy-based inspection?

- A. Web filtering
- B. Antivirus
- C. Web proxy
- D. Application control

207. An employee connects to the https://example.com on the Internet using a web browser. The web server's certificate was signed by a private internal CA. The FortiGate that is inspecting this traffic is configured for full SSL inspection.

This exhibit shows the configuration settings for the SSL/SSH inspection profile that is applied to the policy that is invoked in this instance. All other settings are set to defaults. No certificates have been imported into FortiGate. View the exhibit and answer the question that follows.

New SSL/SSH Inspection Profile

Name: Training

Comments: Write a comment... 0/255

SSL Inspection Options

Enable SSL Inspection of: Multiple Clients Connecting to Multiple Servers

Inspection Method: SSL Certificate Inspection Full SSL Inspection

CA Certificate: Fortinet_CA_SSL Download Certificate

Untrusted SSL Certificates: Allow Block View Trusted CAs List

Which certificate is presented to the employee's web browser?

- A. The web server's certificate.
- B. The user's personal certificate signed by a private internal CA.
- C. A certificate signed by Fortinet_CA_SSL.
- D. A certificate signed by Fortinet_CA_Untrusted.

208. An administrator is attempting to allow access to `https://fortinet.com` through a firewall policy that is configured with a web filter and an SSL inspection profile configured for deep inspection. Which of the following are possible actions to eliminate the certificate error generated by deep inspection? (Choose two.)

- A. Implement firewall authentication for all users that need access to `fortinet.com`.
- B. Manually install the FortiGate deep inspection certificate as a trusted CA.
- C. Configure `fortinet.com` access to bypass the IPS engine.
- D. Configure an SSL-inspection exemption for `fortinet.com`.

209. Which of the following features is supported by web filter in flow-based inspection mode with NGFW mode set to profile-based?

- A. FortiGuard Quotas
- B. Static URL
- C. Search engines
- D. Rating option

210. A team manager has decided that while some members of the team need access to particular website, the majority of the team does not. Which configuration option is the most effective option to support this request?

- A. Implement a web filter category override for the specified website.
- B. Implement web filter authentication for the specified website
- C. Implement web filter quotas for the specified website.
- D. Implement DNS filter for the specified website.

QCM NSE4 : SSL VPN

211. How is traffic routed onto an SSL VPN tunnel from the FortiGate unit side?

- A. A static route must be configured by the administrator using the ssl.root interface as the outgoing interface.
- B. Assignment of an IP address to the client causes a host route to be added to the FortiGate unit's kernel routing table.
- C. A route back to the SSLVPN IP pool is automatically created on the FortiGate unit.
- D. The FortiGate unit adds a route based upon the destination address in the SSL VPN firewall policy.

212. When the SSL proxy is NOT doing man-in-the-middle interception of SSL traffic, which certificate field can be used to determine the rating of a website?

- A. Organizational Unit.
- B. Common name.
- C. Serial Number.
- D. Validity.

213. Regarding the use of web-only mode SSL VPN, which statement is correct?

- A. It support SSL version 3 only.
- B. It requires a Fortinet-supplied plug-in on the web client.
- C. It requires the user to have a web browser that supports 64-bit cipher length.
- D. The JAVA run-time environment must be installed on the client.

214. Which of the following statements are true about Man-in-the-middle SSL Content Inspection? (Choose three.)

- A. The FortiGate device "re-signs" all the certificates coming from the HTTPS servers
- B. The FortiGate device acts as a sub-CA
- C. The local service certificate of the web server must be installed in the FortiGate device
- D. The FortiGate device does man-in-the-middle inspection.
- E. The required SSL Proxy certificate must first be requested to a public certificate authority (CA).

215. Which of the following FSSO agents are required for a DC agent mode solution? (Choose 2)

- A. FSSO agent
- B. DC agent
- C. Collector agent
- D. Radius server

216. What step is required an SSL VPN to access to an internal server using port forward mode?

- A. Configure the virtual IP addresses to be assigned to the SSL VPN users.
- B. Install FortiClient SSL VPN client
- C. Create a SSL VPN realm reserved for clients using port forward mode.
- D. Configure the client application to forward IP traffic to a Java applet proxy.

217. A company needs to provide SSL VPN access to two user groups. The company also needs to display different welcome messages on the SSL VPN login screen for both user groups. What is required in the SSL VPN configuration to meet these requirements?

- A. Two separated SSL VPNs in different interfaces of the same VDOM
- B. Different SSL VPN realms for each group
- C. Different virtual SSLVPN IP addresses for each group
- D. Two firewall policies with different captive portals

218. A user logs into a SSL VPN portal and activates the tunnel mode. The exhibit shows the firewall policy and the user's SSL VPN portal configuration:

Seq.#	Source	Destination	Schedule	Service	Action	NAT
port2 - port1 (1 - 1)						
1	all	all	always	ALL	✓ ACCEPT	Enable
ssl.root (SSL VPN interface) - port2 (2 - 2)						
2	all training	Internal Servers	always	ALL	✓ ACCEPT	Disable
Implicit (3 - 3)						
3	all	all	always	ALL	✗ DENY	

Name: full-access

☒ Enable Tunnel Mode

☒ Enable Split Tunneling

Routing Address: Click to add...

Source IP Pools: SSLVPN_TUNNEL_ADDR1

Client Options: ☐ Save Password ☐ Auto Connect ☐ Always Up (Keep Alive)

Which static route is automatically added to the client's routing table when the tunnel mode is activated?

- A. A route to a destination subnet matching the *Internal Servers* address object.
- B. A route to the destination subnet configured in the tunnel mode widget.
- C. A default route.
- D. A route to the destination subnet configured in the SSL VPN global settings.

219. When a user attempts to connect to an HTTPS site, what is the expected result with this configuration?

View the exhibit.

SSL Inspection Options

Enable SSL Inspection of: Multiple Clients Connecting to Multiple Servers

Inspection Method: Full SSL Inspection

CA Certificate: Fortinet_CA_SSL

Untrusted SSL Certificates: Allow

RPC over HTTPS: ☒

- A. The user is required to authenticate before accessing sites with untrusted SSL certificates.
- B. The user is presented with certificate warnings when connecting to sites that have untrusted SSL certificates.
- C. The user is allowed access all sites with untrusted SSL certificates, without certificate warnings.
- D. The user is blocked from connecting to sites that have untrusted SSL certificates (no exception provided).

220. An administrator needs to inspect all web traffic (including Internet web traffic) coming from users connecting to SSL VPN. How can this be achieved?

- A. Disabling split tunneling
- B. Configuring web bookmarks
- C. Assigning public IP addresses to SSL VPN clients
- D. Using web-only mode

221. How can a browser trust a web-server certificate signed by a third party CA?

- A. The browser must have the CA certificate that signed the web-server certificate installed.
- B. The browser must have the web-server certificate installed.
- C. The browser must have the private key of CA certificate that signed the web-browser certificate installed.
- D. The browser must have the public key of the web-server certificate installed.

222. When browsing to an internal web server using a web-mode SSL VPN bookmark, which IP address is used as the source of the HTTP request?

- A. The FortiGate unit's public IP address
- B. The FortiGate unit's internal IP address
- C. The remote user's virtual IP address
- D. The remote user's public IP address

223. Which statement best describes what SSL.root is?

- A. The name of the virtual network adapter required in each user's PC for SSL VPN Tunnel mode.
- B. The name of a virtual interface in the root VDOM where all the SSL VPN user traffic comes from.
- C. A Firewall Address object that contains the IP addresses assigned to SSL VPN users.
- D. The virtual interface in the root VDOM that the remote SSL VPN tunnels connect to.

224. A FortiGate is configured with the 1.1.1.1/24 address on the wan2 interface and HTTPS Administrative Access, using the default tcp port, is enabled for that interface. Given the SSL VPN settings in the exhibit.

URL Path	Virtual Host	Max Concurrent U
Training		0
students		0

Which of the following SSL VPN login portal URLs are valid? (Choose two.)

- A. http://1.1.1.1:443/Training
- B. https://1.1.1.1:443/STUDENTS
- C. https://1.1.1.1/login
- D. https://1.1.1.1/

225. Which of the following statements are correct regarding SSL VPN Web-only mode? (Choose two.)

- A. It can only be used to connect to web services.
- B. IP traffic is encapsulated over HTTPS.
- C. Access to internal network resources is possible from the SSL VPN portal.
- D. The standalone FortiClient SSL VPN client CANNOT be used to establish a Web-only SSL VPN.
- E. It is not possible to connect to SSH servers through the VPN.

226. Which of the following authentication methods can be used for SSL VPN authentication? (Choose three.)

- A. Remote Password Authentication (RADIUS, LDAP)
- B. Two-Factor Authentication
- C. Local Password Authentication
- D. FSSO
- E. RSSO

227. Which statement best describes what SSL VPN Client Integrity Check does?

- A. Blocks SSL VPN connection attempts from users that has been blacklisted.
- B. Detects the Windows client security applications running in the SSL VPN client's PCs.
- C. Validates the SSL VPN user credential.
- D. Verifies which SSL VPN portal must be presented to each SSL VPN user.
- E. Verifies that the latest SSL VPN client is installed in the client's PC.

228. Which statement is not correct regarding SSL VPN Tunnel mode?

- A. IP traffic is encapsulated over HTTPS.
- B. The standalone FortiClient SSL VPN client can be used to establish a Tunnel mode SSL VPN.
- C. A limited amount of IP applications are supported.
- D. The FortiGate device will dynamically assign an IP address to the SSL VPN network adapter.

229. Which of the following statements describe some of the differences between symmetric and asymmetric cryptography? (Choose two.)

- A. In symmetric cryptography, the keys are publicly available. In asymmetric cryptography, the keys must be kept secret.
- B. Asymmetric cryptography can encrypt data faster than symmetric cryptography
- C. Symmetric cryptography uses one pre-shared key. Asymmetric cryptography uses a pair of keys
- D. Asymmetric keys can be sent to the remote peer via digital certificates. Symmetric keys cannot

230. Which of the following statements best describes what a Public Certificate Authority (CA) is?

- A. A service that provides a digital certificate each time a user is authenticating
- B. An entity that certifies that the information contained in a digital certificate is valid and true.
- C. The FortiGate process in charge of generating digital certificates on the fly for SSL inspection purposes
- D. A service that validates digital certificates for certificate-based authentication purposes

231. Which of the following statements are true about the SSL Proxy certificate that must be used for SSL Content Inspection? (Choose two.)

- A. It cannot be signed by a private CA
- B. It must have either the field "CA=True" or the field "Key Usage=KeyCertSign"
- C. It must be installed in the FortiGate device
- D. The subject field must contain either the FQDN, or the IP address of the FortiGate device

232. Which of the following statements are true about PKI users created in a FortiGate device? (Choose two.)

- A. Can be used for token-based authentication
- B. Can be used for two-factor authentication
- C. Are used for certificate-based authentication
- D. Cannot be members of user groups

233. Which of the following statements best describes what a Certificate Signing Request (CSR) is?

- A. A message sent by the Certificate Authority (CA) that contains a signed digital certificate.
- B. An enquiry submitted to a Certificate Authority (CA) to request a root CA certificate
- C. An enquiry submitted to a Certificate Authority (CA) to request a signed digital certificate
- D. An enquiry submitted to a Certificate Authority (CA) to request a Certificate Revocation List (CRL)

234. Which of the following actions can be used to back up the keys and digital certificates in a FortiGate device? (Choose two.)

- A. Taking a full backup of the FortiGate configuration
- B. Uploading a PKCS#10 file to a USB drive
- C. Manually uploading the certificate information to a Certificate authority (CA)
- D. Uploading a PKCS#12 file to a TFTP server

235. Which of the following statements must be true for a digital certificate to be valid? (Choose two.)

- A. It must be signed by a "trusted" CA
- B. It must be listed as valid in a Certificate Revocation List (CRL)
- C. The CA field must be "TRUE"
- D. It must be still within its validity period

236. Which statement is true regarding SSL VPN timers? (Choose two.)

- A. Allow to mitigate DoS attacks from partial HTTP requests.
- B. SSL VPN settings do not have customizable timers.
- C. Disconnect idle SSL VPN users when a firewall policy authentication timeout occurs.
- D. Prevent SSL VPN users from being logged out because of high network latency.

237. Which of the following conditions must be met in order for a web browser to trust a web server certificate signed by a third-party CA?

- A. The public key of the web server certificate must be installed on the browser.
- B. The web-server certificate must be installed on the browser.
- C. The CA certificate that signed the web-server certificate must be installed on the browser.
- D. The private key of the CA certificate that signed the browser certificate must be installed on the browser.

238. HTTP Public Key Pinning (HPKP) can be an obstacle to implementing full SSL inspection. What solutions could resolve this problem? (Choose two.)

- A. Enable **Allow Invalid SSL Certificates** for the relevant security profile.
- B. Change web browsers to one that does not support HPKP.
- C. Exempt those web sites that use HPKP from full SSL inspection.
- D. Install the CA certificate (that is required to verify the web server certificate) stores of users' computers.

239. Which of the following statements is true regarding SSL VPN settings for an SSL VPN portal?

- A. By default, FortiGate uses WINS servers to resolve names.
- B. By default, the SSL VPN portal requires the installation of a client's certificate.
- C. By default, split tunneling is enabled.
- D. By default, the admin GUI and SSL VPN portal use the same HTTPS port.

240. Which is the correct description of a hash result as it relates to digital certificates?

- A. A unique value used to verify the input data
- B. An output value that is used to identify the person or deduce that authored the input data.
- C. An obfuscation used to mask the input data.
- D. An encrypted output value used to safe-guard the input data

241. An administrator needs to create an SSL-VPN connection for accessing an internal server using the bookmark Port Forward. What step is required for this configuration?

- A. Configure an SSL VPN realm for clients to use the port forward bookmark.
- B. Configure the client application to forward IP traffic through FortiClient.
- C. Configure the virtual IP address to be assigned to the SSL VPN users.
- D. Configure the client application to forward IP traffic to a Java applet proxy.

242. Which users and user groups are allowed access to the network through captive portal?

The screenshot shows the 'Admission Control' configuration page. The 'Security Mode' is set to 'Captive Portal'. The 'Authentication Portal' is set to 'Local'. The 'User Access' is set to 'Restricted to Groups', with an 'Allow all' button.

- A. Only individual users—not groups—defined in the captive portal configuration.
- B. Groups defined in the captive portal configuration
- C. All users
- D. Users and groups defined in the firewall policy

243. Which two statements are true about IPsec VPNs and SSL VPNs? (Choose two.)

- A. SSL VPN creates a HTTPS connection. IPsec does not.
- B. Both SSL VPNs and IPsec VPNs are standard protocols.
- C. Either a SSL VPN or an IPsec VPN can be established between two FortiGate devices.
- D. Either a SSL VPN or an IPsec VPN can be established between an end-user workstation and a FortiGate device

244. Regarding tunnel-mode SSL VPN, which three statements are correct? (Choose three.)

- A. Split tunneling is supported.
- B. It requires the installation of a VPN client.
- C. It requires the use of an Internet browser.
- D. It does not support traffic from third-party network applications.
- E. An SSL VPN IP address is dynamically assigned to the client by the FortiGate unit

245. Which tasks fall under the responsibility of the SSL proxy in a typical HTTPS connection? (Choose two.)

- A. The web client SSL handshake.
- B. The web server SSL handshake.
- C. File buffering.
- D. Communication with the URL filter process.

246. A client can create a secure connection to a FortiGate device using SSL VPN in web-only mode. Which one of the following statements is correct regarding the use of web-only mode SSL VPN?

- A. Web-only mode supports SSL version 3 only.
- B. A Fortinet-supplied plug-in is required on the web client to use web-only mode SSL VPN.
- C. Web-only mode requires the user to have a web browser that supports 64-bit cipher length.
- D. The JAVA run-time environment must be installed on the client to be able to connect to a web-only mode SSL VPN.

247. A client can establish a secure connection to a corporate network using SSL VPN in tunnel mode. Which of the following statements are correct regarding the use of tunnel mode SSL VPN? (Select all that apply.)

- A. Split tunneling can be enabled when using tunnel mode SSL VPN.
- B. Client software is required to be able to use a tunnel mode SSL VPN.
- C. Users attempting to create a tunnel mode SSL VPN connection must be authenticated by at least one SSL VPN policy.
- D. The source IP address used by the client for the tunnel mode SSL VPN is assigned by the FortiGate unit.

248. An issue could potentially occur when clicking Connect to start tunnel mode SSL VPN. The tunnel will start up for a few seconds, then shut down. Which of the following statements best describes how to resolve this issue?

- A. This user does not have permission to enable tunnel mode.
Make sure that the tunnel mode widget has been added to that user's web portal.
- B. This FortiGate unit may have multiple Internet connections.
To avoid this problem, use the appropriate CLI command to bind the SSL VPN connection to the original incoming interface.
- C. Check the SSL adaptor on the host machine.
If necessary, uninstall and reinstall the adaptor from the tunnel mode portal.
- D. Make sure that only Internet Explorer is used. All other browsers are unsupported.

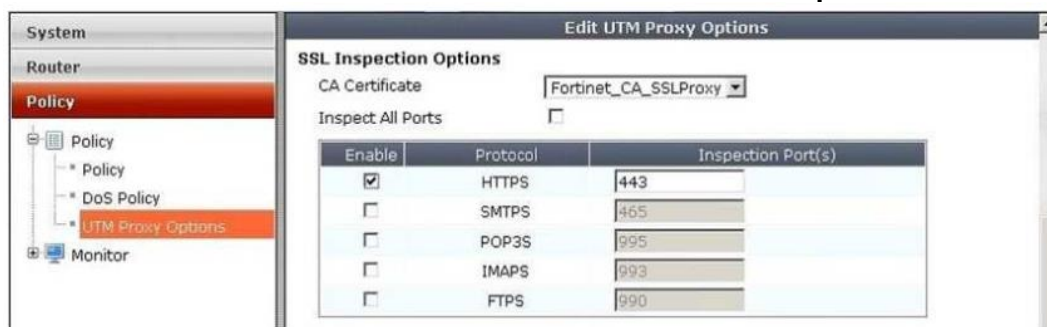
249. A FortiGate unit can create a secure connection to a client using SSL VPN in tunnel mode. Which of the following statements are correct regarding the use of tunnel mode SSL VPN? (Select all that apply.)

- A. Split tunneling can be enabled when using tunnel mode SSL VPN.
- B. Software must be downloaded to the web client to be able to use a tunnel mode SSL VPN.
- C. Users attempting to create a tunnel mode SSL VPN connection must be members of a configured user group on the FortiGate unit.
- D. Tunnel mode SSL VPN requires the FortiClient software to be installed on the user's computer.
- E. The source IP address used by the client for the tunnel mode SSL VPN is assigned by the FortiGate unit.

250. SSL content inspection is enabled on the FortiGate unit. Which of the following steps is required to prevent a user from being presented with a web browser warning when accessing an SSL- encrypted website?

- A. The root certificate of the FortiGate SSL proxy must be imported into the local certificate store on the user's workstation.
- B. Disable the strict server certificate check in the web browser under Internet Options.
- C. Enable transparent proxy mode on the FortiGate unit.
- D. Enable NTLM authentication on the FortiGate unit. NTLM authentication suppresses the certificate warning messages in the web browser.

251. Examine the exhibit shown below then answer the question that follows it.



Within the UTM Proxy Options, the CA certificate Fortinet_CA_SSLProxy defines which of the following:

- A. FortiGate unit's encryption certificate used by the SSL proxy.
- B. FortiGate unit's signing certificate used by the SSL proxy.
- C. FortiGuard's signing certificate used by the SSL proxy.
- D. FortiGuard's encryption certificate used by the SSL proxy.

252. An end user logs into the SSL VPN portal and selects the Tunnel Mode option by clicking on the "Connect" button. The administrator has not enabled split tunneling and so the end user must access the Internet through the SSL VPN Tunnel. Which firewall policies are needed to allow the end user to not only access the internal network but also reach the Internet?

A.

Status	ID	Source	Destination	Schedule	Service	Action
▼ ssl.root -> internal (1)						
☒	2	all	all	always	ANY	ACCEPT
▼ ssl.root -> wan1 (1)						
☒	3	all	all	always	ANY	ACCEPT
▼ wan1 -> internal (1)						
☒	1	all	all	always	ANY	SSL-VPN
► Implicit (1)						

B.

Status	ID	Source	Destination	Schedule
▼ ssl.root -> internal (1)				
☒	2	all	all	always ANY
▼ ssl.root -> wan1 (1)				
☒	3	all	all	always ANY
▼ wan1 -> internal (1)				
☒	1	all	all	always ANY
► Implicit (1)				

C.

Status	ID	Source	Destination	Schedule
▼ wan1 -> internal (1)				
☒	1	all	all	always ANY
▼ wan1 -> wan1 (1)				
☒	2	all	all	always ANY
► Implicit (1)				

D.

Status	ID	Source	Destination	Schedule
▼ wan1 -> internal (1)				
☒	1	all	all	always
▼ wan1 -> wan1 (1)				
☒	2	all	all	always
► Implicit (1)				

253. Which of the following statements are correct regarding the configuration of a FortiGate unit as an SSL VPN gateway? (Select all that apply.) A. Tunnel mode can only be used if the SSL VPN user groups have at least one Host Check option enabled.

B. The specific routes needed to access internal resources through an SSL VPN connection in tunnel mode from the client computer are defined in the routing widget associated with the SSL VPN portal.

C. In order to apply a portal to a user, that user must belong to an SSL VPN user group.

D. The portal settings specify whether the connection will operate in web-only or tunnel mode.

254. When the SSL proxy inspects the server certificate for Web Filtering only in SSL Handshake mode, which certificate field is being used to determine the site rating?

- A. Common Name
- B. Organization
- C. Organizational Unit
- D. Serial Number
- E. Validity

255. In the Tunnel Mode widget of the web portal, the administrator has configured an IP Pool and enabled split tunneling.

Which of the following statements is true about the IP address used by the SSL VPN client?

- A. The IP pool specified in the SSL-VPN Tunnel Mode Widget Options will override the IP address range defined in the SSL-VPN Settings.
- B. Because split tunneling is enabled, no IP address needs to be assigned for the SSL VPN tunnel to be established.
- C. The IP address range specified in SSL-VPN Settings will override the IP address range in the SSL-VPN Tunnel Mode Widget Options.

256. The Host Check feature can be enabled on the FortiGate unit for SSL VPN connections. When this feature is enabled, the FortiGate unit probes the remote host computer to verify that it is "safe" before access is granted.

Which of the following items is NOT an option as part of the Host Check feature?

- A. FortiClient Antivirus software
- B. Microsoft Windows Firewall software
- C. FortiClient Firewall software
- D. Third-party Antivirus software

257. What is required in a FortiGate configuration to have more than one dialup IPsec VPN using aggressive mode?

- A. All the aggressive mode dialup VPNs MUST accept connections from the same peer ID.
- B. Each peer ID MUST match the FQDN of each remote peer.
- C. Each aggressive mode dialup MUST accept connections from different peer ID.
- D. The peer ID setting must NOT be used.

258. An end user logs into the full-access SSL VPN portal and selects the Tunnel Mode option by clicking on the "Connect" button. The administrator has enabled split tunneling.

Given that the user authenticates against the SSL VPN policy shown in the image below, which statement below identifies the route that is added to the client's routing table.

Seq.#	Source	Destination	Schedule	Service	Authentication	Action	UTM Profile	Log	NA
▼ port3 - port1 (1 - 1)									
1	all	all	always	ALL		✓ ACCEPT		⊗	✓
▼ port1 - port3 (2 - 2)									
2	all	WIN2K3				SSL-VPN			⊗
▼ ssl.root (sslvpn tunnel interface) - port3 (3 - 3)									
3	all	all	always	ALL		✓ ACCEPT		⊗	✓
▼ Implicit (4 - 4)									
4	any	any	always	ALL		⊗ DENY		⊗	

- A. A route to destination matching the 'WIN2K3' address object.
- B. A route to the destination matching the 'all' address object.
- C. A default route.
- D. No route is added.

QCM_NSE4_SSO

259. With FSSO DC-agent mode, a domain user could authenticate either against the domain controller running the collector agent and domain controller agent, or a domain controller running only the domain controller agent. If you attempt to authenticate with a domain controller running only the domain controller agent, which statements are correct? (Choose two.)

- A. The login event is sent to a collector agent by the DC agent.
- B. the login event is sent to the FortiGate by the DC agent.
- C. The domain collector agent may perform a DNS lookup for the authenticated client's IP address.
- D. The user cannot be authenticated with the FortiGate in this manner because each domain controller agent requires a dedicated collector agent.

260. Which statement describes what the CLI command `diagnose debug authd fssolist` is used for?

- A. Monitors communications between the FSSO collector agent and FortiGate unit.
- B. Displays which users are currently logged on using FSSO.
- C. Displays are listing of all connected FSSO collector agents.
- D. Lists all DC Agents installed on all domain controllers.

261. Which statement best describes what `SSL.root` is?

- A. The name of the virtual network adapter required in each user's PC for SSL VPN Tunnel mode.
- B. The name of a virtual interface in the root VDOM where all the SSL VPN user traffic comes from.
- C. A Firewall Address object that contains the IP addresses assigned to SSL VPN users.
- D. The virtual interface in the root VDOM that the remote SSL VPN tunnels connect to.

262. A FortiGate is configured with the 1.1.1.1/24 address on the wan2 interface and HTTPS Administrative Access, using the default tcp port, is enabled for that interface. Given the SSL VPN settings in the exhibit.

URL Path	Virtual Host	Max Concurrent U
Training		0
students		0

Which of the following SSL VPN login portal URLs are valid? (Choose two.)

- A. `http://1.1.1.1:443/Training`
- B. `https://1.1.1.1:443/STUDENTS`
- C. `https://1.1.1.1/login`
- D. `https://1.1.1.1/`

263. Which statement best describes what SSL VPN Client Integrity Check does?

- A. Blocks SSL VPN connection attempts from users that has been blacklisted.
- B. Detects the Windows client security applications running in the SSL VPN client's PCs.
- C. Validates the SSL VPN user credential.
- D. Verifies which SSL VPN portal must be presented to each SSL VPN user.
- E. Verifies that the latest SSL VPN client is installed in the client's PC.

264. What are the advantages of FSSO DC mode over polling mode?

- A. Redundancy in the collector agent.
- B. Allows transparent authentication.
- C. DC agents are not required in the AD domain controllers.
- D. Scalability

265. Which of the following statements are correct about NTLM authentication? (Choose three)

- A. NTLM negotiation starts between the FortiGate device and the user's browser.
- B. It must be supported by the user's browser.
- C. It must be supported by the domain controllers.
- D. It does not require a collector agent.
- E. It does not require DC agents.

266. Which of the following statements best describes how the collector agent learns that a user has logged off from the network?

- A. The workstation fails to reply to the polls frequently done by the collector agent.
- B. The DC agent captures the log off event from the event logs, which it forwards to the collector agent.
- C. The work station notifies the DC agent that the user has logged off.
- D. The collector agent gets the logoff events when polling the respective domain controller.

267. Which of the following statements best describes the role of a DC agents in an FSSO DC?

- A. Captures the login events and forward them to the collector agent.
- B. Captures the user IP address and workstation name and forward that information to the FortiGate devices.
- C. Captures the login and logoff events and forward them to the collector agent.
- D. Captures the login events and forward them to the FortiGate devices.

268. Which of the following FSSO modes must be used for Novell eDirectory networks?

- A. Agentless polling
- B. LDAP agent
- C. eDirectory agent
- D. DC agent

269. In a FSSO agentless polling mode solution, where must the collector agent be?

- A. In any Windows server
- B. In any of the AD domain controllers
- C. In the master AD domain controller
- D. The FortiGate device polls the AD domain controllers

270. Which of the following statements are characteristics of a FSSO solution using advanced access mode? (Choose three.)

- A. Protection profiles can be applied to both individual users and user groups
- B. Nested or inherited groups are supported
- C. Usernames follow the LDAP convention: CN=User, OU=Name, DC=Domain
- D. Usernames follow the Windows convention: Domain\username
- E. Protection profiles can be applied to user groups only.

- 271. Which of the following FSSO agents are required for a DC agent mode solution? (Choose two.)**
- A. FSSO agent
 - B. DC agent
 - C. Collector agent
 - D. Radius server
- 272. In a FSSO agent mode solution, how does the FSSO collector agent learn each IP address?**
- A. The DC agents get each user IP address from the event logs and forward that information to the collector agent
 - B. The collector agent does not know, and does not need, each user IP address. Only workstation names are known by the collector agent.
 - C. The collector agent frequently polls the AD domain controllers to get each user IP address.
 - D. The DC agent learns the workstation name from the event logs and DNS is then used to translate those names to the respective IP addresses.
- 273. Which FSSO agents are required for a FSSO agent-based polling mode solution?**
- A. Collector agent and DC agents
 - B. Polling agent only
 - C. Collector agent only
 - D. DC agents only
- 274. What protocol cannot be used with the active authentication type?**
- A. Local
 - B. RADIUS
 - C. LDAP
 - D. RSSO
- 275. Which of the following statements about NTLM authentication are correct? (Choose two.)**
- A. It is useful when users log in to DCs that are not monitored by a collector agent.
 - B. It takes over as the primary authentication method when configured alongside FSSO.
 - C. Multi-domain environments require DC agents on every domain controller.
 - D. NTLM-enabled web browsers are required.
- 276. How does FortiGate verify the login credentials of a remote LDAP user?**
- A. FortiGate sends the user entered credentials to the LDAP server for authentication.
 - B. FortiGate re-generates the algorithm based on the login credentials and compares it against the algorithm stored on the LDAP server.
 - C. FortiGate queries its own database for credentials.
 - D. FortiGate queries the LDAP server for credentials.
- 277. Which of the following statements about advanced AD access mode for FSSO collector agent are true? (Choose two.)**
- A. It is only supported if DC agents are deployed.
 - B. FortiGate can act as an LDAP client configure the group filters.
 - C. It supports monitoring of nested groups.
 - D. It uses the Windows convention for naming, that is, Domain\Username.

- 278. Which of the following statements describe WMI polling mode for FSSO collector agent? (Choose two.)**
- A. The collector agent does not need to search any security event logs.
 - B. WMI polling can increase bandwidth usage with large networks.
 - C. The **NetSessionEnum** function is used to track user logoffs.
 - D. The collector agent uses a Windows API to query DCs for user logins.
- 279. Which authentication scheme is not supported by the RADIUS implementation on FortiGate?**
- A. CHAP
 - B. MSCHAP2
 - C. PAP
 - D. FSSO
- 280. FSSO provides a single sign on solution to authenticate users transparently to a FortiGate unit using credentials stored in Windows active directory. Which of the following statements are correct regarding FSSO in a Windows domain environment when DC-agent mode is used? (Choose two.)**
- A. An FSSO collector agent must be installed on every domain controller.
 - B. An FSSO domain controller agent must be installed on every domain controller.
 - C. The FSSO domain controller agent will regularly update user logon information on the FortiGate unit.
 - D. The FSSO collector agent will receive user logon information from the domain controller agent and will send it to the FortiGate unit.
- 281. Which best describes the authentication timeout?**
- A. How long FortiGate waits for the user to enter his or her credentials.
 - B. How long a user is allowed to send and receive traffic before he or she must authenticate again.
 - C. How long an authenticated user can be idle (without sending traffic) before they must authenticate again.
 - D. How long a user-authenticated session can exist without having to authenticate again.
- 282. Which are valid replies from a RADIUS server to an ACCESS-REQUEST packet from a FortiGate? (Choose two.)**
- A. ACCESS-CHALLENGE
 - B. ACCESS-RESTRICT
 - C. ACCESS-PENDING
 - D. ACCESS-REJECT
- 283. What protocol cannot be used with the active authentication type?**
- A. Local
 - B. RADIUS
 - C. LDAP
 - D. RSSO
- 284. When configuring LDAP on the FortiGate as a remote database for users, what is not a part of the configuration?**
- A. The name of the attribute that identifies each user (Common Name Identifier).
 - B. The user account or group element names (user DN).
 - C. The server secret to allow for remote queries (Primary server secret).
 - D. The credentials for an LDAP administrator (password).
- 285. Which authentication methods does FortiGate support for firewall authentication? (Choose two.)**
- A. Remote Authentication Dial in User Service (RADIUS)
 - B. Lightweight Directory Access Protocol (LDAP)
 - C. Local Password Authentication
 - D. POP3
 - E. Remote Password Authentication

286. Which user group types does FortiGate support for firewall authentication? (Choose three.)

- A. RSSO
- B. Firewall
- C. LDAP
- D. NTLM
- E. FSSO

287. Which two statements are true regarding firewall policy disclaimers? (Choose two.)

- A. They cannot be used in combination with user authentication.
- B. They can only be applied to wireless interfaces.
- C. Users must accept the disclaimer to continue.
- D. The disclaimer page is customizable.

288. Which of the following statements are true about PKI users created in a FortiGate device? (Choose two.)

- A. Can be used for token-based authentication
- B. Can be used for two-factor authentication
- C. Are used for certificate-based authentication
- D. Cannot be members of user groups

289. Which of the following statements about the FSSO collector agent timers is true?

- A. The **workstation verify interval** is used to periodically check if a workstation is still a domain member.
- B. The **IP address change verify interval** monitors the server IP address where the collector agent is installed, and updates the collector agent configuration if it changes.
- C. The **user group cache expiry** is used to age out the monitored groups.
- D. The **dead entry timeout interval** is used to age out entries with an unverified status.

290. Which statement is one disadvantage of using FSSO NetAPI polling mode over FSSO Security Event Log (WinSecLog) polling mode?

- A. It requires a DC agent installed in some of the Windows DCs.
- B. It runs slower.
- C. It might miss some logon events.
- D. It requires access to a DNS server for workstation name resolution.

291. Which are two requirements for DC-agent mode FSSO to work properly in a Windows AD environment? (Choose two.)

- A. DNS server must properly resolve all workstation names.
- B. The remote registry service must be running in all workstations.
- C. The collector agent must be installed in one of the Windows domain controllers.
- D. A same user cannot be logged in into two different workstations at the same time.

292. Which statements are true regarding local user authentication? (Choose two.)

- A. Two-factor authentication can be enabled on a per user basis.
- B. Local users are for administration accounts only and cannot be used to authenticate network users.
- C. Administrators can create the user accounts on a remote server and store the user passwords locally in the FortiGate.
- D. Both the usernames and passwords can be stored locally on the FortiGate.

293. The FortiGate port1 is connected to the Internet. The FortiGate port2 is connected to the internal network. Examine the firewall configuration shown in the exhibit; then answer the question below.

Create New Edit Delete			Section View Global View		Search			
Seq.#	Source	Destination	Schedule	Service	Action	NAT	AV	Web F
port2 - port1 (1 - 1)								
1	all training	all	always	ALL	✓ ACCEPT	Enable		
Implicit (2 - 2)								
2	all	all	always	ALL	⊘ DENY			

Based on the firewall configuration illustrated in the exhibit, which statement is correct?

- A. A user that has not authenticated can access the Internet using any protocol that does not trigger an authentication challenge.
- B. A user that has not authenticated can access the Internet using any protocol except HTTP, HTTPS, Telnet, and FTP.
- C. A user must authenticate using the HTTP, HTTPS, SSH, FTP, or Telnet protocol before they can access all Internet services.
- D. DNS Internet access is always allowed, even for users that has not authenticated.

294. When firewall policy authentication is enabled, which protocols can trigger an authentication challenge? (Choose two)

- A. SMTP
- B. POP3
- C. HTTP
- D. FTP

295. Which statement regarding the firewall policy authentication timeout is true?

- A. It is an idle timeout. The FortiGate considers a user to be "idle" if it does not see any packets coming from the user's source IP.
- B. It is a hard timeout. The FortiGate removes the temporary policy for a user's source IP address after this timer has expired.
- C. It is an idle timeout. The FortiGate considers a user to be "idle" if it does not see any packets coming from the user's source MAC.
- D. It is a hard timeout. The FortiGate removes the temporary policy for a user's source MAC address after this timer has expired.

296. When firewall policy authentication is enabled, only traffic on supported protocols will trigger an authentication challenge. Select all supported protocols from the following:

- A. SMTP
- B. SSH
- C. HTTP
- D. FTP
- E. SCP

297. Which of the following authentication types are supported by FortiGate units? (Select all that apply.)

- A. Kerberos
- B. LDAP
- C. RADIUS
- D. Local Users

298. Which of the following are valid authentication user group types on a FortiGate unit? (Select all that apply.)

- A. Firewall
- B. Directory Service
- C. Local
- D. LDAP
- E. PKI

299. An administrator has configured a FortiGate unit so that end users must authenticate against the firewall using digital certificates before browsing the Internet.

What must the user have for a successful authentication? (Select all that apply.)

- A. An entry in a supported LDAP Directory.
- B. A digital certificate issued by any CA server.
- C. A valid username and password.
- D. A digital certificate issued by the FortiGate unit.
- E. Membership in a firewall user group.

300. The FortiGate unit can be configured to allow authentication to a RADIUS server. The RADIUS server can use several different authentication protocols during the authentication process. Which of the following are valid authentication protocols that can be used when a user authenticates to the RADIUS server? (Select all that apply.)

- A. MS-CHAP-V2 (Microsoft Challenge-Handshake Authentication Protocol v2)
- B. PAP (Password Authentication Protocol)
- C. CHAP (Challenge-Handshake Authentication Protocol)
- D. MS-CHAP (Microsoft Challenge-Handshake Authentication Protocol v1)
- E. FAP (FortiGate Authentication Protocol)

301. Which of the following are valid components of the Fortinet Server Authentication Extensions (FSAE)? (Select all that apply.)

- A. Domain Local Security Agent.
- B. Collector Agent.
- C. Active Directory Agent.
- D. User Authentication Agent.
- E. Domain Controller Agent.

302. The Idle Timeout setting on a FortiGate unit applies to which of the following?

- A. Web browsing
- B. FTP connections
- C. User authentication
- D. Administrator access
- E. Web filtering overrides

303. With FSSO, a domain user could authenticate either against the domain controller running the Collector Agent and Domain Controller Agent, or a domain controller running only the Domain Controller Agent.

If you attempt to authenticate with the Secondary Domain Controller running only the Domain Controller Agent, which of the following statements are correct?

(Select all that apply.)

- A. The login event is sent to the Collector Agent.
- B. The FortiGate unit receives the user information from the Domain Controller Agent of the Secondary Controller.
- C. The Collector Agent performs the DNS lookup for the authenticated client's IP address.
- D. The user cannot be authenticated with the FortiGate device in this manner because each Domain Controller Agent requires a dedicated Collector Agent.

304. Which of the following statements is correct based on the firewall configuration illustrated in the exhibit?

ID	Source	Destination	Schedule	Service	Action	Status	Authentication
internal -> external (6)							
internal -> vlink0 (1)							
internal -> wan2 (1)	8	all	always	ANY	ACCEPT	☒	local
Implicit (1)							

- A. A user can access the Internet using only the protocols that are supported by user authentication.
- B. A user can access the Internet using any protocol except HTTP, HTTPS, Telnet, and FTP. These require authentication before the user will be allowed access.
- C. A user must authenticate using the HTTP, HTTPS, SSH, FTP, or Telnet protocol before they can access any services.
- D. A user cannot access the Internet using any protocols unless the user has passed firewall authentication.

305. Which of the following FSSO agents are required for a DC agent mode solution? (Choose 2)

- A. FSSO agent
- B. DC agent
- C. Collector agent
- D. Radius server

306. Which statement about the firewall policy authentication timeout is true?

- A. It is a hard timeout. The FortiGate removes the temporary policy for a user's source IP address after this times expires.
- B. It is a hard timeout. The FortiGate removes the temporary policy for a user's source MAC address after this times expires.
- C. It is an idle timeout. The FortiGate considers a user to be idle if it does not see any packets coming from the user's source MAC address
- D. It is an idle timeout. The FortiGate considers a user to be idle if it does not see any packets coming from the user's source IP.

307. Which best describes the authentication timeout?

- A. How long FortiGate waits for the user to enter his or her credentials.
- B. How long a user is allowed to send and receive traffic before he or she must authenticate again.
- C. How long an authenticated user can be idle (without sending traffic) before they must authenticate again.
- D. How long a user-authenticated session can exist without having to authenticate again.

308. Which statement about the firewall policy authentication timeout is true?

- A. It is a hard timeout. The FortiGate removes the temporary policy for a user's source IP address after this times expires.
- B. It is a hard timeout. The FortiGate removes the temporary policy for a user's source MAC address after this times expires.
- C. It is an idle timeout. The FortiGate considers a user to be idle if it does not see any packets coming from the user's source MAC address.
- D. It is an idle timeout. The FortiGate considers a user to be idle if it does not see any packets coming from the user's source IP.

QCM_NSE4_antivirus

309. Files reported as "suspicious" were subject to which Antivirus check"?

- A. Grayware
- B. Virus
- C. Sandbox
- D. Heuristic

310. Which antivirus and attack definition update options are supported by FortiGate units? (Choose two.)

- A. Manual update by downloading the signatures from the support site.
- B. FortiGuard pull updates.
- C. Push updates from the FortiGuard Distribution Network.
- D. execute fortiguard-AV-AS command from the CLI.

311. Which statement is correct regarding virus scanning on a FortiGate unit?

- A. Virus scanning is enabled by default.
- B. Fortinet customer support enables virus scanning remotely for you.
- C. Virus scanning must be enabled in a security profile, which must be applied to a firewall policy.
- D. Enabling virus scanning in a security profile enables virus protection for all traffic flowing through the FortiGate.

312. Which antivirus inspection mode must be used to scan SMTP, FTP, POP3 and SMB protocols?

- A. Proxy-based
- B. DNS-based
- C. Flow-based
- D. Man-in-the-middle.

313. Which of the following options can you use to update the virus definitions on a FortiGate unit? (Select all that apply.)

- A. Push update.
- B. Scheduled update
- C. Manual update
- D. FTP update

314. Which of the following antivirus and attack definition update features are supported by FortiGate units? (Select all that apply.)

- A. Manual, user-initiated updates from the FortiGuard Distribution Network.
- B. Hourly, daily, or weekly scheduled antivirus and attack definition and antivirus engine updates from the FortiGuard Distribution Network.
- C. Push updates from the FortiGuard Distribution Network.
- D. Update status including version numbers, expiry dates, and most recent update dates and times.

315. A FortiGate unit can scan for viruses on which types of network traffic? (Select all that apply.)

- A. POP3
- B. FTP
- C. SMTP
- D. SNMP
- E. NetBios

316. Which are the three different types of Conserve Mode that can occur on a FortiGate device? (Choose three.)

- A. Proxy
- B. Operating system
- C. Kernel
- D. System
- E. Device

317. A FortiGate device is configure to perform an AV & IPS scheduled update every hour.

```
Virus Definitions
-----
Version: 21.00487
Contract Expiry Date: Tue Apr 29 00:00:00 2014
Last Updated using scheduled update on Mon Jan
20 01:05:33 2014
Last Update Attempt: Mon Jan 20 10:08:56 2014
Result: Updates Installed
```

```
FG100D3G12800939 # exe time
current time is: 10:35:35
last ntp sync: Mon Jan 20 09:51:59 2014
```

Given the information in the exhibit, when will the next update happen?

- A. 01:00
- B. 02:05
- C. 11:00
- D. 11:08

318. An administrator has enabled proxy-based antivirus scanning and configured the following settings: Which statement about the above configuration is true?

```
config firewall profile-protocol-options
edit default
config http
set oversize-limit 10
set options oversize
end
end
```

- A. Files bigger than 10 MB are not scanned for viruses and will be blocked.
- B. FortiGate scans only the first 10 MB of any file.
- C. Files bigger than 10 MB are sent to the heuristics engine for scanning.
- D. FortiGate scans the files in chunks of 10 MB

319. What is longest length of time allowed on a FortiGate device for the virus scan to complete?

- A. 20 seconds
- B. 30 seconds
- C. 45 seconds
- D. 10 seconds

320. Files that are larger than the oversized limit are subjected to which Antivirus check?

- A. Grayware
- B. Virus
- C. Sandbox
- D. Heuristic

321. Which type of conserve mode writes a log message immediately, rather than when the device exits conserve mode?

- A. Kernel
- B. Proxy
- C. System
- D. Device

322. What is the maximum number of different virus databases a FortiGate can have?

- A. 5
- B. 2
- C. 3
- D. 4

323. Which of the following statements are true about the SSL Proxy certificate that must be used for SSL Content Inspection? (Choose two.)

- A. It cannot be signed by a private CA
- B. It must have either the field "CA=True" or the field "Key Usage=KeyCertSign"
- C. It must be installed in the FortiGate device
- D. The subject field must contain either the FQDN, or the IP address of the FortiGate device

324. Which of the following statements about conserve mode are true? (Choose two.)

- A. FortiGate stops sending files to FortiSandbox for inspection.
- B. FortiGate stops doing RPF checks over incoming packets.
- C. Administrators cannot change the configuration.
- D. Administrators can access the FortiGate only through the console port

325. Examine this FortiGate configuration:

```
config system global
```

```
    set av-failopen pass
```

```
end
```

Examine the output of the following debug command:

```
# diagnose hardware sysinfo conserve
```

```
memory conserve mode: on
```

```
total RAM: 3040 MB
```

```
memory used: 2948 MB 97% of total RAM
```

```
memory freeable: 92 MB 3% of total RAM
```

```
memory used + freeable threshold extreme: 2887 MB 95% of total RAM
```

```
memory used threshold red: 2675 MB 88% of total RAM
```

```
memory used threshold green: 2492 MB 82% of total RAM
```

Based on the diagnostic outputs above, how is the FortiGate handling the traffic for new sessions that require inspection?

- A. It is allowed, but with no inspection
- B. It is allowed and inspected as long as the inspection is flow based
- C. It is dropped.
- D. It is allowed and inspected, as long as the only inspection required is antivirus.

326. Which of the following email spam filtering features is NOT supported on a FortiGate unit?

- A. Multipurpose Internet Mail Extensions (MIME) Header Check
- B. HELO DNS Lookup
- C. Greylisting
- D. Banned Word

327. When a user attempts to connect to an HTTPS site, what is the expected result with this configuration?

View the exhibit.



- A. The user is required to authenticate before accessing sites with untrusted SSL certificates.
- B. The user is presented with certificate warnings when connecting to sites that have untrusted SSL certificates.
- C. The user is allowed access all sites with untrusted SSL certificates, without certificate warnings.
- D. The user is blocked from connecting to sites that have untrusted SSL certificates (no exception provided).

328. Which type of conserve mode writes a log message immediately, rather than when the device exits conserve mode?

- A. Kernel
- B. Proxy
- C. System
- D. Device

329. A firewall policy has been configured for the internal email server to receive email from external parties through SMTP. Exhibits A and B show the antivirus and email filter profiles applied to this policy. What is the correct behavior when the email attachment is detected as a virus by the FortiGate antivirus engine?

Exhibit A

Edit AntiVirus Profile

Name: AV-incoming

Comments: Write a comment... 0/255

Inspection Mode: ☒ Proxy ☐ Flow-based

☐ Inspect Suspicious Files with FortiGuard Analytics

Protocol	Virus Scan and Removal
Web	
HTTP	☐
Email	
SMTP	☒
POP3	☐
IMAP	☐
MAPI	☐
File Transfer	
FTP	☐
IM	
ICQ, Yahoo, MSN Messenger	☐

Apply

View Email Filter Profile

Name: EF-incoming

Comments: Write a comment... 0/255

Inspection Mode: ☒ Proxy ☐ Flow-based

☒ Enable Spam Detection and Filtering

	☒ IMAP	☒ POP3	☒ SMTP
Spam Action	Tagged	Tagged	Discard
Tag Location	Subject	Subject	Subject
Tag Format	Spam	Spam	Spam

FortiGuard Spam Filtering

Local Spam Filtering

Return

- A. The FortiGate unit will remove the infected file and deliver the email with a replacement message to alert the recipient that the original attachment was infected.
- B. The FortiGate unit will reject the infected email and the sender will receive a failed delivery message.
- C. The FortiGate unit will remove the infected file and add a replacement message
- D. Both sender and recipient are notified that the infected file has been removed.
- E. The FortiGate unit will reject the infected email and notify the sender.

QCM HA

330. Which statements about the firmware upgrade process on an active-active high availability (HA) cluster are true? (Choose two.)

- A. The firmware image must be manually uploaded to each FortiGate.
- B. Only secondary FortiGate devices are rebooted.
- C. Uninterruptable upgrade is enabled by default.
- D. Traffic load balancing is temporally disabled while upgrading the firmware

331. Which of the following sequences describes the correct order of criteria used for the selection of a master unit within a FortiGate high availability (HA) cluster when override is disabled?

- A. 1. port monitor, 2. unit priority, 3. up time, 4. serial number.
- B. 1. port monitor, 2. up time, 3. unit priority, 4. serial number.
- C. 1. unit priority, 2. up time, 3. port monitor, 4. serial number.
- D. 1. up time, 2. unit priority, 3. port monitor, 4. serial number.

332. Which of the following statements are correct about the HA command diagnose sys ha reset- uptime? (Choose two.)

- A. The device this command is executed on is likely to switch from master to slave status if override is disabled.
- B. The device this command is executed on is likely to switch from master to slave status if override is enabled.
- C. This command has no impact on the HA algorithm.
- D. This command resets the uptime variable used in the HA algorithm so it may cause a new master to become elected.

333. What are the requirements for a HA cluster to maintain TCP connections after device or link failover? (Choose two.)

- A. Enable session pick-up.
- B. Enable override.
- C. Connections must be UDP or ICMP.
- D. Connections must not be handled by a proxy.

334. In HA, the option Reserve Management Port for Cluster Member is selected as shown in the exhibit below. Which statements are correct regarding this setting? (Choose two.)



High Availability

Mode: Active-Passive

Device Priority: 200

☒ Reserve Management Port for Cluster Member: port7

- A. Interface settings on port7 will not be synchronized with other cluster members.
- B. The IP address assigned to this interface must not overlap with the IP address subnet assigned to another interface.
- C. When connecting to port7 you always connect to the master device.
- D. A gateway address may be configured for port7.

- 335. The exhibit shows the Disconnect Cluster Member command in a FortiGate unit that is part of a HA cluster with two HA members. What is the effect of the Disconnect Cluster Member command as given in the exhibit. (Choose two.)**



The screenshot shows a dialog box titled "Disconnect Cluster Member". It contains the following fields:

- Serial Number: FGVMD10000006268
- Interface: port3 (selected from a dropdown menu)
- IP/Netmask: 10.0.1.251/24

At the bottom right, there are two buttons: "OK" and "Cancel".

- A. Port3 is configured with an IP address for management access.
- B. The firewall rules are purged on the disconnected unit.
- C. The HA mode changes to standalone.
- D. The system hostname is set to the unit serial number.

- 336. An administrator has formed a high availability cluster involving two FortiGate units. [Multiple upstream Layer 2 switches] -- [FortiGate HA Cluster] -- [Multiple downstream Layer 2 switches]**

The administrator wishes to ensure that a single link failure will have minimal impact upon the overall throughput of traffic through this cluster.

Which of the following options describes the best step the administrator can take? The administrator should_____.

- A. Increase the number of FortiGate units in the cluster and configure HA in active-active mode.
- B. Enable monitoring of all active interfaces.
- C. Set up a full-mesh design which uses redundant interfaces.
- D. Configure the HA ping server feature to allow for HA failover in the event that a path is disrupted

- 337. In a high availability cluster operating in active-active mode, which of the following correctly describes the path taken by the SYN packet of an HTTP session that is offloaded to a slave unit?**

- A. Request: internal host; slave FortiGate; master FortiGate; Internet; web server.
- B. Request: internal host; slave FortiGate; Internet; web server.
- C. Request: internal host; slave FortiGate; master FortiGate; Internet; web server.
- D. Request: internal host; master FortiGate; slave FortiGate; Internet; web server.

338. Two devices are in an HA cluster, the device hostnames are STUDENT and REMOTE. Exhibit A shows the command output of diagnose sys session stat for the STUDENT device. Exhibit B shows the command output of diagnose sys session stat for the REMOTE device. Exhibit A and B:

```
STUDENT # diagnose sys session stat
Misc info:      session_count=166 setup_rate=68 exp_count=0 clash=0
                memory_tension_drop=0 ephemeral=0/57344 removeable=0  ha_scan=0
delete=0, flush=0, dev_down=0/0
TCP sessions:
    8 in ESTABLISHED state
    3 in SYN_SENT state
    1 in FIN_WAIT state
   139 in TIME_WAIT state
firewall error stat:
error1=00000000
error2=00000000
error3=00000000
error4=00000000
tt=00000000
cont=00000000
ids_recv=00000000
url_recv=00000000
av_recv=00000000
fqdn_count=00000000
tcp reset stat:
    syncqf=0 acceptqf=0 no-listener=2 data=0 ses=0 ips=0
global: ses_limit=0 ses6_limit=0 rt_limit=0 rt6_limit=0

STUDENT # _
```

```
global: ses_limit=0 ses6_limit=0 rt_limit=0 rt6_limit=0

REMOTE # diagnose sys session stat
Misc info:      session_count=11 setup_rate=0 exp_count=0 clash=4
                memory_tension_drop=0 ephemeral=0/57344 removeable=0  ha_scan=0
delete=0, flush=0, dev_down=0/0
TCP sessions:
    2 in ESTABLISHED state
    1 in SYN_SENT state
firewall error stat:
error1=00000000
error2=00000000
error3=00000000
error4=00000000
tt=00000000
cont=00000000
ids_recv=00000000
url_recv=00000000
av_recv=00000000
fqdn_count=00000000
tcp reset stat:
    syncqf=0 acceptqf=0 no-listener=7 data=0 ses=0 ips=0
global: ses_limit=0 ses6_limit=0 rt_limit=0 rt6_limit=0

REMOTE # _
```

Given the information provided in the exhibits, which of the following statements are correct? (Choose two.)

- A. STUDENT is likely to be the master device.
- B. Session-pickup is likely to be enabled.
- C. The cluster mode is active-passive.
- D. There is not enough information to determine the cluster mode.

339. Which of the following statements correctly describes how a FortiGate unit functions in Transparent mode?

- A. To manage the FortiGate unit, one of the interfaces must be designated as the management interface. This interface may not be used for forwarding data.
- B. An IP address is used to manage the FortiGate unit but this IP address is not associated with a specific interface.
- C. The FortiGate unit must use public IP addresses on the internal and external networks.
- D. The FortiGate unit uses private IP addresses on the internal network but hides them using address translation.

340. What are the requirements for a cluster to maintain TCP connections after device or link failover? (Select all that apply.)

- A. Enable session pick-up.
- B. Only applies to connections handled by a proxy.
- C. Only applies to UDP and ICMP connections.
- D. Connections must not be handled by a proxy.

341. Which of the following statement correct describes the use of the "diagnose sys ha reset-uptime" command?

- A. To force an HA failover when the HA override setting is disabled.
- B. To force an HA failover when the HA override setting is enabled.
- C. To clear the HA counters.
- D. To restart a FortiGate unit that is part of an HA cluster.

342. What are required to be the same for two FortiGate units to form an HA cluster? (Choose two)

- A. Firmware.
- B. Model.
- C. Hostname.
- D. System time zone.

343. Which of the following statements describes the objectives of the gratuitous ARP packets sent by an HA cluster?

- A. To synchronize the ARP tables in all the FortiGate Units that are part of the HA cluster.
- B. To notify the network switches that a new HA master unit has been elected.
- C. To notify the master unit that the slave devices are still up and alive.
- D. To notify the master unit about the physical MAC addresses of the slave units.

344. Which of the following statements are correct regarding a master HA unit? (Choose two)

- A. There should be only one master unit in each HA virtual cluster.
- B. The Master synchronizes cluster configuration with slaves.
- C. Only the master has a reserved management HA interface.
- D. Heartbeat interfaces are not required on a master unit.

345. Which statement describes how traffic flows in sessions handled by a slave unit in an active-active HA cluster?

- A. Packets are sent directly to the slave unit using the slave physical MAC address.
- B. Packets are sent directly to the slave unit using the HA virtual MAC address.
- C. Packets arrived at both units simultaneously, but only the slave unit forwards the session.
- D. Packets are first sent to the master unit, which then forwards the packets to the slave unit.

346. Two FortiGate devices fail to form an HA cluster, the device hostnames are STUDENT and REMOTE. Exhibit A shows the command output of show system ha for the STUDENT device. Exhibit B shows the command output of show system ha for the REMOTE device. Exhibit A:

```
Max number of virtual domains: 10
Virtual domains status: 1 in NAT mode, 0 in TP mode
Virtual domain configuration: disable
FIPS-CC mode: disable
Current HA mode: a-p, master
Branch point: 128
Release Version Information: GA
System time: Thu Jan 24 08:34:19 2013

STUDENT #
STUDENT #
STUDENT #
STUDENT #
STUDENT #
STUDENT # show system ha
config system ha
    set mode a-p
    set password ENC 9FHCYw0JXK9z8w6QkUnUsREWBruUcMJ5NUVE3oU5otyn+4dsq4CnU1GRJB
McEECPiT3Z/3dCmIuYIDgW2sE+1A1kHfAD0U/r5DkaqGnbj15XU/a
    set hbdev "port2" 50
    set override disable
    set priority 200
end

STUDENT # _
```

Exhibit B

```
Log hard disk: Available
Hostname: REMOTE
Operation Mode: NAT
Current virtual domain: root
Max number of virtual domains: 10
Virtual domains status: 1 in NAT mode, 0 in TP mode
Virtual domain configuration: disable
FIPS-CC mode: disable
Current HA mode: a-a, master
Branch point: 128
Release Version Information: GA
System time: Thu Jan 24 08:41:46 2013

REMOTE # show system ha
config system ha
    set mode a-a
    set password ENC 9FHCYw0JXK9z8w6QkUnUsREWBruUcMJ5NUVE3oU5otyn+4ds7YGv12Cir+8
B6Mf/rGXh0u51ygP+yPg15SDnSMEz4J1Nv4E09sk100mBQbcgXhSE
    set hbdev "port2" 50
    set session-pickup enable
    set override disable
    set priority 100
end

REMOTE # _
```

Which one of the following is the most likely reason that the cluster fails to form?

- A. Password
- B. HA mode
- C. Heartbeat
- D. Override

347. Which of the following statements are correct concerning the FortiGate session life support protocol? (Choose two)

- A. By default, UDP sessions are not synchronized.
- B. Up to four FortiGate devices in standalone mode are supported.
- C. only the master unit handles the traffic.
- D. Allows per-VDOM session synchronization.

348. What is the default criteria for selecting the HA master unit in a HA cluster?

- A. port monitor, priority, uptime, serial number
- B. Port monitor, uptime, priority, serial number
- C. Priority, uptime, port monitor, serial number
- D. uptime, priority, port monitor, serial number

349. What information is synchronized between two FortiGate units that belong to the same HA cluster? (Choose three)

- A. IP addresses assigned to DHCP enabled interface.
- B. The master devices hostname.
- C. Routing configured and state.
- D. Reserved HA management interface IP configuration.
- E. Firewall policies and objects.

350. View the exhibit. Based on this output, which statements are correct? (Choose two.)

```
Local-FortiGate # diagnose sys ha checksum cluster

===== FGVM010000058290 =====

is_manage_master()=1, is_root_master()=1
debugzone
global: 85 26 52 f2 f9 6e 3c c9 f5 21 1a 78 69 b6 20 bd
root: 30 51 63 1b 2d ef 77 aa f7 50 00 25 4d 42 a9 7d
all: 38 28 3d e4 24 8f 5b 10 8a 64 30 f2 34 13 c1 35

checksum
global: 85 26 52 f2 f9 6e 3c c9 f5 21 1a 78 69 b6 20 bd
root: 30 51 63 1b 2d ef 77 aa f7 50 00 25 4d 42 a9 7d
all: 38 28 3d e4 24 8f 5b 10 8a 64 30 f2 34 13 c1 35

===== FGVM010000058289 =====

is_manage_master()=0, is_root_master()=0
debugzone
global: 85 26 52 f2 f9 6e 3c c9 f5 21 1a 78 69 b6 20 bd
root: 30 51 63 1b 2d ef 77 aa f7 50 00 25 4d 8a 55 8b
all: 38 28 3d e4 24 8f 5b 10 8a 64 30 f2 34 dc 9a 43

checksum
global: 85 26 52 f2 f9 6e 3c c9 f5 21 1a 78 69 b6 20 bd
root: 30 51 63 1b 2d ef 77 aa f7 50 00 25 4d 8a 55 8b
all: 38 28 3d e4 24 8f 5b 10 8a 64 30 f2 34 dc 9a 43
```

- A. The `all` VDOM is not synchronized between the primary and secondary FortiGate devices.
- B. The `root` VDOM is not synchronized between the primary and secondary FortiGate devices.
- C. The `global` configuration is synchronized between the primary and secondary FortiGate devices.
- D. The FortiGate devices have three VDOMs.

351. Which statements are correct, based on this output? (Choose two.)

```

Local-FortiGate # diagnose sys ha checksum cluster

===== FGVM010000058290 =====

is_manage_master () =1, is_root_master () =1
debugzone
global: 85 26 52 f2 f9 6e 3c c9 f5 21 1a 78 69 b6 20 bd
root: 30 51 63 1b 2d ef 77 aa f7 50 00 25 4d 42 a9 7d
all: 38 28 3d e4 24 8f 5b 10 8a 64 30 f2 34 13 c1 35

checksum
global: 85 26 52 f2 f9 6e 3c c9 f5 21 1a 78 69 b6 20 bd
root: 30 51 63 1b 2d ef 77 aa f7 50 00 25 4d 42 a9 7d
all: 38 28 3d e4 24 8f 5b 10 8a 64 30 f2 34 13 c1 35

===== FGVM010000058289 =====

is_manage_master ()=0, is_root_master ()=0
debugzone
global: 85 26 52 f2 f9 6e 3c c9 f5 21 1a 78 69 b6 20 bd
root: 30 51 63 1b 2d ef 77 aa f7 50 00 25 4d 8a 55 8b
all: 38 28 3d e4 24 8f 5b 10 8a 64 30 f2 34 dc 9a 43

```

- A. The FortiGate have three VDOMs.
- B. The all VDOM is not synchronized between the primary and secondary FortiGate.
- C. The global configuration is synchronized between the primary and secondary FortiGate.
- D. The root VDOM is not synchronized between the primary and secondary FortiGate

QCM NSE4 : Proxy

352. Review the exhibit of an explicit proxy policy configuration.

Create New Edit Delete Expand All Collapse All Search							
Seq.#	To	Source	Destination	Users	Schedule	Action	AV
web (1 - 2)							
1	port1	10.0.1.0/24	all			ACCEPT	
1.1				Student	always		
2	port1	10.0.0.0/8	all		always	ACCEPT	

If there is a proxy connection attempt coming from the IP address 10.0.1.5, and from a user that has not authenticated yet, what action does the FortiGate proxy take?

- A. User is prompted to authenticate. Traffic from the user Student will be allowed by the policy #1. Traffic from any other user will be allowed by the policy #2.
- B. User is not prompted to authenticate. The connection is allowed by the proxy policy #2.
- C. User is not prompted to authenticate. The connection will be allowed by the proxy policy #1.
- D. User is prompted to authenticate. Only traffic from the user Student will be allowed. Traffic from any other user will be blocked.

353. What is valid reason for using session based authentication instead of IP based authentication in a FortiGate web proxy solution?

- A. Users are required to manually enter their credentials each time they connect to a different web site.
- B. Proxy users are authenticated via FSSO.
- C. There are multiple users sharing the same IP address.
- D. Proxy users are authenticated via RADIUS.

354. Examine the following FortiGate web proxy configuration; then answer the question below:

```
config web-proxy explicit
set pac-file-server-status enable
set pac-file-server-port 8080
set pac-file-name wpad.dat
end
```

Assuming that the FortiGate proxy IP address is 10.10.1.1, which URL must an Internet browser use to download the PAC file?

- A. https://10.10.1.1:8080
- B. https://10.10.1.1:8080/wpad.dat
- C. http://10.10.1.1:8080/
- D. http://10.10.1.1:8080/wpad.dat

355. Which statements are true regarding the use of a PAC file to configure the web proxy settings in an Internet browser? (Choose two.)

- A. Only one proxy is supported.
- B. Can be manually imported to the browser.
- C. The browser can automatically download it from a web server.
- D. Can include a list of destination IP subnets where the browser can connect directly to without using a proxy.

356. Which two methods are supported by the web proxy auto-discovery protocol (WPAD) to automatically learn the URL where a PAC file is located? (Choose two.)

- A. DHCP
- B. BOOTP
- C. DNS
- D. IPv6 auto configuration

357. What is a valid reason for using session based authentication instead of IP based authentication in a FortiGate web proxy solution?

- A. Users are required to manually enter their credentials each time they connect to a different web site.
- B. Proxy users are authenticated via FSSO.
- C. There are multiple users sharing the same IP address.
- D. Proxy users are authenticated via RADIUS

358. An Internet browser is using the WPAD DNS method to discover the PAC file's URL. The DNS server replies to the browser's request with the IP address 10.100.1.10. Which URL will the browser use to download the PAC file?

- A. http://10.100.1.10/proxy.pac
- B. https://10.100.1.10/
- C. http://10.100.1.10/wpad.dat
- D. https://10.100.1.10/proxy.pac

359. Which protocol can an Internet browser use to download the PAC file with the web proxy configuration?

- A. HTTPS
- B. FTP
- C. TFTP
- D. HTTP

360. Which of the following must be configured on a FortiGate unit to redirect content requests to remote web cache servers?

- A. WCCP must be enabled on the interface facing the Web cache.
- B. You must enabled explicit Web-proxy on the incoming interface.
- C. WCCP must be enabled as a global setting on the FortiGate unit.
- D. WCCP must be enabled on all interfaces on the FortiGate unit through which HTTP traffic is passing.

361. When using WPAD DNS method, what is the FQDN format that browsers use to query the DNS server?

- A. wpad.<local-domain>
- B. srv_tcp.wpad.<local-domain>
- C. srv_proxy.<local-domain>/wpad.dat
- D. proxy.<local-domain>.wpad

362. Which statements about IP-based explicit proxy authentication are true? (Choose two.)

- A. IP-based authentication is best suited to authenticating users behind a NAT device.
- B. Sessions from the same source address are treated as a single user.
- C. IP-based authentication consumes less FortiGate's memory than session-based authentication.
- D. FortiGate remembers authenticated sessions using browser cookies.

363. Which of the following statements is true regarding the TCP SYN packets that go from a client, through an implicit web proxy (transparent proxy), to a web server listening at TCP port 80? (Choose three.)

- A. The source IP address matches the client IP address.
- B. The source IP address matches the proxy IP address.
- C. The destination IP address matches the proxy IP address.
- D. The destination IP address matches the server IP addresses.
- E. The destination TCP port number is 80.

364. Which of the following statements is true regarding the use of a PAC file to configure the web proxy settings in an Internet browser? (Choose two.)

- A. More than one proxy is supported.
- B. Can contain a list of destinations that will be exempt from the use of any proxy.
- C. Can contain a list of URLs that will be exempted from the FortiGate web filtering inspection.
- D. Can contain a list of users that will be exempted from the use of any proxy.

365. Which of the following are benefits of using web caching? (Choose three.)

- A. Decrease bandwidth utilization
- B. Reduce server load
- C. Reduce FortiGate CPU usage
- D. Reduce FortiGate memory usage
- E. Decrease traffic delay

366. An administrator wants to block HTTP uploads. Examine the exhibit, which contains the proxy address created for that purpose

Edit Address

Category	Address Proxy Address
Name	Training
Color	Change
Type	HTTP Method ▼
Host	all ▼
Request Method	GET HEAD OPTIONS POST ^ ≡ v
Show in Address List	☒
Comments	0/255

Tags

Add Tag Category

OK
Cancel

Where must the proxy address be used?

- A. As the source in a firewall policy.
- B. As the source in a proxy policy.
- C. As the destination in a firewall policy.
- D. As the destination in a proxy policy.

367. Which of the following statements are true when using WPAD with the DHCP discovery method? (Choose two.)

- A. If the DHCP method fails, browsers will try the DNS method.
- B. The browser needs to be preconfigured with the DHCP server's IP address.
- C. The browser sends a DHCPINFORM request to the DHCP server.
- D. The DHCP server provides the PAC file for download.

368. Examine this PAC file configuration.

```
function FindProxyForURL (url, host) {  
  if (shExpMatch (url, "*.fortinet.com/*")) {  
    return "DIRECT";}  
  if (isInNet (host, "172.25.120.0", "255.255.255.0")) {  
    return "PROXY altproxy.corp.com: 8060";}  
  return "PROXY proxy.corp.com: 8090";  
}
```

Which of the following statements are true? (Choose two.)

- A. Browsers can be configured to retrieve this PAC file from the FortiGate.
- B. Any web request to the 172.25.120.0/24 subnet is allowed to bypass the proxy.
- C. All requests not made to Fortinet.com or the 172.25.120.0/24 subnet, have to go through altproxy.corp.com: 8060.
- D. Any web request fortinet.com is allowed to bypass the proxy.

369. In FortiOS session table output, what is the correct 'proto_state' number for an established, non-proxied TCP connection?

- A. 00
- B. 11
- C. 01
- D. 05

QCM NSE4 : Dialup VPN

370. Which of the following statements are correct concerning IPsec dialup VPN configurations for FortiGate devices? (Choose two)

- A. Main mode must be used when there is no more than one IPsec dialup VPN configured on the same FortiGate device.
- B. A FortiGate device with an IPsec VPN configured as dialup can initiate the tunnel connection to any remote IP address.
- C. Peer ID must be used when there is more than one aggressive-mode IPsec dialup VPN on the same FortiGate device.
- D. The FortiGate will automatically add a static route to the source quick mode selector address received from each remote peer.

371. You are tasked to architect a new IPsec deployment with the following criteria:

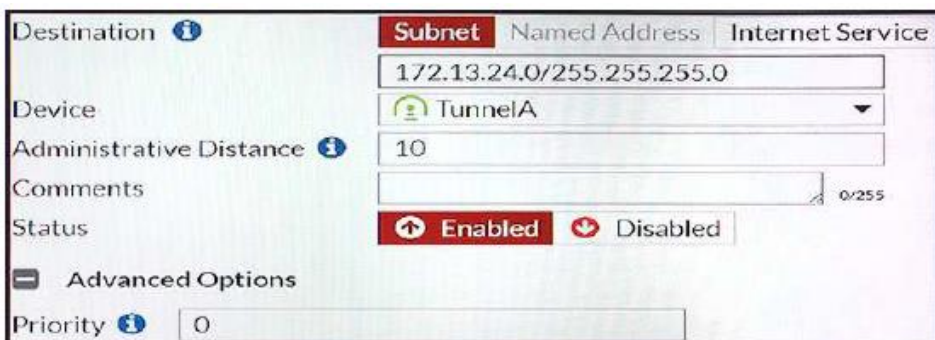
- There are two HQ sites that all satellite offices must connect to.
 - The satellite offices do not need to communicate directly with other satellite offices.
 - No dynamic routing will be used.
 - The design should minimize the number of tunnels being configured.
- Which topology should be used to satisfy all of the requirements?

- A. Redundant
- B. Hub-and-spoke
- C. Partial mesh
- D. Fully meshed

372. Which of the following statements are correct? (Choose two.)



The screenshot shows the configuration for TunnelB. The Destination tab is active, showing the Subnet 172.13.24.0/255.255.255.0. The Device is set to TunnelB, and the Administrative Distance is 5. The Status is Enabled. The Priority is 30.



The screenshot shows the configuration for TunnelA. The Destination tab is active, showing the Subnet 172.13.24.0/255.255.255.0. The Device is set to TunnelA, and the Administrative Distance is 10. The Status is Enabled. The Priority is 0.

- A. This is a redundant IPsec setup.
- B. The TunnelB route is the primary one for searching the remote site. The TunnelA route is used only if the TunnelB VPN is down.
- C. This setup requires at least two firewall policies with action set to IPsec.
- D. Dead peer detection must be disabled to support this type of IPsec setup.

373. Which statements best describe auto discovery VPN (ADVPN). (Choose two.)

- A. It requires the use of dynamic routing protocols so that spokes can learn the routes to other spokes.
- B. ADVPN is only supported with IKEv2.
- C. Tunnels are negotiated dynamically between spokes.
- D. Every spoke requires a static tunnel to be configured to other spokes so that phase 1 and phase 2 proposals are defined in advance.

374. What advantages are there in using a hub-and-spoke IPSec VPN configuration instead of a fully-meshed set of IPSec tunnels? (Select all that apply.)

- A. Using a hub and spoke topology is required to achieve full redundancy.
- B. Using a hub and spoke topology simplifies configuration because fewer tunnels are required.
- C. Using a hub and spoke topology provides stronger encryption.
- D. The routing at a spoke is simpler, compared to a meshed node.

375. Which statements are correct properties of a partial mesh VPN deployment. (Choose two.)

- A. VPN tunnels interconnect between every single location.
- B. VPN tunnels are not configured between every single location.
- C. Some locations are reached via a hub location.
- D. There are no hub locations in a partial mesh.

376. Examine the following spanning tree configuration on a FortiGate in transparent mode: Which statement is correct for the above configuration?

```
config system interface
edit <interface name>
set stp-forward enable
end
```

- A. The FortiGate participates in spanning tree.
- B. The FortiGate device forwards received spanning tree messages.
- C. Ethernet layer-2 loops are likely to occur.
- D. The FortiGate generates spanning tree BPDU frames.

377. What advantages are there in using a fully Meshed IPSec VPN configuration instead of a hub and spoke set of IPSec tunnels?

- A. Using a hub and spoke topology is required to achieve full redundancy.
- B. Using a full mesh topology simplifies configuration.
- C. Using a full mesh topology provides stronger encryption.
- D. Full mesh topology is the most fault-tolerant configuration.

378. Which of the following statements are correct concerning IKE mode config? (Choose two)

- A. It can dynamically assign IP addresses to IPsec VPN clients.
- B. It can dynamically assign DNS settings to IPsec VPN clients.
- C. It uses the ESP protocol.
- D. It can be enabled in the phase 2 configuration.

379. What is required in a FortiGate configuration to have more than one dialup IPsec VPN using aggressive mode?

- A. All the aggressive mode dialup VPNs MUST accept connections from the same peer ID.
- B. Each peer ID MUST match the FQDN of each remote peer.
- C. Each aggressive mode dialup MUST accept connections from different peer ID.
- D. The peer ID setting must NOT be used.

380. A network administrator needs to implement dynamic route redundancy between a FortiGate unit located in a remote office and a FortiGate unit located in the central office.

The remote office accesses central resources using IPSec VPN tunnels through two different Internet providers.

What is the best method for allowing the remote office access to the resources through the FortiGate unit used at the central office?

- A. Use two or more route-based IPSec VPN tunnels and enable OSPF on the IPSec virtual interfaces.
- B. Use two or more policy-based IPSec VPN tunnels and enable OSPF on the IPSec virtual interfaces.
- C. Use route-based VPNs on the central office FortiGate unit to advertise routes with a dynamic routing protocol and use a policy-based VPN on the remote office with two or more static default routes.
- D. Dynamic routing protocols cannot be used over IPSec VPN tunnels.

381. Which of the following statements are correct concerning IPsec dialup VPN configurations for FortiGate devices? (Choose two)

- A. Main mode must be used when there is no more than one IPsec dialup VPN configured on the same FortiGate device.
- B. A FortiGate device with an IPsec VPN configured as dialup can initiate the tunnel connection to any remote IP address.
- C. Peer ID must be used when there is more than one aggressive-mode IPsec dialup VPN on the same FortiGate device.
- D. The FortiGate will automatically add a static route to the source quick mode selector address received from each remote peer.

QCM_NSE4_Transparent

382. Which statements correctly describe transparent mode operation? (Choose three.)

- A. The FortiGate acts as transparent bridge and forwards traffic at Layer-2.
- B. Ethernet packets are forwarded based on destination MAC addresses, NOT IP addresses.
- C. The transparent FortiGate is clearly visible to network hosts in an IP trace route.
- D. Permits inline traffic inspection and firewalling without changing the IP scheme of the network.
- E. All interfaces of the transparent mode FortiGate device must be on different IP subnets.

383. Examine the following spanning tree configuration on a FortiGate in transparent mode:

```
config system interface
edit <interface name>
set stp-forward enable
end
```

Which statement is correct for the above configuration?

- A. The FortiGate participates in spanning tree.
- B. The FortiGate device forwards received spanning tree messages.
- C. Ethernet layer-2 loops are likely to occur.
- D. The FortiGate generates spanning tree BPDU frames.

384. Which of the following statements are correct differences between NAT/route and transparent mode? (Choose two.)

- A. In transparent mode, interfaces do not have IP addresses.
- B. Firewall policies are only used in NAT/ route mode.
- C. Static routers are only used in NAT/route mode.
- D. Only transparent mode permits inline traffic inspection at layer 2.

385. Which of the following statements is true regarding a FortiGate device operating in transparent mode? (Choose three.)

- A. It acts as a layer 2 bridge
- B. It acts as a layer 3 router
- C. It forwards frames using the destination MAC address.
- D. It forwards packets using the destination IP address.
- E. It can perform content inspection (antivirus, web filtering, etc)

386. Which of the following statements are correct concerning layer 2 broadcast domains in transparent mode VDOMs?(Choose two)

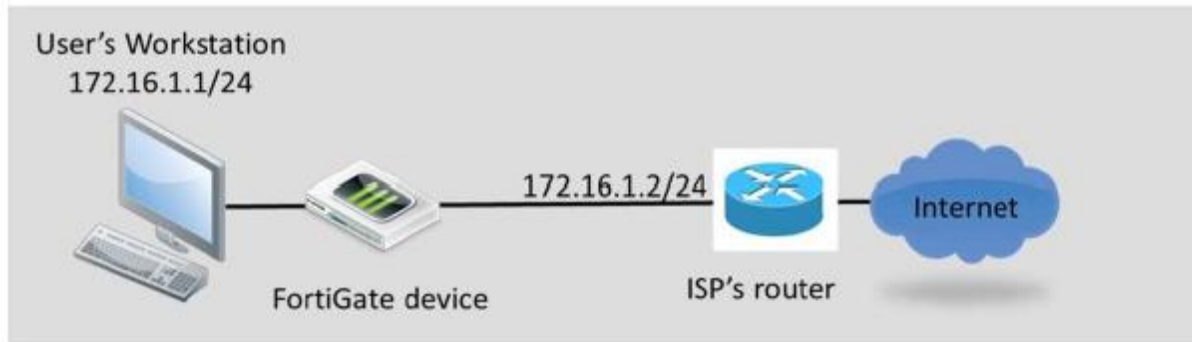
- A. The whole V
- B. VDOM is a single broadcast domain even when multiple VLAN are used.
- B. Each VLAN is a separate broadcast domain.
- C. Interfaces configured with the same VLAN ID can belong to different broadcast domains.
- D. All the interfaces in the same broadcast domain must use the same VLAN ID.

387. Which of the following statements is correct regarding FortiGate interfaces and spanning tree protocol? (Choose Two)

- A. Only FortiGate switch interfaces Participate in spanning tree.
- B. All FortiGate interfaces in transparent mode VDOMs participate in spanning tree.
- C. All FortiGate interfaces in NAT/route mode VDOMs Participate in spanning tree.
- D. All FortiGate interfaces in transparent mode VDOMs may block or forward BPDUs.

388. Review to the network topology in the exhibit.

The workstation, 172.16.1.1/24, connects to port2 of the FortiGate device, and the ISP router, 172.16.1.2, connects to port1. Without changing IP addressing, which configuration changes are required to properly forward users traffic to the Internet?(Choose two)



- A. At least one firewall policy from port2 to port1 to allow outgoing traffic.
- B. A default route configured in the FortiGuard devices pointing to the ISP's router.
- C. Static or dynamic IP addresses in both FortiGate interfaces port1 and port2.
- D. The FortiGate devices configured in transparent mode.

389. Which of the following statements correctly describes how a FortiGate unit functions in Transparent mode?

- A. To manage the FortiGate unit, one of the interfaces must be designated as the management interface. This interface may not be used for forwarding data.
- B. An IP address is used to manage the FortiGate unit but this IP address is not associated with a specific interface.
- C. The FortiGate unit must use public IP addresses on the internal and external networks.
- D. The FortiGate unit uses private IP addresses on the internal network but hides them using address translation.

390. Examine the following spanning tree configuration on a FortiGate in transparent mode:

```
config system interface
edit <interface name>
set stp-forward enable
end
```

Which statement is correct for the above configuration?

- A. The FortiGate participates in spanning tree.
- B. The FortiGate device forwards received spanning tree messages.
- C. Ethernet layer-2 loops are likely to occur.
- D. The FortiGate generates spanning tree BPDU frames.

391. Which statements correctly describe transparent mode operation? (Choose three.)

- A. The FortiGate acts as transparent bridge and forwards traffic at Layer-2.
- B. Ethernet packets are forwarded based on destination MAC addresses, NOT IP addresses.
- C. The transparent FortiGate is clearly visible to network hosts in an IP trace route.
- D. Permits inline traffic inspection and firewalling without changing the IP scheme of the network.
- E. All interfaces of the transparent mode FortiGate device must be on different IP subnets.

392. In transparent mode, forward-domain is an CLI setting associate with _____.

- A. static route
- B. a firewall policy
- C. an interface
- D. a virtual domain

393. Which of the following statements are TRUE for Port Pairing and Forwarding Domains? (Select all that apply.)

- A. They both create separate broadcast domains.
- B. Port Pairing works only for physical interfaces.
- C. Forwarding Domains only apply to virtual interfaces.
- D. They may contain physical and/or virtual interfaces.
- E. They are only available in high-end models.

394. In Transparent Mode, forward-domain is an attribute of _____.

- A. an interface
- B. a firewall policy
- C. a static route
- D. a virtual domain

QCM NSE4 IPS

395. What FortiGate feature can be used to block a ping sweep scan from an attacker?

- A. Web application firewall (WAF)
- B. Rate based IPS signatures
- C. One-arm sniffer
- D. DoS policies

396. Your Linux email server runs on a non-standard port number, port 2525. Which statement is true?

- A. IPS cannot scan that traffic for SMTP anomalies because of the non-standard port number. You must reconfigure the server to run on port 2.
- B. To apply IPS to traffic to that server, you must configured FortiGate SMTP proxy to listen on port 2525
- C. IPS will apply all SMTP signatures, regardless of whether they apply to clients or servers.
- D. Protocol decoders automatically detect SMTP and scan for matches with appropriate IPS signature.

397. Examine the following log message for IPS and identify the valid responses below. (Select all that apply.)

```
2012-07-01 09:54:28 oid=2 log_id=18433 type=ips subtype=anomaly
pri=alert vd=root severity="critical" src="192.168.3.168"
dst="192.168.3.170" src_int="port2" serial=0 status="detected" proto=1
service="icmp" count=1 attack_name="icmp_flood" icmp_id="0xa8a4"
icmp_type="0x08" icmp_code="0x00" attack_id=16777316 sensor="1"
ref="http://www.fortinet.com/ids/VID16777316" msg="anomaly:
icmp_flood,
51 > threshold 50"
```

- A. The target is 192.168.3.168.
- B. The target is 192.168.3.170.
- C. The attack was detected and blocked.
- D. The attack was detected only.
- E. The attack was TCP based.

398. Identify the statement which correctly describes the output of the following command:

```
diagnose ips anomaly list
```

- A. Lists the configured DoS policy.
- B. List the real-time counters for the configured DoS policy.
- C. Lists the errors captured when compiling the DoS policy.
- D. Lists the IPS signature matches.

399. By default, the Intrusion Protection System (IPS) on a FortiGate unit is set to perform which action?

- A. Block all network attacks.
- B. Block the most common network attacks.
- C. Allows all traffic
- D. Allow and log all traffic

400. Review the IPS sensor filter configuration shown in the exhibit ; Based on the information in the exhibit, which statements are correct regarding the filter? (Choose two.)

Pattern Based Signatures and Filters

Create New Edit Delete				
Severity	Target	OS	Action	Packet Logging
Critical	Server	Linux	Block	

- A. It does not log attacks targeting Linux servers.
- B. It matches all traffic to Linux servers.
- C. Its action will block traffic matching these signatures.
- D. It only takes effect when the sensor is applied to a policy

401. In which of the following report templates would you configure the charts to be included in the report?

- A. Layout Template
- B. Data Filter Template
- C. Output Template
- D. Schedule Template

402. An administrator is examining the attack logs and notices the following entry:

```
type=ips subtype=signature pri=alert vd=root serial=1995
attack_id=103022611 src=69.45.64.22 dst=192.168.1.100 src_port=80
dst_port=4887 src_int=wlan dst_int=internal sta-tus=detectedproto=6
service=4887/tcp user=N/A group=N/A msg=web_client:
IE.IFRAME.BufferOverflow.B
```

Based on the information displayed in this entry, which of the following statements are correct? (Select all that apply.)

- A. This is an HTTP server attack.
- B. The attack was detected and blocked by the FortiGate unit.
- C. The attack was against a FortiGate unit at the 192.168.1.100 IP address.
- D. The attack was detected and passed by the FortiGate unit

403. Review the CLI configuration below for an IPS sensor and identify the correct statements regarding this configuration from the choices below. (Select all that apply.)

```

config ips sensor
edit "LINUX_SERVER"
set comment ''
set replacemsg-group ''
set log enable
config entries
edit 1
set action default
set application all
set location server
set log enable
set log-packet enable s
set os Linux
set protocol all
set quarantine none
set severity all
set status default
next
end
next
end

```

- A. The sensor will log all server attacks for all operating systems.
- B. The sensor will include a PCAP file with a trace of the matching packets in the log message of any matched signature.
- C. The sensor will match all traffic from the address object `LINUX\_SERVER`.
- D. The sensor will reset all connections that match these signatures.
- E. The sensor only filters which IPS signatures to apply to the selected firewall policy

404. Which of the following report templates must be used when scheduling report generation?

- A. Layout Template
- B. Data Filter Template
- C. Output Template
- D. Chart Template

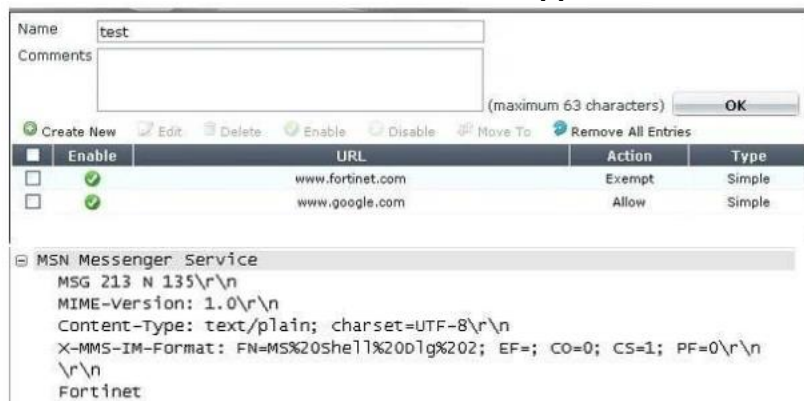
405. Which best describe the mechanism of a TCP SYN flood?

- A. The attacker keeps open many connections with slow data transmission so that other clients cannot start new connections.
- B. The attacker sends a packet designed to "sync" with the FortiGate.
- C. The attacker sends a specially crafted malformed packet, intended to crash the target by exploiting its parser.
- D. The attacker starts many connections, but never acknowledges to fully form them.

406. Acme Web Hosting is replacing one of their firewalls with a FortiGate. It must be able to apply port forwarding to their back-end web servers while blocking virus uploads and TCP SYN floods from attackers. Which operation mode is the best choice for these requirements?

- A. NAT/route
- B. NAT mode with an interface in one-arm sniffer mode
- C. Transparent mode
- D. No appropriate operation mode exists

407. Which of the following describes the best custom signature for detecting the use of the word "Fortinet" in chat applications?



- A. The sample packet trace illustrated in the exhibit provides details on the packet that requires detection. F-SBID(--protocol tcp; --flow from_client; --pattern "X-MMS-IM-Format"; --pattern "fortinet"; -- no_case;)
- B. F-SBID(--protocol tcp; --flow from_client; --pattern "fortinet"; --no_case;)
- C. F-SBID(--protocol tcp; --flow from_client; --pattern "X-MMS-IM-Format"; --pattern "fortinet"; -- within 20; --no_case;)
- D. F-SBID(--protocol tcp; --flow from_client; --pattern "X-MMS-IM-Format"; --pattern "fortinet"; -- within 20;)

408. Which statement is correct concerning creating a custom signature?

- A. It must start with the name
- B. It must indicate whether the traffic flow is from the client or the server.
- C. It must specify the protocol. Otherwise, it could accidentally match lower-layer protocols.
- D. It is not supported by Fortinet Technical Support.

409. Which operating system vulnerability can you protect when selecting signatures to include in an IPS sensor? (Choose three)

- A. Irix
- B. QNIX
- C. Linux
- D. Mac OS
- E. BSD

410. Which statement concerning IPS is false?

- A. IPS packages contain an engine and signatures used by both IPS and other flow-based scans.
- B. One-arm topology with sniffer mode improves performance of IPS blocking.
- C. IPS can detect zero-day attacks.
- D. The status of the last service update attempt from FortiGuard IPS is shown on System>Config>FortiGuard and in output from 'diag autoupdate version'

411. Which of the following statements is correct regarding the NAC Quarantine feature?

- A. With NAC quarantine, files can be quarantined not only as a result of antivirus scanning, but also for other forms of content inspection such as IPS and DLP.
- B. NAC quarantine does a client check on workstations before they are permitted to have administrative access to FortiGate.
- C. NAC quarantine allows administrators to isolate clients whose network activity poses a security risk.
- D. If you chose the quarantine action, you must decide whether the quarantine type is NAC quarantine or File quarantine

412. An organization wishes to protect its SIP Server from call flooding attacks. Which of the following configuration changes can be performed on the FortiGate unit to fulfill this requirement?

- A. Apply an application control list which contains a rule for SIP and has the "Limit INVITE Request" option configured.
- B. Enable Traffic Shaping for the appropriate SIP firewall policy.
- C. Reduce the session time-to-live value for the SIP protocol by running the configure system session-ttl CLI command.
- D. Run the set udp-idle-timer CLI command and set a lower time value.

413. On your FortiGate 60D, you've configured firewall policies. They port forward traffic to your Linux Apache web server. Select the best way to protect your web server by using the IPS engine.

- A. Enable IPS signatures for Linux servers with HTTP, TCP and SSL protocols and Apache applications. Configured DLP to block HTTP GET request with credit card numbers.
- B. Enable IPS signatures for Linux servers with HTTP, TCP and SSL protocols and Apache applications. Configure DLP to block HTTP GET with credit card numbers. Also configure a DoS policy to prevent TCP SYN floods and port scans.
- C. None. FortiGate 60D is a desktop model, which does not support IPS.
- D. Enable IPS signatures for Linux and windows servers with FTP, HTTP, TCP, and SSL protocols and Apache and PHP applications.

414. An administration wants to throttle the total volume of SMTP sessions to their email server. Which of the following DoS sensors can be used to achieve this?

- A. tcp_port_scan
- B. ip_dst_session
- C. udp_flood
- D. ip_src_session

415. Which changes to IPS will reduce resource usage and improve performance? (Choose three)

- A. In custom signature, remove unnecessary keywords to reduce how far into the signature tree that FortiGate must compare in order to determine whether the packet matches.
- B. In IPS sensors, disable signatures and rate based statistics (anomaly detection) for protocols, applications and traffic directions that are not relevant.
- C. In IPS filters, switch from 'Advanced' to 'Basic' to apply only the most essential signatures.
- D. In firewall policies where IPS is not needed, disable IPS.
- E. In firewall policies where IPS is used, enable session start logs.

416. Which profile could IPS engine use on an interface that is in sniffer mode? (Choose three)

- A. Antivirus (flow based)
- B. Web filtering (PROXY BASED)
- C. Intrusion Protection
- D. Application Control
- E. Endpoint control

417. You are creating a custom signature. Which has incorrect syntax?

- A. F-SBID(--attack_id 1842,--name "Ping.Death";--protocol icmp; --data_size>32000;)
- B. F-SBID(--name "Block.SMTP.VRFY.CMD";--pattern "vrfy";-- service SMTP; --no_case;--context header;)
- C. F-SBID(--name "Ping.Death";--protocol icmp;--data_size>32000;)
- D. F-SBID(--name "Block".HTTP.POST"; --protocol tcp;-- service HTTP;-- flow from_client; --pattern "POST"; -- context uri;--within 5,context;)

418. An administrator has created a custom IPS signature. Where does the custom IPS signature have to be applied?

- A. In an IPS sensor
- B. In an interface.
- C. In a DoS policy.
- D. In an application control profile.

419. Which one of the following processes is involved in updating IPS from FortiGuard?

- A. FortiGate IPS update requests are sent using UDP port 443.
- B. Protocol decoder update requests are sent to `service.fortiguard.net`.
- C. IPS signature update requests are sent to `update.fortiguard.net`.
- D. IPS engine updates can only be obtained using push updates.

420. Examine the IPS sensor configuration shown in the exhibit, and then answer the question below.

IPS Sensor

Name: [\[View IPS Signatures\]](#)

Comments: 0 / 255

IPS Signatures

+ Add Signatures
🗑️ Delete
✎ Edit IP Exemptions

Name	Exempt IPs	Severity	Target	Service	OS	Action	Packet Logging
No matching entries found							

IPS Filters

+ Add Filter
✎ Edit Filter
🗑️ Delete

Filter Details	Action	Packet Logging
Location: server OS: Windows	🚫 Block	❌

Apply

Forward Traffic Logs

🔄
📄
⚙️ Add Filter

#	Date/Time	Source	Destination	Application Name	Result	Policy
1	10:09:03	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
2	10:09:03	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
3	10:09:02	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
4	10:09:02	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
5	10:09:01	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
6	10:08:59	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
7	10:08:57	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
8	10:08:57	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
9	10:08:57	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)
10	10:08:57	10.200.1.254	10.200.1.200	🔒 HTTPS	✓ 1.30kB/2.65 kB	2(Web-Server-Access-IPS)

An administrator has configured the WINDOS_SERVERS IPS sensor in an attempt to determine whether the influx of HTTPS traffic is an attack attempt or not. After applying the IPS sensor, FortiGate is still not generating any IPS logs for the HTTPS traffic.

What is a possible reason for this?

- A. The IPS filter is missing the Protocol: HTTPS option.
- B. The HTTPS signatures have not been added to the sensor.
- C. A DoS policy should be used, instead of an IPS sensor.
- D. A DoS policy should be used, instead of an IPS sensor.
- E. The firewall policy is not using a full SSL inspection profile.

421. What types of traffic and attacks can be blocked by a web application firewall (WAF) profile? (Choose three.)

- A. Traffic to botnetservers
- B. Traffic to inappropriate web sites
- C. Server information disclosure attacks
- D. Credit card data leaks
- E. SQL injection attacks

422. Acme Web Hosting is replacing one of their firewalls with a FortiGate. It must be able to apply port forwarding to their back-end web servers while blocking virus uploads and TCP SYN floods from attackers. Which operation mode is the best choice for these requirements?

- A. NAT/route
- B. NAT mode with an interface in one-arm sniffer mode
- C. Transparent mode
- D. No appropriate operation mode exists

423. Which UTM feature sends a UDP query to FortiGuard servers each time FortiGate scans a packet (unless the response is locally cached)?

- A. Antivirus
- B. VPN
- C. IPS
- D. Web Filtering

QCM_NSE4_Fabric

424. You are configuring the root FortiGate to implement the security fabric. You are configuring port10 to communicate with a downstream FortiGate. View the default Edit Interface in the exhibit below:

Edit Interface

Interface Name: port10(00:0C:29:53:DE:D7)
Alias:
Link Status: Up
Type: Physical Interface

Tags
Role: Undefined

Address
Addressing mode: **Manual** | DHCP | One-Arm-Sniffer | Dedicated to FortiSwitch
IP/Network Mask:

Administrative Access
IPv4 ☐ ☐ HTTPS ☐ HTTP ☐ PING ☐ FMG-Access
☐ CAPWAP ☐ SSH ☐ SNMP ☐ FTM
☐ RADIUS Accounting ☐ FortiTelemetry

☐ DHCP Server

Networked Devices
Device Detection ☐

OK **Cancel**

When configuring the root FortiGate to communicate with a downstream FortiGate, which settings are required to be configured? (Choose two.)

- A. Device detection enabled.
- B. Administrative Access: FortiTelemetry.
- C. IP/Network Mask.
- D. Role: Security Fabric.

425. Which define device identification? (Choose two.)

- A. Device identification is enabled by default on all interfaces.
- B. Enabling a source device in a firewall policy enables device identification on the source interfaces of that policy.
- C. You cannot combine source user and source device in the same firewall policy.
- D. FortiClient can be used as an agent based device identification technique.
- E. Only agentless device identification techniques are supported.

QCM_NSE4_Hardware acceleration

426. Which statement best describes what the FortiGate hardware acceleration processors main task is?

- A. Offload traffic processing tasks from the main CPU.
- B. Offload management tasks from the main CPU.
- C. Compress and optimize the network traffic.
- D. Increase maximum bandwidth available in a FortiGate interface.

427. Which statement best describes the objective of the SYN proxy feature available in SP processors?

- A. Accelerate the TCP 3-way handshake
- B. Collect statistics regarding traffic sessions
- C. Analyze the SYN packet to decide if the new session can be offloaded to the SP processor
- D. Protect against SYN flood attacks.

428. For FortiGate devices equipped with Network Processor (NP) chips, which are true? (Choose three.)

- A. For each new IP session, the first packet always goes to the CPU.
- B. The kernel does not need to program the NPU. When the NPU sees the traffic, it determines by itself whether it can process the traffic
- C. Once offloaded, unless there are errors, the NP forwards all subsequent packets. The CPU does not process them.
- D. When the last packet is sent or received, such as a TCP FIN or TCP RST signal, the NP returns this session to the CPU for tear down.
- E. Sessions for policies that have a security profile enabled can be NP offloaded.

429. Two FortiGate units with NP6 processors form an active-active cluster. The cluster is doing security profile (UTM) inspection over all the user traffic.

What statements are true regarding the sessions that the master unit is offloading to the slave unit for inspection? (Choose two.)

- A. They are offloaded to the NP6 in the master unit.
- B. They are not offloaded to the NP6 in the master unit.
- C. They are offloaded to the NP6 in the slave unit.
- D. They are not offloaded to the NP6 in the slave unit.

430. Which of the following Fortinet hardware accelerators can be used to offload flow-based antivirus inspection? (Choose two.)

- A. SP3
- B. CP8
- C. NP4
- D. NP6

431. Which traffic inspection features can be executed by a security processor (SP)? (Choose three.)

- A. TCP SYN proxy
- B. SIP session helper
- C. Proxy-based antivirus
- D. Attack signature matching
- E. Flow-based web filtering

432. An administrator is using the FortiGate built-in sniffer to capture HTTP traffic between a client and a server, however, the sniffer output shows only the packets related with TCP session setups and disconnections. Why?

- A. The administrator is running the sniffer on the internal interface only.
- B. The filter used in the sniffer matches the traffic only in one direction.
- C. The FortiGate is doing content inspection.
- D. TCP traffic is being offloaded to an NP6.

433. Which of the following statements best describe the main requirements for a traffic session to be offload eligible to an NP6 processor? (Choose three.)

- A. Session packets do NOT have an 802.1Q VLAN tag.
- B. It is NOT multicast traffic.
- C. It does NOT require proxy-based inspection.
- D. Layer 4 protocol must be UDP, TCP, SCTP or ICMP.
- E. It does NOT require flow-based inspection.

434. Which statement best describes what a Fortinet System on a Chip (SoC) is?

- A. Low-power chip that provides general purpose processing power
- B. Chip that combines general purpose processing power with Fortinet's custom ASIC technology
- C. Light-version chip (with fewer features) of an SP processor
- D. Light-version chip (with fewer features) of a CP processor

435. Which of the following statements are true regarding traffic accelerated by an NP processor? (Choose two.)

- A. TCP SYN packets are always handled by the NP Processor
- B. The initial packets go to the NP Processor, where a decision is taken on if the session can be offloaded or not.
- C. Packets for a session termination are always handled by the CPU.
- D. The initial packets go to the CPU, where a decision is taken on if the session can be offloaded or not.

436. Which is one of the conditions that must be met for offloading the encryption and decryption of IPsec traffic to an NP6 processor?

- A. no protection profile can be applied over the IPsec traffic.
- B. Phase-2 anti-replay must be disabled.
- C. Phase 2 must have an encryption algorithm supported by the NP6.
- D. IPsec traffic must not be inspected by any FortiGate session helper.

437. Which statement best describes the objective of the SYN proxy feature available in SP processors?

- A. Accelerate the TCP 3-way handshake
- B. Collect statistics regarding traffic sessions
- C. Analyze the SYN packet to decide if the new session can be offloaded to the SP processor
- D. Protect against SYN flood attacks.

438. Which of the following traffic shaping functions can be offloaded to a NP processor? (Choose two.)

- A. Queue prioritization
- B. Traffic cap (bandwidth limit)
- C. Differentiated services field rewriting
- D. Guarantee bandwidth

439. Which statement best describes what a Fortinet System on a Chip (SoC) is?

- A. Low-power chip that provides general purpose processing power
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- D. Light-version chip (with fewer features) of a CP processor

440. Which statements are true about offloading antivirus inspection to a Security Processor (SP)? (Choose two.)

- A. Both proxy-based and flow-based inspection are supported.
- B. A replacement message cannot be presented to users when a virus has been detected.
- C. It saves CPU resources.
- D. The ingress and egress interfaces can be in different SPs.

441. Which IP packets can be hardware-accelerated by a NP6 processor? (Choose two.)

- A. Fragmented packets.
- B. Multicast packet.
- C. SCTP packet.
- D. GRE packet.

QCM NSE4 : SD-WAN

442. Which of the following statements are true regarding WAN Link Load Balancing? (Choose two).

- A. There can be only one virtual WAN Link per VDOM.
- B. FortiGate can measure the quality of each link based on latency, jitter, or packets percentage.
- C. Link health checks can be performed over each link member if the virtual WAN interface.
- D. Distance and priority values are configured in each link member if the virtual WAN interface.

443. When using SD-WAN, how do you configure the next-hop gateway address for a member interface so that FortiGate can forward Internet traffic?

- A. It must be configured in a static route using the **sdwan** virtual interface.
- B. It must be provided in the SD-WAN member interface configuration.
- C. It must be configured in a policy-route using the **sdwan** virtual interface.
- D. It must be learned automatically through a dynamic routing protocol.

444. Which of the following statements are true regarding WAN Link Load Balancing? (Choose two).

- A. There can be only one virtual WAN Link per VDOM.
- B. FortiGate can measure the quality of each link based on latency, jitter, or packets percentage.
- C. Link health checks can be performed over each link member in the virtual WAN interface.
- D. Distance and priority values are configured in each link member in the virtual WAN interface.

QCM NSE4 Divers

445. Examine this output from a debug flow:

```
id=2 line=4677 msg="vd-root received a packet (proto=6, 66.171.121.44:80 ->10.200.1.1:49886)
[S.], seq 3567496940, ack 2176715502, win 5840"
id=2 line=4739 msg="Find an existing session, id=00007fc0, reply direction"
id=2 line=2733 msg="DNAT 10.200.1.1:49886 -> 10.0.1.10:49886"
id=2 line=2582 msg="find a route: flag=00000000 gw=10.0.1.10 via port3"
```

Which statements about the output are correct? (Choose two.)

- A. The packet was allowed by the firewall policy with the ID 00007fc0.
- B. FortiGate routed the packet through **port3**.
- C. FortiGate received a TCP SYN/ACK packet.
- D. The source IP address of the packet was translated to 10.0.1.10.

446. View the exhibit. Why is the administrator getting the error shown in the exhibit?

```
Login as: admin
Local-FortiGate #
Local-FortiGate # config vdom

Local-FortiGate (vdom) # edit root
current vf=root : 0

Local-FortiGate (root) # config system global

command parse error before 'global'
Command fail. Return code 1

Local-FortiGate (root) #
```

- A. The administrator `admin` does not have the privileges required to configure global settings.
- B. The global settings cannot be configured from the `root` VDOM context.
- C. The command `config system global` does not exist in FortiGate.
- D. The administrator must first enter the command `edit global`

447. In which process states is it impossible to interrupt/kill a process? (Choose two.)

- A. S-Sleep
- B. R-Running
- C. D-Uninterruptable Sleep
- D. Z-Zombie

448. Examine the following output from the `diagnose sys session list` command:

```
session info: proto=6 proto_state=65 duration=3 expire=9 timeout=3600
flags=00000000 sockflag=00000000 sockport=443 av_idx=9 use=5
origin-shaper=guarantee-100kbps prio=2 guarantee 12800Bps max
134217728Bps traffic 13895Bps
reply-shaper=guarantee-100kbps prio=2 guarantee 12800Bps max
134217728Bps traffic 13895Bps
state=redir local may_dirty ndr npu nlb os rs
statistic(bytes/packets/allow_err): org=864/8/1 reply=2384/7/1 tuples=3
origin->sink: org pre->post, reply pre->post dev=7->6/6->7
gwy=172.17.87.3/10.1.10.1
hook=post dir=org act=snat
192.168.1.110:57999->74.201.86.29:443(172.17.87.16:57999)
hook=pre dir=reply act=dnat 74.201.86.29:443-
>172.17.87.16:57999(192.168.1.110:57999)
hook=post dir=reply act=noop
74.201.86.29:443->192.168.1.110:57999(0.0.0.0:0)
misc=0 policy_id=1 id_policy_id=0 auth_info=0 chk_client_info=0 vd=0
npu info: flag=0x00/0x00, offload=0/0, ips_offload=0/0, epid=0/0,
ipid=0/0, vlan=0/0
```

Which statements are true regarding the session above? (Choose two.)

- A. Session Time-To-Live (TTL) was configured to 9 seconds.
- B. FortiGate is doing NAT of both the source and destination IP addresses on all packets coming from the 192.168.1.110 address.
- C. The IP address 192.168.1.110 is being translated to 172.17.87.16.
- D. The FortiGate is not translating the TCP port numbers of the packets in this session.

449. The exhibit shows a part output of the diagnostic command 'diagnose debug application ike 255', taken during establishment of a VPN. Which of the following statement are correct concerning this output? (Choose two)

```
Ike 0:Remote:7:22: responder received first quick-mode message
ike 0:Remote:7:22: peer proposal is: peer:0:0.0.0.0-255.255.255.255:0, me:0:0.0.0.0-255.255.255.255:0
ike 0:Remote:7: sent IKE msg (quick_risend): 172.20.186.222:500->172.20.187.114:500, len=356
ike 0: comes 172.20.187.114:500->172.20.186.222:500,ifindex=2....
ike 0:Remote:7:P2:22: replay protection enabled
ike 0:Remote:7:P2:22: SA life soft seconds=1750.
ike 0:Remote:7:P2:22: SA life hard seconds=1800.
ike 0:Remote:7:P2:22: IPsec SA selectors #src=1 #dst=1
ike 0:Remote:7:P2:22: src 0 7 0:0.0.0.0-255.255.255.255:0
ike 0:Remote:7:P2:22: dst 0 7 0:0.0.0.0-255.255.255.255:0
ike 0:Remote:7:P2:22: add IPsec SA: SPIs=6e13ca19/8f1ce9ae
ike 0:Remote:7:P2:22: added IPsec SA: SPIs=6e13ca19/8f1ce9ae
ike 0:Remote:7:P2:22: sending SNMP tunnel UP trap
```

- A. The quick mode selectors negotiated between both IPsec VPN peers is 0.0.0.0/32 for both source and destination addresses.
- B. The output corresponds to a phase 2 negotiation
- C. NAT-T enabled and there is third device in the path performing NAT of the traffic between both IPsec VPN peers.
- D. The IP address of the remote IPsec VPN peer is 172.20.187.114

450. A network administrator connects his PC to the INTERNAL interface on a FortiGate unit.

The administrator attempts to make an HTTPS connection to the FortiGate unit on the VLAN1 interface at the IP address of 10.0.1.1, but gets no connectivity.

The following troubleshooting commands are executed from the DOS prompt on the PC and from the CLI.

```
C:\>ping 10.0.1.1
Pinging 10.0.1.1 with 32 bytes of data:
Reply from 10.0.1.1: bytes=32 time<1ms TTL=255
Reply from 10.0.1.1: bytes=32 time<1ms TTL=255
Reply from 10.0.1.1: bytes=32 time<1ms TTL=255
Reply from 10.0.1.1: bytes=32 time<1ms TTL=255 user1 # get system interface
== [ internal ]
name. internal mode. static ip: 10.0.1.254 255.255.255.128 status: up
netbios-forward. disable type. physical mtu-override. disable
== [ vlan1 ]
name. vlan1 mode. static ip: 10.0.1.1 255.255.255.128 status: up netb
ios-forward. disable type. vlan mtu-override. disable
user1 # diagnose debug flow trace start 100
user1 # diagnose debug ena
user1 # diagnose debug flow filter daddr 10.0.1.1 10.0.1.1
id=20085 trace_id=274 msg="vd-root received a packet(proto=6,
10.0.1.130:47927->10.0.1.1:443) from internal."
id=20085 trace_id=274 msg="allocate a new session-00000b1b"
id=20085 trace_id=274 msg="find SNAT: IP-10.0.1.1, port-43798"
id=20085 trace_id=274 msg="iprope_in_check() check failed, drop"
```

Based on the output from these commands, which of the following explanations is a possible cause of the problem?

- A. The Fortigate unit has no route back to the PC.
- B. The PC has an IP address in the wrong subnet.
- C. The PC is using an incorrect default gateway IP address.
- D. The FortiGate unit does not have the HTTPS service configured on the VLAN1 interface.
- E. There is no firewall policy allowing traffic from INTERNAL-> VLAN1.

**451. Which are outputs for the command `diagnose hardware deviceinfo nic'?
(Choose two.)**

- A. ARP cache
- B. Physical MAC address
- C. Errors and collisions
- D. Listening TCP ports

452. What are examples of correct syntax for the session table diagnostics command? (Choose two.)

- A. diagnose sys session filter clear
- B. diagnose sys session src 10.0.1.254
- C. diagnose sys session filter
- D. diagnose sys session filter list dst.

453. The diag sys session list command is executed in the CLI. The output of this command is shown in the exhibit.

```
session info: proto=6 proto_state=11 duration=539 expire=3571 timeout=3600
flags=00000000 sockflag=00000000 sockport=80 av_idx=0 use=5
origin-shaper=guarantee-100kbps prio=1 guarantee 12288/sec max 134217728/sec
traffic 123/sec
reply-shaper=low-priority prio=3 guarantee 0/sec max 134217728/sec traffic 115/sec
per_ip_shaper=
ha_id=0 hakey=1335
policy_dir=0 tunnel=/
state=redir local may_dirty ndr os rs rem
statistic(bytes/packets/allow_err): org=3201/59/1 reply=2672/58/1 tuples=3
origin->sink: org pre->post, reply pre->post dev=9->3/3->9
gwy=76.27.192.1/192.168.203.2
hook=post dir=org act=snat 192.168.203.2:3196-
>128.100.58.53:80(76.27.195.147:58618)
hook=pre dir=reply act=dnat 128.100.58.53:80-
>76.27.195.147:58618(192.168.203.2:3196)
hook=post dir=reply act=noop 128.100.58.53:80->192.168.203.2:3196(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=10 id_policy_id=0 auth_info=0 chk_client_info=0 vd=0
serial=00115cae tos=ffff app_list=2000 app=0
dd_type=0 dd_rule_id=0
per_ip_bandwidth meter: addr=192.168.203.2, bps=1351
```

Based on the output from this command, which of the following statements is correct?

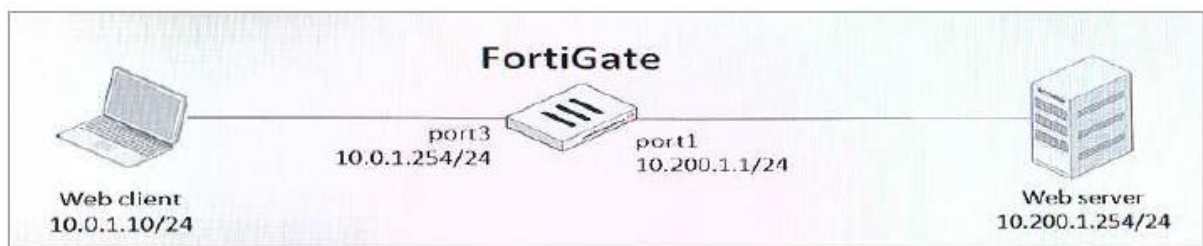
- A. This is a UDP session.
- B. Traffic shaping is being applied to this session.
- C. This is an ICMP session.
- D. This traffic has been authenticated.
- E. This session matches a firewall policy with ID 5.

454. In the debug command output shown in the exhibit, which of the following best described the MAC address 00:09:0f:69:03:7e?

```
# diagnose ip arp list
index=2 ifname=port1 172.20.187.150 00:09:0f:69:03:7e
state=00000004 use=4589 confirm=4589 update=2422 ref=1
```

- A. It is one of the secondary MAC addresses of the port1 interface.
- B. It is the primary MAC address of the port interface.
- C. It is the MAC address of another network devices located in the same LAN segment as the FortiGate unit's port1 interface.
- D. It is the HA virtual MAC address.

455. View the exhibit. The client cannot connect to the HTTP web server. The administrator run the FortiGate built-in sniffer and got the following output:



```
FortiGate # diagnose sniffer packet any "port 80" 4
interfaces=[any]
filters=[port 80]
11.510058 port3 in 10.0.1.10.49255 -> 10.200.1.254.80: syn 697263124
11.760531 port3 in 10.0.1.10.49255 -> 10.200.1.254.80: syn 868017830
14.505371 port3 in 10.0.1.10.49255 -> 10.200.1.254.80: syn 697263124
14.755510 port3 in 10.0.1.10.49255 -> 10.200.1.254.80: syn 868017830
```

What should be done next to troubleshoot the problem?

- A. Execute another sniffer in the FortiGate, this time with the filter "host 10.0.1.10".
- B. Run a sniffer in the web server.
- C. Capture the traffic using an external sniffer connected to port1.
- D. Execute a debug flow.

456. Examine this output from the diagnose sys top command:

```
# diagnose sys top 1
Run Time: 11 days, 3 hours and 29 minutes
ON, ON, 1S, 99I; 971T, 528F, 160KF
  sshd          123      S      1.9      1.2
  ipseengine    61       S <      0.0      5.2
  miglogd       45       S      0.0      4.9
  pyfcgid       75       S      0.0      4.5
  pyfcgid       73       S      0.0      3.9
```

Which statements about the output are true? (Choose two.)

- A. sshd is the process consuming most memory
- B. sshd is the process consuming most CPU
- C. All the processes listed are in sleeping state
- D. The sshd process is using 123 pages of memory

457. An administrator is using the FortiGate built-in sniffer to capture HTTP traffic between a client and a server, however, the sniffer output shows only the packets related with TCP session setups and disconnections. Why?

- A. The administrator is running the sniffer on the internal interface only.
- B. The filter used in the sniffer matches the traffic only in one direction.
- C. The FortiGate is doing content inspection.
- D. TCP traffic is being offloaded to an NP6.

458. Review the exhibit

Review the IPsec diagnostics output of the command `diagnose vpn tunnel list` shown in the exhibit below.

```
STUDENT # diagnose vpn tunnel list
list all ipsec tunnel in vd 0
-----
name=FCClient_0 ver=1 serial=3 10.200.1.1:4500->10.200.3.1:64916 lgwy=static tun=intf mode=dial_inst
parent=FCClient index=0
proxyid_num=1 child_num=0 refcnt=8 ilast=2 olast=2
stat: rxp=59 txp=0 rxb=15192 txb=0
dpd: mode=active on=1 idle=5000ms retry=3 count=0 seqno=10
natt: mode=keepalive draft=32 interval=10 remote_port=64916
proxyid=FCClient proto=0 sa=1 ref=2 auto_negotiate=0 serial=1
src: 0:0.0.0.0-255.255.255.255:0
dst: 0:172.20.1.1-172.20.1.1:0
SA: ref=3 options=00000006 type=00 soft=0 mtu=1280 expire=1717 replaywin=1024 seqno=1
life: type=01 bytes=0/0 timeout=1791/1800
dec: spi=a29046e9 esp=3des key=24 0525830c6fd67ca37e9d6dad174d175e24f97c3b87f428fa
    ah=sha1 key=20 982f8ba194f3f797773efc605c8321b728dabf1d
enc: spi=19be4052 esp=3des key=24 da597cb7fec913528f8598d1aa7ecd17156a2a7a4afeeb4c
    ah=sha1 key=20 9e2c5d0fc055fa0149bc66024732e9a85bbe8016
-----
```

Which statements are correct regarding this output (Choose two.)

- ☐ A. The connecting client has been allocated address 172.20.1.1.
- ☐ B. In the Phase 1 settings, dead peer detection is enabled.
- ☐ C. The tunnel is idle.
- ☐ D. The connecting client has been allocated address 10.200.3.1.

459. What FortiGate components are tested during the hardware test? (Choose three.)

- A. Administrative access
- B. HA heartbeat
- C. CPU
- D. Hard disk
- E. Network interfaces

460. Examine this output from a debug flow:

```
id=20085 trace_id=1 func=print_pkt_detail line=5363 msg="vd-root received a packet(proto=1,
10.0.1.10:1->10.200.1.254:2048)
from port3. type=8, code=0, id=1, seq=33."
id=20085 trace_id=1 func=init_ip_session_common line=5519 msg="allocate a new session=00000340"
id=20085 trace_id=1 func=vf_ip_route_input_common line=2583 msg="find a route: flag=04000000 gw=10.200.1.254
port1"
id=20085 trace_id=1 func=fw_forward_handler line=586 msg="Denied by forward policy check (policy 0)"
```

Why did the FortiGate drop the packet?

- A. The next-hop IP address is unreachable.
- B. It failed the RPF check.
- C. It matched an explicitly configured firewall policy with the action **DENY**.
- D. It matched the default implicit firewall policy.

461. Examine the exhibit, which contains a session diagnostic output.

```
session info: proto=6 proto state=01 duration=26 expire=3594 timeout=3600 flags=00000000 sockflag=
00000000 sockport=0 av_idx=0 use=4
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy-dir=0 tunnel=/ vlan_cos=0/255
state=may_dirty
statistic(bytes/packets/allow_err): org=1490/14/1 reply=10479/13/1 tuples=2
tx speed (Bps/kbps):56/0 rx speed(Bps/kbps):397/3
origin->sink: org pre->post, reply pre->post dev=5->3/3->5 gwy=10.200.1.254/10.0.1.10
hook=post dir=org act=snat 10.0.1.10:60267->52.84.125.124:443(10.200.1.100:60267)
hook=pre dir=reply act=dnat 52.84.125.124:443->10.200.1.100:60267(10.0.1.10:60267)
pos/ (before,after) 0/(0,0), 0/(0,0)
misc=0 policy id=1 auth info=0 chk client info=0 vd=0
serial=00009bd8 tos=ff/ff app_list=0 app=0 url_cat=0
dd type=0 dd_mode=0
total session 129
```

Which of the following statements about the session diagnostic output is true?

- A. The session is in ESTABLISHED state.
- B. The session is in LISTEN state.
- C. The session is in TIME_WAIT state.
- D. The session is in CLOSE_WAIT state.

462. Which of the following statements about backing up logs from the CLI and downloading logs from the GUI are true? (Choose two.)

- A. Log downloads from the GUI are limited to the current filter view
- B. Log backups from the CLI cannot be restored to another FortiGate.
- C. Log backups from the CLI can be configured to upload to FTP as a scheduled time
- D. Log downloads from the GUI are stored as LZ4 compressed files.

463. Which commands are appropriate for investigating high CPU? (Choose two.)

- A. diag sys top
- B. diag hardware sysinfo mem
- C. diag debug flow
- D. get system performance status

464. In a Crash log, what does a status of 0 indicate?

- A. Abnormal termination of a process
- B. A process closed for any reason
- C. Scanunitd process crashed
- D. Normal shutdown with no abnormalities
- E. DHCP process crashed

465. Examine the exhibit, which shows the partial output of an IKE real-time debug.

```
ike 0: comes 172.20.187.114:500->172.20.186.222:500,ifindex=2....
ike 0: IKEv1 exchange=Identity Protection id=4497f0b077c742b5/0000000000000000 len=296
ike 0:4497f0b077c742b5/0000000000000000:8: responder: main mode get 1st message...
...
ike 0:4497f0b077c742b5/0000000000000000:8: SA proposal chosen, matched gateway Remote
ike 0: found Remote 172.20.186.222 2 -> 172.20.187.114:500
...
ike 0:Remote:8: sent IKE msg (ident_r1send): 172.20.186.222:500->172.20.187.114:500, len=160
ike 0: comes 172.20.187.114:500->172.20.186.222:500,ifindex=2....
ike 0:Remote:8: responder:main mode get 2nd message...
....
ike 0:Remote:8: sent IKE msg (ident_r2send): 172.20.186.222:500->172.20.187.114:500, len=292
ike 0:Remote:8: ISAKMP SA 4497f0b077c742b5/fbbb59b259a0fc3e key 24:DCD18FBE7CFA138E27B06F
ike 0: comes 172.20.187.114:500->172.20.186.222:500,ifindex=2....
ike 0:Remote:8: responder: main mode get 3rd message...
...
ike 0:Remote:8: PSK authentication succeeded
ike 0:Remote:8: authentication OK
ike 0:Remote:8: established IKE SA 4497f0b077c742b5/fbbb59b259a0fc3e
```

Which of the following statement about the output is true?

- A. The VPN is configured to use pre-shared key authentication.
- B. Extended authentication (XAuth) was successful.
- C. Remote is the host name of the remote IPsec peer.
- D. Phase 1 went down.

466. Which statement correctly describes the output of the command `diagnose ips anomaly list`?

- A. Lists the configured DoS policy.
- B. List the real-time counters for the configured DoS policy.
- C. Lists the errors captured when compiling the DoS policy.
- D. Lists the IPS signature matches.

467. Review the IKE debug output for IPsec shown in the exhibit below. Which statements is correct regarding this output?

```
STUDENT # ike 0: comes 10.200.3.1:500->10.200.1.1:500,ifindex=2....
ike 0: IKEv1 exchange=Informational id=9e2606ac7ae83d7a/612da78d3ab3f945:15b10705 len=92
ike 0: in 9E2606AC7AE83D7A612DA78D3AB3F9450810050115B107050000005C26E2A7EC8461AC15E989C705B6C1F667A41957AE11FB7003C07A1
07B0934D036E1A1074348E08FD6839146C618525C6EC51E2F26885B688E035F52B4
ike 0:Remote_1:10: dec 9E2606AC7AE83D7A612DA78D3AB3F9450810050115B107050000005C0B000018E281874EECF170EE522216A4E3A927C71-
C000000200000000101108D299E2606AC7AE83D7A612DA78D3AB3F945000009C17511ED6EE549507
ike 0:Remote_1:10: notify msg received: R-U-THERE
ike 0:Remote_1:10: enc 9E2606AC7AE83D7A612DA78D3AB3F94508100501734C5CDF000000540B0000181C047F014CEEF1BCEC8DA915F3B18AEB0C
A000000200000000101108D299E2606AC7AE83D7A612DA78D3AB3F945000009C
ike 0:Remote_1:10: out 9E2606AC7AE83D7A612DA78D3AB3F94508100501734C5CDF0000005CB3CC431065A1737144B02F1AAACE79C1BE7125E4255
BBB4E5FA7A967FE99C7B731057FF33728BE42AA983E79C919DA9B64EBCE087EFOA02666C1FBD2C62F
ike 0:Remote_1:10: sent IKE msg (R-U-THERE-ACK): 10.200.1.1:500->10.200.3.1:500, len=92, id=9e2606ac7ae83d7a/612da78d3ab3f
734c5cdf
ike 0:Remote_1: link is idle 2 10.200.1.1->10.200.3.1:500 dpd=1 seqno=34
```

- A. The output is a phase 1 negotiation.
- B. The output is a phase 2 negotiation.
- C. The output captures the dead peer detection messages.
- D. The output captures the dead gateway detection packets.

468. An administrator observes that the `port1` interface cannot be configured with an IP address. What can be the reasons for that? (Choose three.)

- A. The interface has been configured for one-arm sniffer.
- B. The interface is a member of a virtual wire pair.
- C. The operation mode is transparent.
- D. The interface is a member of a zone.
- E. Captive portal is enabled in the interface.

469. What is not true of configuring disclaimers on the FortiGate?

- A. Disclaimers can be used in conjunction with captive portal.
- B. Disclaimers appear before users authenticate.
- C. Disclaimers can be bypassed through security exemption lists.
- D. Disclaimers must be accepted in order to continue to the authentication login or originally intended destination.

470. Which of the following spam filtering methods are supported on the FortiGate unit? (Select all that apply.)

- A. IP Address Check
- B. Open Relay Database List (ORDBL)
- C. Black/White List
- D. Return Email DNS Check
- E. Email Checksum Check

471. Under what circumstance would you enable LEARN as the Action on a firewall policy?

- A. You want FortiGate to compile security feature activity from various security-related logs, such as virus and attack logs.
- B. You want FortiGate to monitor a specific security profile in a firewall policy, and provide recommendations for that profile.
- C. You want to capture data across all traffic and security vectors, and receive learning logs and a report with recommendations.
- D. You want FortiGate to automatically modify your firewall policies as it learns your networking behavior.

472. Examine the following CLI configuration: What statement is true about the effect of the above configuration line?

```
config system session-ttl
set default 1800
end
```

- A. Sessions can be idle for more than 1800 seconds.
- B. The maximum length of time a session can be open is 1800 seconds.
- C. After 1800 seconds, the end user must re-authenticate.
- D. After a session has been open for 1800 seconds, the FortiGate sends a keepalive packet to both client and server.

473. File blocking rules are applied before which of the following?

- A. Firewall policy processing
- B. Virus scanning
- C. Web URL filtering
- D. White/Black list filtering

474. Which of the following Session TTL values will take precedence?

- A. Session TTL specified at the system level for that port number
- B. Session TTL specified in the matching firewall policy
- C. Session TTL dictated by the application control list associated with the matching firewall policy
- D. The default session TTL specified at the system level

475. Which TCP states does the global setting 'tcp-half-open-timer' applies to? (Choose two.)

- A. SYN SENT
- B. SYN & SYN/ACK
- C. FIN WAIT
- D. TIME WAIT

476. In FortiOS session table output, what are the two possible 'proto_state' values for a UDP session? (Choose two.)

- A. 00
- B. 11
- C. 01
- D. 05

477. Which is true of FortiGate's session table?

- A. NAT/PAT is shown in the central NAT table, not the session table.
- B. It shows TCP connection states.
- C. It shows IP, SSL, and HTTP sessions.
- D. It does not show UDP or ICMP connection state codes, because those protocols are connectionless.

478. In the debug command output shown in the exhibit, which of the following best described the MAC address 00:09:0f:69:03:7e ?

```
# diagnose ip arp list
index=2 ifname=port1 172.20.187.150 00:09:0f:69:03:7e
state=00000004 use=4589 confirm=4589 update=2422 ref=1
```

- A. It is one of the secondary MAC addresses of the port1 interface.
- B. It is the primary MAC address of the port interface.
- C. It is the MAC address of another network devices located in the same LAN segment as the FortiGate unit's port1 interface.
- D. It is the HA virtual MAC address.

479. Which of the following is an advantage of using SNMP v3 instead of SNMP v1/v2 when querying the FortiGate unit?

- A. Packet encryption
- B. MIB-based report uploads
- C. SNMP access limits through access lists
- D. Running SNMP service on a non-standard port is possible

480. An administrator wishes to generate a report showing Top Traffic by service type, but wants to exclude SMTP traffic from the report.

Which of the following statements best describes how to do this?

- A. In the Service field of the Data Filter, type 25/smtp and select the NOT checkbox.
- B. Add the following entry to the Generic Field section of the Data Filter: service="!smtp".
- C. When editing the chart, uncheck mlog to indicate that Mail Filtering data is being excluded when generating the chart.
- D. When editing the chart, enter 'dns' in the Exclude Service field.

481. An administrator has configured the following settings:

```
config system settings
set ses-denied-traffic enable
end
```

```
config system global
set block-session-timer 30
end
```

What does the configuration do? (Choose two.)

- A. Reduces the amount of logs generated by denied traffic.
- B. Enforces device detection on all interfaces for 30 minutes.
- C. Blocks denied users for 30 minutes.
- D. Creates a session for traffic being denied.

QCM_NSE4_Application Control

482. How do you configure a FortiGate to apply traffic shaping to P2P traffic, such as BitTorrent?

- A. Apply a traffic shaper to a BitTorrent entry in an application control list, which is then applied to a firewall policy.
- B. Enable the *shape* option in a firewall policy with *service* set to BitTorrent.
- C. Define a DLP rule to match against BitTorrent traffic and include the rule in a DLP sensor with traffic shaping enabled.
- D. Apply a traffic shaper to a protocol options profile.

483. The exhibit is a screen shot of an Application Control profile. Different settings are circled and numbered. Select the number identifying the setting which will provide additional information about YouTube access, such as the name of the video watched.

Categories

All Categories

- Proxy (106)
- Storage.Backup (296, 16)
- VoIP (31)
- Remote.Access (91)
- Update (48)
- Web.Client (18)
- Social.Media (150, 31)
- Video/Audio (206, 13) 1**
- Unknown Applications 2**

Application Overrides

+ Add Signatures Edit Parameters Delete

Application Signature	Category	Action
YouTube	Video/Audio	Monitor 3
YouTube_Search.Video	Video/Audio	Monitor
YouTube_Video.Access	Video/Audio	Monitor
YouTube_Video.Play	Video/Audio	Monitor

Options

Allow and Log DNS Traffic ☒

QUIC ☒ Allow Block 4

Replacement Messages for HTTP-based Applications ☒ 5

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

484. How do application control signatures update on a FortiGate device?

- A. Through FortiGuard updates.
- B. Upgrade the FortiOS firmware to a newer release.
- C. By running the Application Control auto-learning feature.
- D. Signatures are hard coded to the device and cannot be updated.

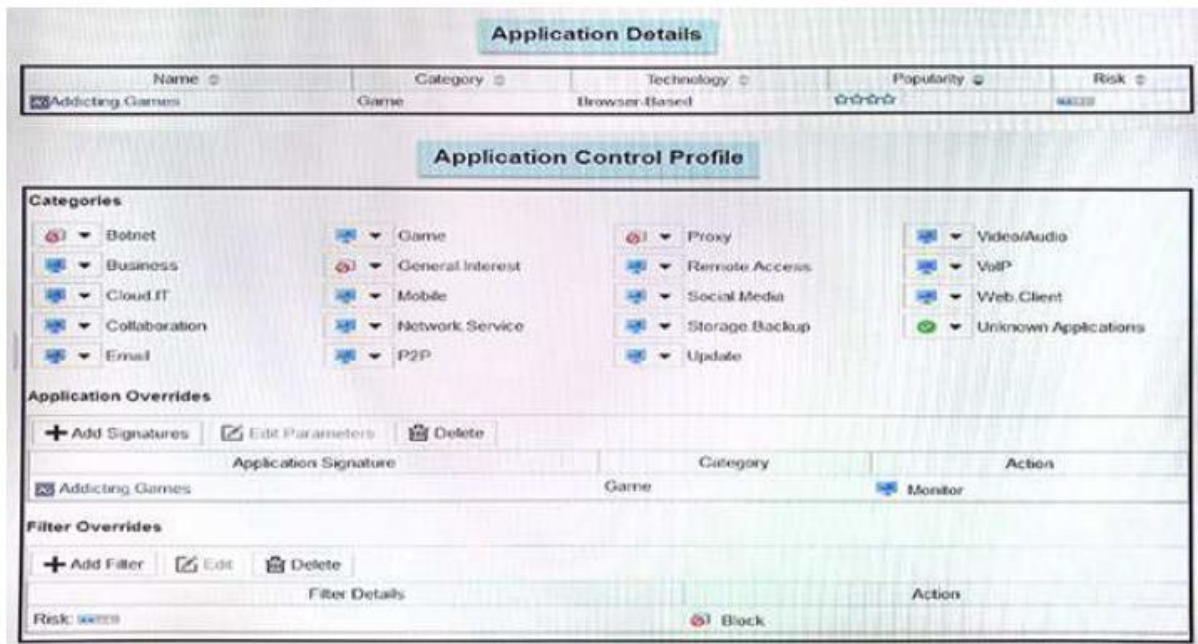
485. Which answer best describes what an "Unknown Application" is?

- A. All traffic that matches the internal signature for unknown applications.
- B. Traffic that does not match the RFC pattern for its protocol.
- C. Any traffic that does not match an application control signature
- D. A packet that fails the CRC check.

486. What actions are possible with Application Control? (Choose three.)

- A. Warn
- B. Allow
- C. Block
- D. Traffic Shaping
- E. Quarantine

487. A user behind the FortiGate is trying to go to <http://www.addictinggames.com> (Addicting.Games). Based on this configuration, which statement is true?



- A. **Addicting.Games** is allowed based on the **Application Overrides** configuration.
- B. **Addicting.Games** is blocked based on the **Filter Overrides** configuration.
- C. **Addicting.Games** can be allowed only if the **Filter Overrides** actions is set to **Exempt**.
- D. **Addicting.Games** is allowed based on the **Categories** configuration.

488. Which statements about application control are true? (Choose two.)

- A. Enabling application control profile in a security profile enables application control for all the traffic flowing through the FortiGate.
- B. It cannot take an action on unknown applications.
- C. It can inspect encrypted traffic.
- D. It can identify traffic from known applications, even when they are using non-standard TCP/UDP ports.

489. Which statements are true regarding traffic shaping that is applied in an application sensor, and associated with a firewall policy? (Choose two.)

- A. Shared traffic shaping cannot be used.
- B. Only traffic matching the application control signature is shaped.
- C. Can limit the bandwidth usage of heavy traffic applications.
- D. Per-IP traffic shaping cannot be used.

490. Which of the following statements are true regarding application control? (Choose two.)

- A. Application control is based on TCP destination port numbers.
- B. Application control is proxy based.
- C. Encrypted traffic can be identified by application control.
- D. Traffic shaping can be applied to the detected application traffic.

491. The exhibit is a screen shot of an Application Control profile. Different settings are circled and numbered. Select the number identifying the setting which will provide additional information about YouTube access, such as the name of the video watched.



- A. 1
B. 2
C. 3
D. 4
E. 5

492. Which action can be applied to each filter in the application control profile?

- A. Block, monitor, warning, and quarantine
B. Allow, monitor, block and learn
C. Allow, block, authenticate, and warning
D. Allow, monitor, block, and quarantine

493. View the exhibit. Based on the configuration shown in the exhibit, what statements about application control behavior are true? (Choose two.)

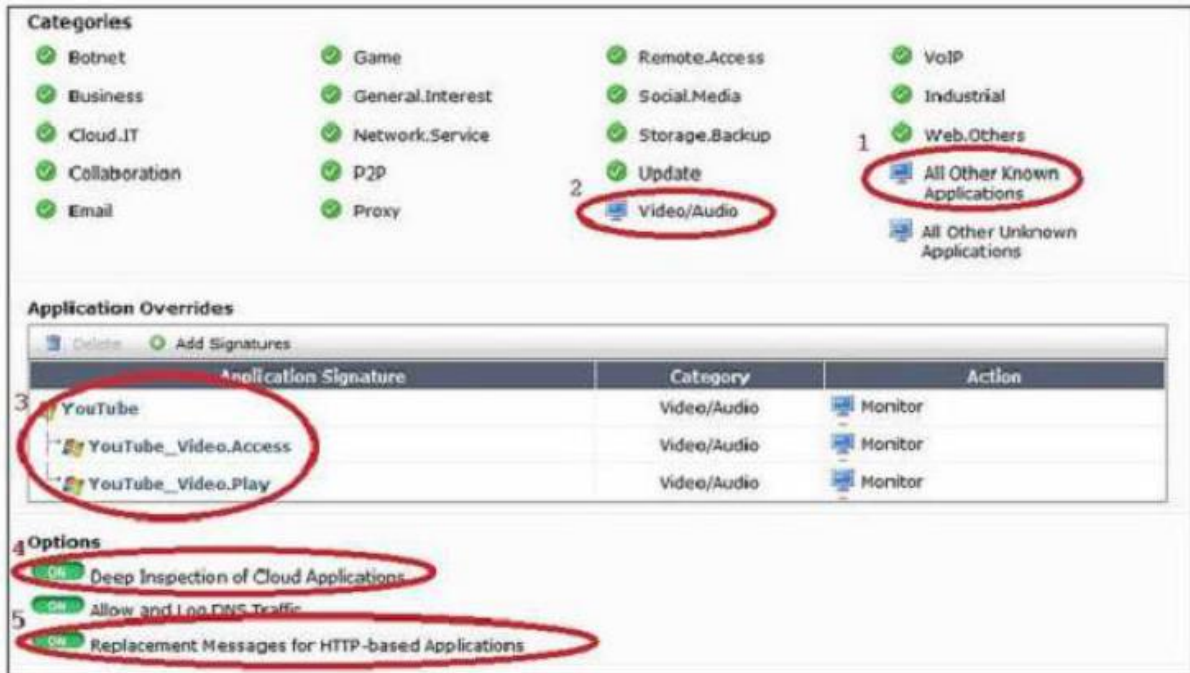
ID	Name	Source	Destination	Schedule	Service	Applications	URL Category	Action	NAT	Security Profiles	Log
port 3 → port 1											
2	Video/Audio	all	all	always	ALL	Video/Audio	+	DENY		SSL certificate-inspection	All
4	Social_Media	all	all	always	ALL	Social Media	Social Networking	DENY		SSL certificate-inspection	All
3	ALLOW_ALL	all	all	always	ALL			ACCEPT	Custom		UTM
Implicit											
0	Implicit Deny	all	all	always	ALL			DENY			Disabled

- A. Access to all unknown applications will be allowed.
B. Access to browser-based **Social.Media** applications will be blocked.
C. Access to mobile social media applications will be blocked.
D. Access to all applications in **Social.Media** category will be blocked.

**494. Which of the following statements are true regarding application control?
(Choose two.)**

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- C. Encrypted traffic can be identified by application control.
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Different settings are circled and numbered. Select the number identifying the setting which will provide additional information about YouTube access, such as the name of the video watched.

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- C. 3
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- E. 5

496. How do application control signatures update on a FortiGate device?

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- D. Signatures are hard coded to the device and cannot be updated.

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- B. Traffic that does not match the RFC pattern for its protocol.
- C. Any traffic that does not match an application control signature
- D. A packet that fails the CRC check.

QCM_Answers

1	D	44	C	87	AD	130	AC	173	A	216	D
2	CDE	45	A	88	D	131	A ?	174	BD	217	B
3	B	46	C	89	C	132	B	175	C	218	A
4	BCE	47	AC	90	D	133	D	176	B	219	B
5	CDE	48	BCD	91	AC	134	B	177	B	220	A
6	A	49	D	92	B	135	ADE	178	A	221	A
7	ABD	50	D	93	A	136	BC	179	A	222	B
8	AB	51	ABC	94	C	137	D	180	A	223	B
9	CD	52	BCD	95	A	138	AB	181	A	224	BD
10	A	53	D	96	C	139	CD	182	AB	225	CD
11	B	54	A	97	A	140	BD	183	D	226	ABD
12	D	55	A	98	C	141	AD	184	C	227	B
13	C	56	AE	99	AC	142	BD	185	BC	228	C
14	C	57	A	100	C	143	AB	186	CDF	229	CD
15	A	58	A	101	D	144	CD	187	CD	230	D
16	C	59	CD	102	CD	145	D	188	D	231	BC
17	ACE	60	A	103	ABE	146	C	189	AD	232	AB
18	BD	61	A	104	B	147	B	190	BC	233	B
19	BC	62	A	105	D	148	B	191	A	234	AD
20	AD	63	B	106	BC	149	D	192	BCE	235	AD
21	BC	64	AD	107	ABD	150	ADE	193	CD	236	AD
22	AD	65	A	108	AB	151	BCEF	194	ABC	237	C
23	B	66	A	109	A	152	AD	195	AB	238	BC
24	ABC	67	D	110	C	153	A	196	CD	239	D
25	BD	68	AB	111	AB	154	AC	197	ABD	240	A
26	A	69	D	112	AC	155	B	198	CDE	241	D
27	B	70	C	113	D ?	156	AB	199	D	242	D
28	C	71	D	114	AB	157	AC	200	E	243	AD
29	ABC	72	B	115	CD	158	AB	201	BCD	244	ABE
30	ABCD	73	BD	116	ABE	159	AC	202	D	245	AB
31	EXECUTE	74	B	117	ABC	160	AB	203	A	246	C
32	D	75	B	118	A	161	BC	204	B ?	247	ABCD
33	A	76	AC	119	C	162	ABE	205	B	248	B
34	ABC	77	ACD	120	B	163	D	206	B	249	ABCE
35	ABC	78	AD	121	AB	164	D	207	A	250	A
36	C	79	A	122	A	165	BEG	208	BD	251	B
37	B	80	CD	123	C	166	C	209	D	252	A
38	AB	81	D	124	BC	167	A	210	A	253	CD
39	BCE	82	C	125	ADE	168	BDE	211	B	254	A
40	AD	83	AD	126	A	169	BDE	212	B	255	A
41	D	84	D	127	CD	170	A	213	C	256	B
42	ACE	85	C	128	C	171	B	214	BCE	257	C
43	E	86	B	129	B	172	C	215	BC	258	A

259	CD	303	AC	347	AD	391	AC	435	AD	479	A
260	B	304	D	348	B	392	C	436	C	480	A
261	B	305	BC	349	ACE	393	AD	437	D	481	CD
262	BD	306	D	350	BC	394	A	438	CD	482	A
263	B	307	C	351	CD	395	D	439	B	483	D
264	D	308	D	352	D	396	D	440	BC	484	A
265	ABE	309	D	353	C	397	BD	441	BC	485	C
266	A	310	AC	354	D	398	B	442	AC	486	BCE
267	A	311	c	355	CD	399	C	443	B	487	A
268	C	312	C	356	AC	400	CD	444	AC	488	CD
269	D	313	ABC	357	C	401	A	445	BD	489	BC
270	BCE	314	ABCD	358	C	402	CD	446	B	490	CD
271	BC	315	ABC	359	D	403	BE	447	CD	491	D
272	D	316	ACD	360	A	404	A ?	448	CD	492	D
273	C	317	D	361	A	405	D	449	BD	493	AB
274	D	318	A	362	BC	406	A	450	D	494	CD
275	AD	319	B	363	ADE	407	A	451	BC	495	D
276	A	320	C	364	BC	408	D	452	AC	496	A
277	BD	321	B	365	ABE	409	CDE	453	B	497	C
278	CD	322	C	366	B	410	B	454	C	498	
279	D	323	CD	367	AC	411	C	455	D	499	
280	BD	324	AC	368	AD	412	A	456	BC	500	
281	C	325	A	369	C	413	D	457	D	501	
282	AD	326	C	370	CD	414	B	458	AB	502	
283	D	327	B	371	B	415	ABD	459	CDE	503	
284	C	328	A	372	AB	416	ABD	460	D	504	
285	AC	329	B	373	AC	417	Aucune ?	461	A	505	
286	ABE	330	BD	374	BD	418	A	462	AB	506	
287	CD	331	B	375	BC	419	C	463	AD	507	
288	BC	332	AD	376	B	420	E	464	D	508	
289	D	333	AD	377	D	421	ACE	465	A	509	
290	C	334	AD	378	AB	422	A	466	B	510	
291	AB	335	AC	379	C	423	D	467	C	511	
292	AD	336	C	380	A	424	BC	468	ABC	512	
293	D	337	D	381	CD	425	BD	469	C	513	
294	CD	338	AD	382	ABD	426	A	470	ABCDE	514	
295	A	339	B	383	B	427	D	471	B	515	
296	CD	340	AD	384	AC	428	ACD	472	A	516	
297	BCD	341	A	385	ACE	429	BC	473	B	517	
298	A	342	AB	386	BC	430	AB	474	C	518	
299	BE	343	B	387	BD	431	ADE	475	AD	519	
300	ABCD	344	AB	388	AD	432	D	476	AC	520	
301	BE	345	D	389	B	433	CDE	477	B	521	
302	D	346	B	390	ABD	434	B	478	C	522	

QCM_empty_grid

1		44		87		130		173		216	
2		45		88		131		174		217	
3		46		89		132		175		218	
4		47		90		133		176		219	
5		48		91		134		177		220	
6		49		92		135		178		221	
7		50		93		136		179		222	
8		51		94		137		180		223	
9		52		95		138		181		224	
10		53		96		139		182		225	
11		54		97		140		183		226	
12		55		98		141		184		227	
13		56		99		142		185		228	
14		57		100		143		186		229	
15		58		101		144		187		230	
16		59		102		145		188		231	
17		60		103		146		189		232	
18		61		104		147		190		233	
19		62		105		148		191		234	
20		63		106		149		192		235	
21		64		107		150		193		236	
22		65		108		151		194		237	
23		66		109		152		195		238	
24		67		110		153		196		239	
25		68		111		154		197		240	
26		69		112		155		198		241	
27		70		113		156		199		242	
28		71		114		157		200		243	
29		72		115		158		201		244	
30		73		116		159		202		245	
31		74		117		160		203		246	
32		75		118		161		204		247	
33		76		119		162		205		248	
34		77		120		163		206		249	
35		78		121		164		207		250	
36		79		122		165		208		251	
37		80		123		166		209		252	
38		81		124		167		210		253	
39		82		125		168		211		254	
40		83		126		169		212		255	
41		84		127		170		213		256	
42		85		128		171		214		257	
43		86		129		172		215		258	

259		303		347		391		435		479	
260		304		348		392		436		480	
261		305		349		393		437		481	
262		306		350		394		438		482	
263		307		351		395		439		483	
264		308		352		396		440		484	
265		309		353		397		441		485	
266		310		354		398		442		486	
267		311		355		399		443		487	
268		312		356		400		444		488	
269		313		357		401		445		489	
270		314		358		402		446		490	
271		315		359		403		447		491	
272		316		360		404		448		492	
273		317		361		405		449		493	
274		318		362		406		450		494	
275		319		363		407		451		495	
276		320		364		408		452		496	
277		321		365		409		453		497	
278		322		366		410		454		498	
279		323		367		411		455		499	
280		324		368		412		456		500	
281		325		369		413		457		501	
282		326		370		414		458		502	
283		327		371		415		459		503	
284		328		372		416		460		504	
285		329		373		417		461		505	
286		330		374		418		462		506	
287		331		375		419		463		507	
288		332		376		420		464		508	
289		333		377		421		465		509	
290		334		378		422		466		510	
291		335		379		423		467		511	
292		336		380		424		468		512	
293		337		381		425		469		513	
294		338		382		426		470		514	
295		339		383		427		471		515	
296		340		384		428		472		516	
297		341		385		429		473		517	
298		342		386		430		474		518	
299		343		387		431		475		519	
300		344		388		432		476		520	
301		345		389		433		477		521	
302		346		390		434		478		522	